

HelixCode CLI-Agent Fusion — Programme Continuation Guide

Last updated: 2026-05-29 (close-out¹⁵⁰ — PARALLEL STREAMS landed (operator “work several streams in parallel”). Stream A / HXC-029: 3 more API banks (entity-management 35 PASS / admin-operations 22 / security-validation 16) self-driving vs the LIVE server, each 3× deterministic + mutation-proof; helix_qa 3e7d339 pushed all remotes; 4/18 banks now verified. Stream B / HXC-034 wave 1: §11.4.102 cascaded into ALL 10 dependencies/HelixDevelopment/* submodules (debate_orchestrator/doc_processor/helix_llm/helix_memory/helix_specifier/llm_orchestrator/llm_provider/llms_verifier/models/vision_engine), committed + pushed + ls-remote-verified (2 diverged mirrors cleanly rebased, no force); 10 meta pointers bumped this commit. Stream C: github catch-up detached. HXC-035 FILED (High): the bank verification surfaced a real product defect — POST /api/v1/auth/register → 400 internal_auth_failed_create_user on the live DB (blocks all authenticated paths); systematic-debugging dispatched per §11.4.102. Open set 5: HXC-029 (14 banks left), HXC-031, HXC-034 (waves 2+: vasic-digital/* + top-level), HXC-035 (debugging), + note debate_orchestrator needs §11.4.98/99/101 backfill. PRIOR close-out¹⁴⁹ — CONSTITUTION §11.4.102 ADDED + HXC-029 first bank verified + github-throughput root-caused. (1) §11.4.102 (operator mandate): mandatory superpowers:systematic-debugging activation on ANY issue (Iron Law — no fix without root-cause investigation; composes §11.4.6/§11.4.7) + using-superpowers always-loaded + plugin/dependency-availability — added to constitution Constitution.md/CLAUDE.md/AGENTS.md/QWEN.md (+exports), committed 656b43a, pushed+ls-remote-verified to the constitution upstream (over :443). Meta pointer bumped 8ec2b38→656b43a (this commit). Cascade into 68 owned submodules + CM-COVENANT-114-102-PROPAGATION gate filed as **HXC-034** (large follow-up). (2) **HXC-029** first bank: helixcode-full-qa-api.json (helix_qa f5c6182) self-driving vs the LIVE server (PG+Redis+helixcode :8080 stood up) — 3 deterministic runs 32 PASS/0 FAIL/2 honest-SKIP + mutation-proof; committed 26acc959, pushed. (3) **github-push root cause FOUND** via instrumented push (systematic-debugging per the new §11.4.102): upload throughput to github collapsed to ~**173 B/s** (gitlab fast; reachability+MTU+auth+port all fine) → large**

binary-export packs time out = the broken-pipe; environmental (ISP/route), handled via gitlab-primary + detached-retry. Open set 3: HXC-029 (17 banks remain), HXC-031, HXC-034. PRIOR close-out¹⁴⁸ — HXC-030 §11.4.99 OPERATOR-DOC SWEEP COMPLETE + CLOSED. 8 doc batches (subagent-driven per §11.4.70 — worktree then serial-non-worktree once worktree stale-base re-work surfaced) footered **38/38 (100%)** operator-facing instruction/guide/manual/setup/troubleshooting/tutorial docs, each via LIVE WebFetch of official sources (go.dev/postgresql.org/redis/ollama/opentelemetry/platform.claude.com) — never training data. Real stale fixes: Go 1.24→1.26.3, golang:1.21→1.26 Dockerfiles, go1.21.5→1.26.3, postgres 14→15, Anthropic/OpenAI doc-redirect URLs; honest §11.4.99(B) negative findings where sources 403'd; CONST-036 model IDs flagged-not-guessed. New scripts/gates/sources_verified_gate.sh wired **G13 ENFORCING** (was advisory; flipped at 100%) — future operator doc without a footer FAILs the sweep; scope correctly excludes qa_evidence/helix_qa/architecture/coverage/materials (evidence/internal, not instructions). git push: gitlab reliable; github intermittently broken-pipes (host↔github) but a DETACHED persistent-retry loop converges it every time (caught up at attempt 7) — all commits on BOTH remotes. Open set now 2: HXC-029 (18 auto-executable bank rewrites, needs infra), HXC-031 (Codex/Cline ports). PRIOR close-out¹⁴⁷ — HXC-033 RESOLVED + CLOSED: codegraph 0.9.7 own-org index restored. ROOT CAUSE (operator's data-compat hypothesis CONFIRMED): 0.9.7 requires explicit codegraph init before index; the pre-0.9.7 DB was incompatible. Fix (operator-directed wipe+reindex): wiped gitignored DB (codegraph.db/-wal/-shm; 1.7 GB stale WAL) + codegraph init (tracked config.json preserved) + codegraph index .. Result Files 75,663 / Nodes 1,272,492 (≈ HXC-017 baseline; edges finalize async). **§11.4.79 probe PASS via codegraph query:** NewBundleTranslator→dependencies/HelixDevelopment/llm_orchestrator + vasic-digital (10 own-org hits); third-party LLama_CPP excluded. Corrected two of my own mis-diagnoses per §11.4.6: the “crash” was a faulty pgrep pattern (process was healthy, just slow); “own-org unreachable” was the wrong CLI verb (search→query in 0.9.7) + a stale MCP-server DB. Bonus: de-bloated docs/codegraph/Status.md 3.66 MB→8 KB (ANSI-spinner garbage from codegraph_sync.sh) + generated its HTML/PDF siblings. Non-blocking follow-ups: restart codegraph MCP server (tools/codegraph) to serve the fresh DB; fix codegraph_sync.sh ANSI dump (§11.4.26 constitution-submodule). **Parallel streams this window (operator “multiple background streams”):** **HXC-029 bank-classification (f99ae5b5 — 18/19 auto-executable, 0 genuinely-manual, NO live §11.4.98 violation) + HXC-030 §11.4.99 doc batch 2 (f23585b7 — PROVIDER_FEATURES/SECURITY/DEPLOYMENT WebFetch-verified) landed concurrently.** github push lagging on broken-pipe (gitlab current; will reconcile). Open set 3: HXC-029/030/031. PRIOR close-out¹⁴⁶ — codegraph 0.9.7 → HXC-033 FILED (Operator-blocked). Operator installed codegraph 0.9.7; §11.4.80

post-update sync FAILED: the install reset the gitignored index DB and every rebuild attempt died — full codegraph index . KILLED mid-run twice (no diagnostic; 54,207 then --force 4,630 files vs HXC-017's 76,044), codegraph sync . exit 1 (8,461). MCP codegraph_search BundleTranslator resolves ONLY helix_code, NOT the own-org llm_orchestrator symbol → own-org submodules dropped from the index = §11.4.79 regression. Host memory ample (not §12.6 OOM); tracked .codegraph/config.json intact (own-org includes + §11.4.10 credential excludes present — not a config regression). Root cause UNCONFIRMED per §11.4.6 (0.9.7 crashes with no actionable diagnostic). Self-resolution exhausted (quiet/force/sync/memory/flags/config all checked); per §11.4.101 surfaced to operator (downgrade / upstream-bug / accept-degraded — autonomous downgrade would violate §11.4.80's latest-installed). Documented in docs/codegraph/Status.md; evidence in qa-results/codegraph_.log. Open set now 4: HXC-029/030/031 + HXC-033(blocked). PRIOR close-out¹⁴⁵ — HXC-032 FIXED + CLOSED (operator “do everything now”). Resolved the LLMOrchestrator broken-merge build-breaker: all 26 committed conflict hunks across 5 Go files (translator.go/ translator_test.go/multi_pool.go/cmd/orchestrator/main.go/ challenges/runner/main.go) resolved to the HEAD (i18n-migrated) side — the consumer-consistent lineage (Pkg()/SetPkgTranslator()/ TPlural). Made the package coherent: bundle.go BundleTranslator gained an honest TPlural (interpolates count; no CLDR plural-form selection claimed per §11.9); automation_test.go TestAutomation_UpstreamsScripts aligned to the lowercase upstreams/ rename (4350384) that caused the conflict. **Verification:** **submodule go build+go vet+go test ./... 10/10 packages PASS (was build failure), zero conflict markers, downstream helix_agent go build ./... exit 0 (was broken). Submodule committed d3956ad, pushed origin/master 1e198e3..d3956ad (FF, no force), verified via ls-remote; meta .gitmodules pointer bumped same meta commit.** HXC-032 closed Fixed (→ Fixed.md) per CONST-057. Open set now 3: HXC-029 (§11.4.98 long-tail), HXC-030 (§11.4.99 full sweep), HXC-031 (Codex/Cline ports). DB 145 items validate OK + WAL-checkpointed; G11/G12/clarity PASS. PRIOR close-out¹⁴⁴ — §11.4.99 DOC RECONCILIATION + HXC-031 RECLASSIFIED + HXC-032 FILED (operator “do it all”). (1) HXC-030 §11.4.99 increment (commit 68443b8c): reconciled stale Go 1.24→1.26.3 (verified vs go.dev/dl: go1.26.3 latest stable; 1.24 past support) + PostgreSQL 14→15 across 8 operator-facing docs (DEPLOYMENT_GUIDE, troubleshooting/guide, user_manual/README, general/DEVELOPER_GUIDE + SLASH_COMMANDS, 3 tutorials), each with a ## Sources verified 2026-05-29 footer; README already done in HXC-028; full §11.4.99 30-doc sweep continues. (2) HXC-031 RECLASSIFIED: the “deferred build-breaking CONST-052 renames” were stale prose — investigation found NO owned-org renames remain (HelixDevelopment/ leaves already lowercase; only third-party HuggingFace_Hub/LLama_CPP/Ollama + 1 third-party Upstreams/ remain, all CONST-052-exempt; parent org-dirs

are intentional carve-outs per HXC-001). Only the Codex/Cline reference-agent ports genuinely remain. (3) HXC-032 FILED (Bug, High): scripts/const052_verify_refs.sh surfaced committed git conflict markers in 5 LLMOrchestrator Go files (26 hunks: translator.go/translator_test.go/multi_pool.go/cmd/orchestrator/main.go/challenges/runner/main.go) breaking helix_agent build — a §11.4 build-layer bluff, ALREADY on origin/master; root cause = i18n-migration lineage (8032035) merged with CONST-052 rename 4350384 markers-unresolved; verified-correct side = HEAD (i18n-migrated — consumers need Pkg()/SetPkgTranslator()/TPlural); deferred to a dedicated build+test-verified pass per §11.4.101 rather than rush-push 26 hunks to the remote (submodule left at its coherent committed state). DB 144 items validate OK + WAL-checkpointed; G11/G12/clarity PASS; tracker exports regenerated. PRIOR close-out¹⁴³ — HXC-029 §11.4.98 CONFIRMED-VIOLATION FIXES (operator “do it all now, yes”, local-only). Fixed both hard findings from the HXC-029 audit: (1) helix_code/tests/regression/server_timeout_test.go TestServerStability was t.Skip("run manually") over an unimplemented body — a §11.4 / §11.4.98 PASS-bluff — now a REAL self-driving test (real net/http server with config timeout semantics, real HTTP I/O, asserts idle-survival; PASS -count=3 ~1.75s each); also repaired the sibling TestServerTimeoutConfiguration CWD-dependent require.NoError(config.Load) hard-fail → skip-with-reason; whole tests/regression package now GREEN (was RED at HEAD, pre-existing). (2) helix_code/tests/e2e/scripts/clean.sh interactive read -p → --force/CLEAN_FORCE=1/TTY-gated self-driving (keeps results non-interactively, no hang; bash -n clean). HXC-029 → In progress; remaining = 19 HelixQA manual-review-required banks + ~24 integration files long-tail. DB resynced (143 items, validate OK, WAL-checkpointed); summaries+exports regenerated; G11/G12/clarity gates PASS. PRIOR close-out¹⁴² — DOC-SYNC + FORWARD-WORK FILING (operator “do it all”, local-only, no push). **Items 1-3 (safe):** (1) closed the long-standing §11.4.91 gap — the previously hand-maintained summaries had a “TODO: create” generator; built scripts/generate_issues_summary.sh + scripts/generate_fixed_summary.sh (mechanical, parse Issues.md/Fixed.md headings + Status/Type/Discovered, --check mode) and regenerated the stale summaries (were “1 open / round-464”; now **3 open / 34 closed**, Fixed_Summary Bug=25/Feature=69/Task=12/total=106); added scripts/gates/summary_sync_gate.sh wired as **G12** in verify-all-constitution-rules.sh (CM-{ISSUES,FIXED}-SUMMARY-SYNC freshness; paired §1.1 mutation proven PASS→stale→FAIL→restore→PASS) — this is the HXC-013-follow-up auto-sync seam; existing G11 (CM-WORKABLE-ITEMS-MD-DB-IN-SYNC) + §11.4.91 clarity gate both re-confirmed PASS (clarity 36/0). (2) resolved the helix_qa working-tree drift — 8 third-party tools/opensource/* nested-submodule gitlinks restored to recorded SHAs (stale index.lock cleared per §11.4.82(H)); meta tree clean. (3) confirmed Issues.md/Fixed.md are byte-identical to the DB (no tracker drift — earlier

“drift” was a stale *summary*, now fixed). **Items 4-6 (filed per operator):** HXC-029 (§11.4.98 forward — full-automation compliance sweep of every live/integration/e2e/Challenge test), HXC-030 (§11.4.99 forward — latest-source doc cross-reference sweep across all operator docs) **dispatched as background subagents** (§11.4.70/89) to produce first-deliverable audit ledgers under docs/qa/HXC-029|030/; HXC-031 (deferred build-breaking CONST-052 renames + Codex/Cline ports) filed **EXECUTION HELD** per §11.4.101 (high-blast-radius, awaiting per-batch operator go-ahead). DB resynced 140→143 items, validate OK, byte-identical round-trip, WAL-checkpointed (§11.4.95). PRIOR close-out¹⁴¹ — ALL TRACKER ITEMS CLOSED; open set was EMPTY (137 items: 72 Implemented + 41 Fixed + 24 Completed, DB-validated). This close-out lands **HXC-013** (SQLite single-source-of-truth, §11.4.93/95) + **HXC-025** (constitution §11.4.98/99/101 cascade). **HXC-013:** the constitution submodule’s non-functional workable-items scaffold is now a FUNCTIONAL Go binary (built+pushed UPSTREAM per §11.4.74 at constitution e460a5d) — sync md-to-db/db-to-md/validate/diff, byte-identical round-trip (0 diff on the real 137-item tracker via body_md blobs), go test passing; HelixCode now ships the TRACKED docs/workable_items.db (§11.4.95, WAL-checkpointed, .gitignore !docs/workable_items.db) + constitution pin → e460a5d. Operator-decisions resolved with the scope’s recommended safe-defaults per §11.4.101. Follow-up: auto-sync wiring + CM-WORKABLE-ITEMS gate. **HXC-025:** constitution fetched 15cd4bc→6017af9 (new universal anchors §11.4.98 full-automation-anti-bluff, §11.4.99 latest-source-doc-xref, §11.4.101 autonomous-decision; §11.4.100 ATMOSphere-demoted, not cascaded); cascaded all 3 into HelixCode’s 5 root govfiles + ALL 68 owned submodules (6 parallel subagent waves + pilot; each submodule committed+pushed to its org remotes, append-only, non-ff union-merged, no force-push); verify-governance-cascade.sh extended to 27 anchors → 0 failures (root + 68 submodules); CONST-055 sweep G1-G10 → 0 failures. New constitution mandates §11.4.98 (live-test full-automation audit) + §11.4.99 (doc latest-source xref) imply forward work. **CUMULATIVE SESSION (close-outs 139-141): 34 real production bugs fixed across 35 packages; HXC-013/014/014b/015/018/019/024/025 all closed; ~45 commits across HelixCode + constitution + 68 submodules, all pushed. Test infra (podman PG×2/Redis/Ollama) left running for re-runs.** PRIOR close-out¹⁴⁰ — operator authorised the infra batch + HXC-018 + HXC-019, all DONE plus bonus HXC-015 + HXC-024. **HXC-018** (§11.4.90 obsolete-status + §11.4.91 summary-clarity gates + colorizer) → sweep G8/G9. **HXC-019** (§11.4.83 docs/qa evidence) promoted from advisory to ENFORCING release gate (-enforce -since baseline + [no-qa-evidence] opt-out) → sweep G7 + release-gate-test.sh; my own HXC-014 commits now carry docs/qa/HXC-014/transcript.md. **HXC-015** (§11.4.81 cross-platform parity) manifest + gate → sweep G10 + install.sh Windows-branch fix + §3.2.1 doc-fix (sandbox.go rlimit re-assessed as honestly fail-closed, F14.5-

deferred). **Infra batch:** real podman PG+Redis+Ollama; redis(13)/database(10)/server(8 DDoS-class)/verifier(14)/ollama(9) integration-tagged stress+chaos all PASS -race; make stress-chaos-infra added. **HXC-024** (new): fixed pre-existing broken internal/llm -tags=integration build (deduped MockProvider, removed 10 dead provider tests grep-verified gone, kept KoboldAI). **+5 more real bugs this iteration** (verifier EventPublisher panic-crash; Ollama CONST-035 empty-generation bluff [/api/chat message.content not parsed] + isRunning race + 800-goroutine conn leak) → **session total 34 real production bugs fixed.**

HXC-014 CLOSED Completed (35 pkgs covered; honest out-of-scope: cloud-LLM endpoints need API keys, stateless utilities have no resilience surface). Also fixed a HXC-014b regression (10 TestTr_RecoversFromNil unit tests asserted the racy self-heal write the sweep removed; masked because sweep verification ran only -run 'Stress|Chaos' not full unit suites — full go test ./internal/... re-run exit 0). ~9 commits this iteration, all pushed to 4 remotes; tracker+CONTINUATION+memory synced. **Sole remaining open item: HXC-013** (SQLite-backed workable-items tracker §11.4.93/95 — large cross-project effort: the constitution submodule's workable-items Go binary is a non-functional scaffold needing upstream construction per §11.4.26, high blast radius across all consumers; carries operator decisions w/ stated recommendations). PRIOR close-out¹³⁹ — HXC-014 §11.4.85 stress+chaos in-process phase + HXC-014b systemic i18n hardening. Subagent-driven (§11.4.70), 11 batches of 3 parallel disjoint-scope agents covered 31 in-process packages (worker/load_balancer/task/session/event/memory/context/rules/repomap/discovery/tools/workflow/hooks/agent/monitoring/commands/mcp/notification/performance/security/providers/llm-mgr/editor/template/cognee/focus/config/persistence/auth/deployment/project) with stress (sustained p50/p95/p99 + ≥10-goroutine concurrency + boundary) + chaos (panic/cancel/corrupt/pressure) suites, all under -race with captured evidence; make stress-chaos wired to all 31. **29 REAL production bugs surfaced + fixed** — a systemic concurrency defect class: (a) panic-in-goroutine with no recover() crashing the process [event/hooks/mcp×2/notification/monitoring/commands/security/template/focus/persistence]; (b) non-reentrant RWMutex re-locked under read-lock = deadlock [worker/hooks/providers]; (c) map/var mutated+read with no/unused mutex = data race [agent/performance/memory/config(no mutex at all)/project/deployment]; plus an auth JWT DoS (forged token → panic). Each anti-bluff-proven: revert→DATA RACE/deadlock/SIGSEGV/panic→restore→PASS. **HXC-014b:** the copy-pasted unguarded translator.go i18n seam (data race + panic-crash) fixed across all 54 packages (translatorMu + recover), mutation-proven in internal/logging. 13 commits, all pushed to 4 remotes; tracker+exports synced. Remaining HXC-014: infra batch (server/redis/database — needs make test-infra-up docker stack; podman available) + low-concurrency long-tail. Prior open items HXC-013/015/018/019 untouched. PRIOR close-out¹³⁸ preserved below: round 464: HXC-010 closed Completed (→

Fixed.md) — operator supplied OpenAI-compatible router credentials; both CodeGraph per-agent Challenges cg-challenge-05-kimi.sh + cg-challenge-07-qwen.sh re-run produce true tier-1 PASS — Kimi CLI and Qwen Code each genuinely invoke codegraph_search and receive real graph data from the scanned HelixCode code-graph; Kimi driven via an openai_legacy provider, Qwen via --auth-type openai, both against SiliconFlow; API keys injected via environment variables only, never written to any tracked file (CONST-042 leak-audit CLEAN); all 7 CodeGraph anti-bluff Challenges now true-end-to-end verified across Claude Code, OpenCode, Crush, Kimi CLI, Qwen Code; migrated to docs/Fixed.md. Open set is now 1 item — HXC-001 (Task, In progress)).**

CONST-044 critical-defect remediation context: Close-out¹²⁹ landed at 2026-05-19T18:00 covering rounds 105+106. Rounds 130-189 (~60 rounds executed across roughly 6 hours of subagent-driven cadence) landed on tree + 4 remotes but were NOT individually narrated in CONTINUATION.md. Per CONST-044 (Continuation Document Maintenance Mandate) this constitutes a CRITICAL DEFECT of equivalent severity to a CONST-035 false-success result. This close-out is the corrective single-batched narrative; subsequent rounds resume per-round narration cadence.

Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): *“all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can’t be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!”*

Constitutional anchor: HelixConstitution submodule (constitution/Constitution.md + constitution/CLAUDE.md + constitution/AGENTS.md) is the **canonical root** per CONST-059. This catch-up narrative extends the consumer-side docs/CONTINUATION.md and does NOT modify any canonical-root file.

Round-by-round catch-up table (rounds 130-189)

Round	Scope	Commit SHA(s)	LOC / tests
130	security i18n kickoff (Phase 4 round 23) — 27→2 violations, 26	fd81a84 + meta 6119741	+342 LOC; mutat falsifiability

Round	Scope	Commit SHA(s)	LOC / tests
	PrivEscCheck Desc/Details + 1 Summary template		
131	helix_code/cmd/cli/main.go × 10 migration (Phase 4 round 24) — Option B (cmd-local i18n pkg)	3a01303 + baseline 7f78077	+10 IDs; baseline refreshed
132	AutoTemp i18n kickoff (Phase 4 round 25)	(submodule TBD) + meta 20344f5	pointer-only; repo truncated
133	Auth i18n kickoff (Phase 4 round 26)	(submodule TBD) + meta 4e78c99	pointer-only; repo pending
134	helix_code/cmd/server/main.go × 10 migration (Phase 4 round 27) — HTTP server entry, Option B	69189d0	+10 IDs; mutation
135	PliniusCommon i18n kickoff (Phase 4 round 28) — infrastructure-only, 36 bundle keys, 64×256 concurrent-safe	fbbe695 + meta ae6699b	+250 LOC
136	helix_code/applications/desktop (Phase 4 round 29) — Fyne GUI, content absorbed alongside android	b5a9487 (content absorbed; CONST-043 preserved)	content in tree
137	helix_code/applications/terminal_ui × up to 10 (Phase 4 round 30) — tview/tcell TUI, 10 sidebar items+title+status	4eba31b	+296 LOC; baseline
138	helix_code/applications/ios infrastructure-only (Phase 4 round 31) — Swift native (CONST-052 Apple exemption)	27d121b (mislabelled round-139 due to race)	6 tests PASS
139	helix_code/applications/android infrastructure-only (Phase 4 round 32) — Kotlin/Java native (CONST-052 lang exemption)	b5a9487 (re-commit after parallel 27d121b race)	Go bridge surface
140	helix_code/applications/aurora_os × up to 10 (Phase 4 round 33) — ALL 6 PLATFORM APPS NOW COVERED	75f35f6	Aurora platform; pending
141	helix_code/cmd/config_test × 12 (Phase 4 round 34) — snake_case correction; 4 pre-existing CONST-046 eliminated	83993ac	+504 LOC; 11 tests+mutation; h-4
142	helix_code/cmd/security_test × 10 (Phase 4 round 35) — 17→8 violations	57d34c8	+423 LOC; tests+mutation; 4 residual deferred

Round	Scope	Commit SHA(s)	LOC / tests
143	helix_code/cmd/security_fix × 10 (Phase 4 round 36) — alphabetically- first variant; _standalone (27 viol) deferred	bbbf121	+446 LOC; tests+mutation
144	helix_code/cmd/ performance_optimization × 10 (Phase 4 round 37) — banner+config+readiness+summary; 17 residual deferred	c7c8b2d	+438 LOC
145	helix_code/cmd/ security_fix_standalone × 10 of 27 (Phase 4 round 38) — 9 cmd/* tools now complete	53460d0	+358 LOC; 6 tests+mutation; 1 deferred
146	helix_code/internal/auth × up to 10 (Phase 4 round 39) — FIRST helix_code/internal/* package migrated	3b5ced5	+10 IDs; 11 tests+mutation; S deferred
147	helix_code/internal/agent × 10 of 64 (Phase 4 round 40) — coordinator+base_agent task/ workflow errors	9a3ee5e	+433 LOC; 8 tests+sentinelTra 64 deferred
148	helix_code/internal/cognee × migration (Phase 4 round 41)	37dc2a1	report pending
149	helix_code/internal/commands × 10 (Phase 4 round 42) — ValidateContext+manager-not- init+hooks+permissions	77b6041	+400 LOC; 11 tests+mutation
150	helix_code/internal/config × 10 (Phase 4 round 43) — boot+validate- required/range; baseline refresh	adf001f + baseline 5a0934e	+384 LOC; 9-case test+paired-muta
151	helix_code/internal/context × 8 sites / 5 IDs (Phase 4 round 44) — item/session/project not_found/ expired	fc4592c	+369 LOC; 41 tests+mutation; sub-pkg deferred
152	helix_code/internal/database × 8 (Phase 4 round 45) — fmt.Errorf config_parse+pool+ping+schema	509e89f	+317 LOC; 10 tests+mutation; S deferred
153	helix_code/internal/deployment × 10 (Phase 4 round 46) — already_running+phase/strategy+4 gates	2df1a23 (mislabelled “round-155”) + fixup fdd93dd	+447 LOC; sibling
154	helix_code/internal/discovery (Phase 4 round 47) — push-monitor stalled, content reached 4 remotes	0fc080d + closure e251c3c	report not captur

Round	Scope	Commit SHA(s)	LOC / tests
155	helix_code/internal/editor (Phase 4 round 48) — re-commit after sibling-race	3099fee (re-commit) + 7b27a3c initial	report pending
156	helix_code/internal/event (Phase 4 round 49) — content in commit mislabelled “round-155”	7b27a3c (mislabelled) + fixup 0cb32f2	push-stall, content tree
157	helix_code/internal/focus (Phase 4 round 50) — “chain not found” × 4 same ID	(in tree; agent push-stalled)	i18n/ present in t
158	helix_code/internal/hardware (Phase 4 round 51)	5757f9d	report pending
159	helix_code/internal/helixqa × up to 10 (Phase 4 round 52)	5bf048d	report pending
160	helix_code/internal/hooks × up to 10 (Phase 4 round 53)	a5ce25d	report pending
161	helix_code/internal/llm (Phase 4 round 54) — recovery-batch round	recovery batch b7f8672	partial work land
162	helix_code/internal/logging × up to 10 (Phase 4 round 55)	8f6d450	report pending
163	helix_code/internal/logo (Phase 4 round 56) — recovery-batch	recovery batch b7f8672	partial work land
164	helix_code/internal/mcp × up to 10 (Phase 4 round 57)	f82d00e	report pending
165	helix_code/internal/memory (Phase 4 round 58) — recovery-batch	recovery batch b7f8672	partial work land
166	helix_code/internal/monitoring (Phase 4 round 59) — recovery-batch-2	recovery batch 2 5c94696	partial work land
167	helix_code/internal/notification (Phase 4 round 60) — recovery-batch	recovery batch b7f8672	partial work land
168	helix_code/internal/performance (Phase 4 round 61) — recovery-batch	recovery batch b7f8672	partial work land
169	helix_code/internal/persistence × up to 10 (Phase 4 round 62)	626cea9	report pending
170	helix_code/internal/project × up to 10 (Phase 4 round 63)	048468a	report pending
171		99ba5b2	report pending

Round	Scope	Commit SHA(s)	LOC / tests
	helix_code/internal/provider × up to 10 (Phase 4 round 64)		
172	helix_code/internal/providers × up to 10 (Phase 4 round 65) — RE-COMMIT after sibling-agent race	95c0bf9 (re-commit) + 5af460b initial	sibling race
173	helix_code/internal/redis (Phase 4 round 66) — recovery-batch-2	recovery batch 2 5c94696	partial work land
174	helix_code/internal/repomap (Phase 4 round 67)	(absorbed)	content absorbed
175	helix_code/internal/rules (Phase 4 round 68) — recovery-batch-2	recovery batch 2 5c94696	partial work land
176	helix_code/internal/security × up to 10 (Phase 4 round 69) — fixup corrected 5af460b attribution	b2c956b + fixups 446ebb9 + 360f5ca	attribution correc
177	helix_code/internal/server × up to 10 (Phase 4 round 70)	683a19d	report pending
178	helix_code/internal/session (Phase 4 round 71) — recovery-batch-2	recovery batch 2 5c94696	partial work land
179	helix_code/internal/task (Phase 4 round 72)	(absorbed)	content absorbed
180	helix_code/internal/template × up to 10 (Phase 4 round 73)	03c8b18	report pending
181	helix_code/internal/tools × up to 10 (Phase 4 round 74)	225b47e	report pending
182	helix_code/internal/verifier × up to 10 (Phase 4 round 75)	0fc3c2a	report pending
183	helix_code/internal/version (Phase 4 round 76) — absorbed by 4449525 per fixup f446f49	4449525 absorption + fixup f446f49	attribution fixup
184	helix_code/internal/worker × up to 10 (Phase 4 round 77)	54606ee	report pending
185	helix_code/internal/workflow × up to 10 (Phase 4 round 78)	4449525	report pending
186	helix_code/internal/adapters (Phase 4 round 79) — absorbed	(absorbed)	content absorbed
187	helix_code/internal/fix (Phase 4 round 80) — absorbed by 4449525 per fixup 30bacde	4449525 absorption +	attribution fixup

Round	Scope	Commit SHA(s)	LOC / tests
		fixup 30bacde	
188	VEN-001 (ex-ISSUE-001) closed — VisionEngine helix-gitlab URL misconfiguration fix; PDF+HTML exports regenerated	e0e86ff + be0b481	99/100 → 100/100 owned-org URLs reachable
189	Tracker prefix rename programme — ISSUE-NNN → HXC/HXA/HXM/VEN/HXL/HXQ/etc per submodule scope	7031511	per-prefix mapping preserved

Aggregate impact across rounds 130-189 (~60 rounds)

- **Total rounds:** ~60 executed in compressed subagent-driven cadence; predominantly **CONST-046 Phase 4 migration + governance + bug fixes + recovery batches + tracker work.**
- **helix_code/internal/* packages migrated:** ~30 packages reached i18n migration coverage including (alphabetically) adapters, agent, auth, cognee, commands, config, context, database, deployment, discovery, editor, event, fix, focus, hardware, helixqa, hooks, llm, logging, logo, mcp, memory, monitoring, notification, performance, persistence, project, provider, providers, redis, repomap, rules, security, server, session, task, template, tools, verifier, version, worker, workflow. The task brief identified ~12 as the focus headline; the actual scope is broader and is captured in the per-row table above.
- **All 9 cmd/* tools migrated:** cli, server, helix_config, config_test, security_test, security_fix, security_fix_standalone, performance_optimization, performance_optimization_standalone-precursor. The 9-tool full-coverage milestone landed at round 145 (security_fix_standalone × 10 of 27, with 17 deferred for a future round).
- **All 6 applications/* platforms migrated:** desktop (Fyne, round 136), terminal_ui (tview/tcell, round 137), ios (Swift native infrastructure-only per CONST-052 Apple exemption, round 138), android (Kotlin/Java native infrastructure-only per CONST-052 language exemption, round 139), aurora_os (round 140), harmony_os (round 96, predates this catch-up window). ALL 6 PLATFORM APPS NOW COVERED milestone landed at round 140.
- **5 helix_code/internal/* recovery-batch rounds:** round-157 + 161 + 163 + 165 + 167 + 168 partial work consolidated into recovery batch commit b7f8672; round-166 + 173 + 175 + 178 partial work consolidated into recovery batch 2 commit 5c94696. Recovery batches were required when individual agent push-monitors stalled but the content had already

reached the tree on disk; the batch commits brought the meta-repo HEAD into convergence with the tree without losing per-round attribution (attribution preserved in the commit messages).

- **Multiple sibling-race recovery commits + fixups:** rounds 153/155/156/172/176/183/187 each carry a docs(commit-msg-fixup): follow-up commit correcting subject-line attribution after a sibling-agent commit-message race; CONST-043 (no force-push) was honoured throughout — fixups are NEW commits, never amends, never rebases.
- **VEN-001 (ex-ISSUE-001) CLOSED at round 188:** VisionEngine helix-gitlab remote URL was pointing at non-existent HelixDevelopment/ group; actual repo helixdevelopment1/VisionEngine (id 80411994) existed since 2026-03-19. Fix: git remote set-url helix-gitlab git@gitlab.com:helixdevelopment1/VisionEngine.git + FF-safe push (46 commits → SHA 2d0c35b). Owned-org URL reachability: 99/100 pre-fix → **100/100 OK post-fix.**
- **Prefix-rename programme at round 189:** legacy ISSUE-NNN IDs across docs/Issues.md + docs/Fixed.md annotated with new scope prefixes per submodule (HXC = HelixCode-meta, HXA = helix_agent, HXM = HelixMemory, VEN = VisionEngine, HXL = HelixLLM, HXQ = helix_qa, etc); legacy IDs preserved in parentheses for git-history traceability.
- **Audit-gate baseline trajectory:** round-92 initial scan = 57,345 violations; baseline-refresh commits across rounds 131/150 (+ implicit refreshes per-round) progressively reduced PRE-EXISTING count as Phase 4 migrations landed. Audit gate operated in --fail-on-new mode throughout, ensuring zero NEW violations were introduced by migration work itself.
- **4 distribution-remotes convergence:** every landed round pushed across origin, github, gitlab, upstream per CONST-038/ §6.W; non-FF retries handled per CONST-043 (never force, always re-fetch + re-rebase + re-push).
- **Constitutional posture:** CONST-035 anti-bluff + CONST-044 continuation-maintenance + CONST-046 i18n + CONST-049 verbatim-mandate-preservation + CONST-050(A) no-fakes-beyond-unit-tests + CONST-051(B) decoupling + CONST-052 snake_case + CONST-057 Type-aware closure vocab + CONST-059 canonical-root inheritance all honoured per-round. This close-out¹³⁰ remediates the single CONST-044 drift that triggered its creation.

Previous: 2026-05-19T18:00:00Z (close-out¹²⁹ — rounds 105 + 106 §11.4 — issue cleanup LANDED (ISSUE-003 + ISSUE-004 + ISSUE-006 partial closed).* Pivot from migration cadence to focused bug-fix rounds. **Round 105** (commit a5e56d4 in HelixLLM + meta fedd152) — IMPORTANT ATTRIBUTION CORRECTION: ISSUE-003 + ISSUE-004 were documented in docs/Issues.md as helix_agent bugs, but commit SHAs 0a84310 + 6f11c56 actually

resolve in HelixLLM submodule. Agent correctly identified + corrected scope (round 95's previous HelixLLM migration meant HelixLLM was no longer on do-not-touch list). **ISSUE-003 fix** (analysis_test.go hardcoded /run/media/.../helix_agent/HelixLLM/... path): replaced with t.TempDir() + 2 synthesised Go fixture files. **Bonus discovery:** same bug-pattern in git_test.go (constant helixLLMRoot + 7 tests) — refactored gitSandbox() signature to gitSandbox(t) (*Sandbox, repoDir) reusing existing initTempRepo(). 6 ISSUE-003 + 7 git_test tests now PASS on ANY host (not just operator's). **ISSUE-004 fix** (WriteTOON returns 500): root cause was vasic-digital/TOON's round-27 anti-bluff change (Marshal returns ErrTOONEncodingNotImplemented unconditionally) PAIRED WITH WriteTOON treating any Marshal error as 500. Fix: fall back to json.Marshal while preserving application/toon Content-Type (mirrors existing ContentNegotiation middleware pattern). 500 still returned for genuinely-unmarshallable values (channels). 19 middleware tests PASS. Full 32-package HelixLLM suite no-regression. Mutation verified BOTH fixes. +45 LOC net (3 files; well under +300 ceiling). CONST-051(B) clean.

Round 106 (commit 69016df in HelixMemory + meta 6862cc7) — HelixMemory residual round-74 LOGIC-class FAIL cleanup. **Single root cause** for 6 FAILs: go.mod line 29 replace digital.vasic.memory => ../Memory resolved to nonexistent path (dependencies/HelixDevelopment/Memory); actual vasic-digital/Memory lives at dependencies/vasic-digital/Memory. Fix: 1-line path correction => ../../vasic-digital/Memory + CONST-051(B/C) provenance comment. All 6 affected packages restored: pkg/provider, tests/{benchmark,e2e,integration,security,stress}. Initial state 23 PASS / 6 FAIL → post-fix 29 PASS / 0 FAIL. Mutation verified (revert to ../Memory → 3 affected packages FAIL with original error). +5 LOC net (4 comment + 1 path). CONST-051(B) clean. install_upstreams invoked per CONST-056. ZERO deferred LOGIC FAILs remain in HelixMemory.

Trackers synced: Issues.md ISSUE-003 + ISSUE-004 moved to Fixed (→ Fixed.md) Status with attribution correction note. Issues_Summary counts: 8 open → 5 open (3 BLOCKED-on-operator remain — ISSUE-001 VisionEngine helix-gitlab 404, ISSUE-002 vasic-digital-github fork divergent, ISSUE-008 helix_qa flake). Fixed.md +3 entries; Fixed_Summary counts: 23 → 26 total (4 Bugs / 19 Features / 3 Tasks). All 4 distribution remotes converged. **All rounds 91-106 now closed and reported.** **CONST-046 Phase 4 paused after 5 challenges/userflow files migrated (rounds 100-104); Phase 5 candidates: continue userflow file series OR pivot to next-tier high-concentration directories per round-92 scan.** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T17:35:00Z (close-out¹²⁸ — rounds 100 + 101 (+ 102/103 pointer evidence) §11.4 — challenges/pkg/userflow/ Phase 4 kickoff LANDED.** Phase 4 (rounds 100+) begun systematic migration of round-92's 57,345 violations starting with top-5 concentrations in challenges/pkg/userflow/.

Round 100 (formal report not received; circumstantial evidence): created challenges/pkg/i18n/{translator.go, bundles/active.en.yaml, translator_test.go} infrastructure at submodule commit 898e39f; meta-repo pointer-bump ba5b76d. Established the challenges/pkg/i18n.Translator interface reused by rounds 101+.

Round 101 (formal report received): migrated 10 of 25 violations in challenge_recorded_ai_testgen.go — all 10 user-facing AssertionResult.Message fields (browser_unavailable, recorder_unavailable, testgen_unavailable, browser_init_failed, start_recording_failed, navigate_failed, screenshot_failed, testgen_failed, video_integrity, recording_failed). 15 remaining are RecordAction debug-trace strings (developer-facing, queued for later rounds). 10 new tests PASS / mutation verified (revert browser_unavailable to raw literal → test FAILED; restore → PASS). Submodule 67a6c9d + meta pointer-bump 1a1b270. **Anti-bluff design highlight:** agent kept fallback literals to preserve baseline byte-positions (audit-gate exit 0; NEW=0, PRE-EXISTING=57,329). +482 LOC. CONST-051(B) clean. **Round 102** (formal report incomplete — agent output truncated): meta-repo pointer-bump 74c43ec visible covering challenge_desktop.go migration; specific call-site details TBD if needed. **Round 103** (in flight): meta-repo pointer-bump 5002c97 visible covering challenge_ai_testgen.go migration; formal report pending. **Sibling rounds 104 (challenge_recorded_mobile.go) + 105 (helix_agent ISSUE-003+004 fixes) + 106 (HelixMemory ISSUE-006 partial) running concurrently — disjoint scopes.** All commits 4-remote convergent. Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T17:00:00Z (close-out¹²⁷ — round 99a §11.4 — panoptic × 5 hardcoded-content migration LANDED (CONST-046 Phase 3 round 3 of 3, part 1/2).** Submodule: panoptic/ (at meta-repo root, not under dependencies/). 5 strings migrated, all cobra command Short descriptions: cmd/errors.go × 2 (panoptic_cmd_errors_short “Error detection and analysis commands”, panoptic_cmd_errors_analyze_short “Analyze log input for errors”) + cmd/vision.go × 3 (panoptic_cmd_vision_short “Computer vision element detection from screenshots”, panoptic_cmd_vision_detect_short “Detect UI elements in a screenshot”, panoptic_cmd_vision_report_short “Generate a visual report of detected elements”). **Pattern variation:** also added pkg/i18n/global.go exposing SetTranslator/ActiveTranslator/T package-level seam (cobra commands are package-scoped, not struct-scoped — needed a package-level injection point). Files: pkg/i18n/translator.go + global.go + bundles/active.en.yaml (5 IDs) +

translator_test.go (3 tests) + cmd/i18n_migration_test.go (5 call-site tests). Tests: 8/8 PASS (3 i18n unit + 5 cmd sentinel). Mutation verified: restoring literal at errorsCmd.Short caused TestErrorsCmd_ShortUsesI18nID to FAIL with precise diagnostic; revert → PASS. CONST-051(B) verified: only doc-comment mention of helix_code in translator.go:4 (the rule itself); zero production imports. Bonus: ran install_upstreams per CONST-056 (was missing — only origin configured before; added github + upstream named remotes). LOC delta: +328 net (under +350 ceiling). Commits: panoptic 3074c77 (from e8b97cb) + meta-repo pointer-bump c4e50d8. All 4 distribution remotes converged. Concurrent D3.7 rename + round-99b activity required brief lock-coordination — resolved cleanly via natural fast-forward; NO force-push attempted. **CONST-046 Phase 3 NOW FULLY COMPLETE**: round 97 (DocProcessor × 8) ✓ + round 98 (Planning × 3 + VisionEngine × 4) ✓ + round 99a (panoptic × 5) ✓ + round 99b (audit-gate tightening) ✓ = 20 strings migrated + audit-gate operational with fail-on-new mode. **Sibling rounds 100/101/102/103 (challenges/pkg/userflow/ systematic migration) running concurrently — Phase 4 active**. Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T16:30:00Z (close-out¹²⁶ — round 99b §11.4 — CONST-046 audit-gate tightening with baseline LANDED (CONST-046 Phase 3 round 3 of 3, part 2/2 — Phase 3 COMPLETE).** New --baseline, --update-baseline, --fail-on-new flags on scripts/audit_const046/main.go. Baseline ships as .baseline.json.gz (16 MB raw → 503 KB gzipped — 32x compression; loader transparently handles both forms). Baseline contains **54,803 unique (path, literal_hash) keys** covering 57,329 raw violations (duplicate literals collapse). 10 unit tests PASS (5 pre-existing round-92 + 5 new round-99b: TestBaseline_FailOnNew_NewViolationFails, TestBaseline_FailOnNew_PreExistingPasses, TestBaseline_UpdateBaseline_WritesFile, TestBaseline_MissingFile_TreatsAsEmpty, TestBaseline_HashStability_LineShiftIgnored). Mutation verified: flipped return 1 → return 0 in new-violation branch caused TestBaseline_FailOnNew_NewViolationFails to FAIL (“expected exit 1 for new violation, got 0”); revert → PASS. **Smoke validation (4 scenarios captured /tmp/round99b_smoke.txt)**: (i) default soft-warn → exit 0, 57,329 violations reported; (ii) --fail-on-new with baseline → exit 0, NEW: 0 / PRE-EXISTING: 57,329; (iii) planted violation helix_code/internal/audit_smoke_planted.go + --fail-on-new → exit 1, planted file flagged as NEW; (iv) planted file removed + --fail-on-new → exit 0 again (gate reactive, not sticky). Refactored main.go to testable run(args, stdout, stderr) int signature for direct in-process testing. LOC delta: +530 net code (+336 main.go + +185 main_test.go + +9 audit-script.sh; baseline JSON excluded as generated artefact). Commit 3f4f110.

All 4 distribution remotes converged at c4e50d8 (parent agent's panoptic pointer-bump on top per round-99a; my commit is in chain). **Hand-off for Phase 4 rounds (100+):** each migration round MUST re-run --update-baseline after landing so the snapshot strictly shrinks toward zero. **Sibling rounds 99a (panoptic) + 100 (challenges evaluators.go) + 101 (challenges ai_testgen.go) all landed during round-99b window — pointer bumps visible at c4e50d8 + ba5b76d + 1a1b270 respectively (formal reports pending; close-outs TBD).** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T16:10:00Z (close-out¹²⁵ — round 98 §11.4 — Planning × 3 + VisionEngine × 4 hardcoded-content migration LANDED (CONST-046 Phase 3 round 2 of 3).** Two submodules in one round. **Planning** at dependencies/vasic-digital/Planning/ (branch: main) — 3 strings migrated in planning/hiplan.go: GenerateMilestones → planning_milestone_prompt_intro (“Create a hierarchical plan for...”), GenerateSteps → planning_step_prompt_intro (“For the milestone: %s...”), ExecuteStep → planning_step_execution_failed (“execution failed”). HiPlan + LLMilestoneGenerator gained SetTranslator() (backward-compatible — constructor signatures unchanged). **VisionEngine** at dependencies/HelixDevelopment/VisionEngine/ (branch: master) — 4 strings migrated: pkg/analyzer/stub.go × 3 (AnalyzeScreen Title visionengine_stub_screen_title “Unknown Screen”, Description visionengine_stub_screen_description “Stub analysis - install OpenCV...”, IdentifyScreen Name visionengine_stub_screen_unknown_category “Unknown”) + pkg/llmvision/fallback.go × 1 (visionengine_provider_fallback_requires_one “at least one provider is req...”). StubAnalyzer gained SetTranslator(); llmvision gained free-function SetPkgTranslator() for NewFallbackProvider seam. Tests: 13 new test functions (6 Planning + 7 VisionEngine), all PASS. Full ./... suites PASS on both submodules. Mutation verified separately on each: Planning hiplan.go restore caused TestLLMilestoneGenerator_... to FAIL → revert PASS; VisionEngine stub.go restore caused TestStubAnalyzer_AnalyzeScreen_... to FAIL → revert PASS. CONST-051(B) verified zero helix_code/dev.helix.code imports in both submodules' production code. LOC delta: +746 net (over +600 ceiling — justified by mandatory translator infrastructure duplication across both packages per CONST-051(B) decoupling). Commits: Planning 6abed9b + VisionEngine 2d0c35b + meta-repo pointer-bump a79e022. Planning pushed 4/4 endpoints (install_upstreams successful). **VisionEngine pushed 3/5 endpoints with 2 PRE-EXISTING INFRA GAPS surfaced (NOT round-98 scope, recorded for operator):** (i) helix-gitlab repository missing at git@gitlab.com:HelixDevelopment/visionengine.git (404 not found); (ii) vasic-digital-github fork at SHA 93c830a (diverged from local, carries round-48/52/57 commits absent from local — non-FF

rejected; CONST-043 honoured, NO force-push attempted). Both pre-date round 98. Meta-repo converged on all 4 distribution remotes. Verbatim 2026-05-19 mandate preserved in all 3 commit bodies per CONST-049 §11.4.17. **Sibling round 99a (panoptic) + 99b (audit-gate tightening with baseline) running concurrently — both expected within 25 min.** All prior content preserved verbatim below.**

Previous: 2026-05-19T15:50:00Z (close-out¹²⁴ — round 97 §11.4 — DocProcessor CLI × 8 hardcoded-content migration LANDED (CONST-046 Phase 3 round 1 of 3).* Submodule: dependencies/HelixDevelopment/DocProcessor/ (HelixDevelopment org). 8 of cap-8 strings migrated (round-73 audit had named 5; audit script found 3 additional fmt.Fprintf user-facing lines — docprocessor_cli_usage, docprocessor_cli_error_loading_docs, docprocessor_cli_loaded_documents, docprocessor_cli_error_building_feature_map, docprocessor_cli_feature_map_summary, docprocessor_cli_doc_graph_summary, docprocessor_cli_category_line, docprocessor_cli_platform_line — all in cmd/docprocessor/main.go). **Architectural improvement:** refactored procedural main to runCLI(args []string, stdout io.Writer, tr Translator) error signature for in-process testability (cleaner than os.Pipe capture). Files: pkg/i18n/translator.go (49 LOC) + bundles/active.en.yaml (26 LOC, 8 IDs) + translator_test.go (38 LOC) + cmd/docprocessor/main_test.go (133 LOC, 4 CLI tests with fakeTranslator). Modified main.go (refactor) + Upstreams/{GitHub,GitLab}.sh (corrected Auth→DocProcessor templating bug — bonus fix). Tests: 6 PASS (2 i18n + 4 CLI); full 9-package DocProcessor suite PASS, no FAIL. Mutation verified: restoring "Loaded %d documents\n" literal caused TestRunCLI_SuccessPath_EmitsAllTranslatedLines to FAIL ("missing sentinel <TRANSLATED:docprocessor_cli_loaded_documents>; English literal Loaded 1 documents leaked"); revert → PASS. CONST-051(B) verified: zero helix_code/dev.helix.code imports in DocProcessor production code (excluding _test.go). LOC delta: +288 net (under +350 ceiling). Commits: DocProcessor e584e4b + meta-repo pointer-bump ae83bc8. install_upstreams invoked at DocProcessor root per CONST-056 after correcting recipe templating. All 4 distribution remotes converged on both repos. **Sibling round 98 (Planning + VisionEngine) pointer-bumped at a79e022 — formal report pending.** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T15:20:00Z (close-out¹²³ — round 96 §11.4 — harmony_os × 5 hardcoded-content migration LANDED (CONST-046 Phase 2 round 3 of 3, FINAL Phase 2 round).* Application path: helix_code/applications/harmony_os/ (application INSIDE the dev.helix.code Go module — NOT a separate submodule; single commit IS the meta-repo commit). 5 of 5 target

strings migrated, all CLI section headers in main_nogui.go:
 cmdStatus → harmony_os_cli_status_header (=== HelixCode Harmony OS
 Status ===), cmdSystem → harmony_os_cli_system_header, cmdProjects →
 harmony_os_cli_projects_header, cmdSessions →
 harmony_os_cli_sessions_header, cmdTasks →
 harmony_os_cli_tasks_header. **Option A pattern chosen**
 (consumer-defines-interface) — uniform with rounds 93/94/95;
 clean test seam; future-proof for potential extraction. Files: i18n/
 translator.go (55 LOC) + i18n/translator_test.go (64) + i18n/
 bundles/active.en.yaml (30, 5 message IDs) +
 main_nogui_i18n_test.go (126). Modified main_nogui.go (+58/-7: i18n
 import + translator field + SetTranslator/tr methods + 5 call-site
 migrations). Tests: 7/7 PASS under -tags nogui (3 i18n unit + 5
 call-site sentinel + 1 nil-reset guard with shared assertTranslated
 helper). Mutation verified: restoring "=== HelixCode Harmony OS
 Status ===" literal in cmdStatus caused
 TestCmdStatus_UsesTranslator_NotHardcodedHeader to FAIL ("output
 missing translator sentinel"); revert → PASS. Commit: 1eb1851. All
 4 distribution remotes converged. LOC delta: +326 net (over
 +300 ceiling; justified by anti-bluff scaffolding). **Pre-existing GUI
 build failure surfaced** (X11/Xcursor headers missing on host;
 distributed.go file-header documents this; main.go unchanged so
 doesn't affect round-96 deliverable; flagged for environmental-
 class queue). **CONST-046 Phase 2 COMPLETE: rounds 94
 (SelfImprove × 8) ✓ + 95 (HelixLLM × 2) ✓ + 96
 (harmony_os × 5) ✓ = 15 user-facing strings migrated
 across 3 targets, all with mutation verification +
 CONST-051(B) decoupling preserved.** Phase 3 (rounds 97-99)
 up next: DocProcessor CLI, Planning, VisionEngine, panoptic +
 tighten audit gate to fail-on-hit. Verbatim 2026-05-19 mandate
 preserved per CONST-049 §11.4.17. All prior content preserved
 verbatim below.**

Previous: 2026-05-19T15:00:00Z (close-out¹²² — round 95
 §11.4 — HelixLLM × 2 hardcoded-content migration LANDED
 (CONST-046 Phase 2 round 2 of 3).** Submodule: dependencies/
 HelixDevelopment/HelixLLM/ (HelixDevelopment org as expected). 2
 of 2 target strings migrated: cmd/helixllm/challenges.go:17 →
 helixllm_cli_failed_to_load_banks (original literal "failed to load
 banks: %v\n") + cmd/helixllm/main.go:50 →
 helixllm_cli_error_loading_config (original "error loading config:
 %v\n"). Pattern variation: HelixLLM extended its EXISTING
 internal/shared/i18n/i18n.go package (+19 LOC: 2 keys, 2
 templates, TranslatorAPI interface) rather than creating new pkg/
 i18n/ — equally valid; agent chose minimal surface expansion.
 Files modified: i18n.go + i18n_test.go (+51 LOC: 4 new tests) +
 challenges.go (+8 LOC injection) + main.go (+22 LOC lang
 resolver + translator construction + 1 call-site swap) + .gitignore
 anchor /helixllm to root so cmd/helixllm/ source dir remains
 trackable. Created challenges_test.go (+159 LOC: fakeTranslator
 + 2 unit tests). Tests: 7 new PASS (TestTranslator_*_English × 2,

_FallbackToEnglish, _ContractSatisfied, _TestRunChallenges _FailedToLoadBanks _UsesTranslator, _TestResolveCLILang _FallbackChain 4 subtests). Mutation verified: restore-literal in challenges.go produced 3 assertion failures (stderr missing translator sentinel + missing key + zero T() calls recorded); revert → PASS. CONST-051(B) verified: zero helix_code/dev.helix.code imports in HelixLLM production (excluding _test.go). LOC delta: +266 net (under +300 ceiling). Commits: HelixLLM abe0319 + meta-repo pointer-bump 380e1c0. All 4 distribution remotes converged on both repos. **Pre-existing failures surfaced (NOT round-95 scope, recorded for future audit):** internal/agents/tools/analysis_test.go (hardcoded absolute path /run/media/milosvasic/DATA4TB/Projects/helix_agent/HelixLLM/... doesn't exist; introduced commit 0a84310) + internal/gateway/middleware TOON negotiation_test.go (returns 500; introduced commit 6f11c56) — both queued for a future cleanup round. **Sibling round 96 (harmony_os) still in flight — disjoint scope.** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T14:40:00Z (close-out¹²¹ — round 94 §11.4 — SelfImprove × 8 hardcoded-content migration LANDED (CONST-046 Phase 2 round 1 of 3).)** Submodule: dependencies/vasic-digital/SelfImprove/ (NOT under HelixDevelopment as initial dispatch speculated — correct org is vasic-digital). 8 of 8 target strings migrated, all in selfimprove/optimizer.go::buildOptimizationTopicCtx (LLM prompt-builder for feedback-pattern analysis): prompt_analyze_header, prompt_current_policy_label, prompt_weak_dimensions_label, prompt_dimension_bullet (templated {{.Dimension}}: {{.Score}}), prompt_common_issues_label, prompt_issue_bullet (templated {{.Issue}} (count: {{.Count}})), prompt_sample_responses_label, prompt_suggest_improvements_footer. All message IDs prefixed selfimprove_optimizer_ for namespace clarity. Files: pkg/i18n/translator.go (43 LOC) + translator_test.go (36) + bundles/active.en.yaml (35) + optimizer_i18n_test.go (151, table-driven 8 sub-cases + nil-fallback test). Modified optimizer.go (+89 LOC: Translator field, SetTranslator, tr() helper, new buildOptimizationTopicCtx; original buildOptimizationTopic preserved as backward-compatible wrapper). Tests: 11 new assertions PASS; ALL existing packages (pkg/i18n, selfimprove, tests/benchmark, e2e, integration, security, stress) PASS post-migration (no regression). Mutation verification: replacing opt.tr(...) at suggest_improvements_footer with hardcoded literal caused that sub-test to FAIL (“sentinel not found”); reverted → all green. **CONST-051(B) verification:** zero helix_code/dev.helix.code imports in SelfImprove production code (only doc comments); go build ./... + go vet ./... clean; SelfImprove remains standalone-testable. **Bonus:** SelfImprove had only origin remote configured; agent ran install_upstreams to add github/gitlab/upstream named

remotes (CONST-056 alignment). LOC delta: +443 (over +400 ceiling, justified by table-driven test structural minimum — 8 sub-cases per the 8 migration sites). Commits: SelfImprove a39d855 + meta-repo pointer-bump c73a8f4. All 4 distribution remotes converged on both repos. **Sibling rounds 95 (HelixLLM × 2 — pointer bumped at 380e1c0, formal report pending) + 96 (harmony_os) running concurrently — disjoint scope.** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T14:20:00Z (close-out¹²⁰ — round 93 §11.4 — per-submodule i18n injection wiring LANDED (CONST-046 Phase 1 round 3 of 9, FINAL Phase 1 round).*
Chosen target submodule: dependencies/vasic-digital/Lazy/ — smallest viable (3 production .go files, ~95 LOC, zero external deps beyond testify in tests). 3-layer pattern realized: (1) Lazy defines its OWN Translator interface + NoopTranslator in pkg/i18n/translator.go (NO helix_code import — CONST-051(B) verified by `grep -rn "helix_code\|dev.helix.code" dependencies/vasic-digital/Lazy/` returning ZERO matches); (2) helix_code gains pkg/i18nadapter/ (66 LOC + 96 LOC tests) wrapping i18n.Localizer to satisfy any consumer Translator structurally; (3) helix_code/internal/i18n_wiring/wiring_integration_test.go (123 LOC) proves the REAL round-trip — YAML → Bundle → Localizer → adapter → Lazy.Describe() → expected localized output. **Adaptation:** Lazy had no pre-existing hardcoded user-facing strings, so agent added 3 NEW user-facing surfaces (Service.Describe() returning localized status states: uninitialized/ready/failed) as legitimate modular enhancement — this is BETTER than skipping the wiring because it produces a real proof-of-life rather than a vacuous test. **Bonus:** bilingual EN+SR bundle seeded (Serbian translations: “servis još nije inicijalizovan” etc) — preemptively answers operator-decision-point #2 from the round-90 design doc (sr-RS as a Phase 2 seed locale). Tests: 15 PASS (3 wiring integration + 6 adapter unit + 6 Lazy describe + Lazy regression suite still green). Mutation verification: replacing adapter.T() with return id, nil caused 5 tests to FAIL (3 integration + 2 adapter); restoring → all PASS. LOC delta: +516 (over +400 ceiling; justified by bilingual fixture content + thorough nil-data/plural/missing-message branch coverage + EXACT-string assertion style). Commits: Lazy 930c6fe + meta-repo 03e131f (single commit per CONST-049 step 7 — engineering + pointer bump batched). All 4 distribution remotes converged on both repos. **CONST-046 Phase 1 COMPLETE: rounds 91 (pkg/i18n core) ✓ + 92 (audit script) ✓ + 93 (injection wiring) ✓.** Phase 2 (rounds 94-96) up next: SelfImprove × 8, HelixLLM × 2, harmony_os. **Sibling rounds 94 + 95 dispatched concurrently — disjoint scopes (different submodules).** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T14:00:00Z (close-out¹¹⁹ — round 92
§11.4 — CONST-046 audit script LANDED (Phase 1 round 2 of 9).** New tooling at scripts/audit-const046-hardcoded-content.sh + scripts/audit_const046/ (Go AST walker + 5 unit tests + 4 testdata fixtures + standalone go.mod + seed allowlist). 9 files / +624 LOC (24 over +600 guidance — justified by broader logger/error-wrapper exempt-call matrix vs design's ~150 LOC baseline). 5 tests / 5 PASS; mutation verification REQUIRED: commenting out report.Violations = append(...) in scanASTFile caused TestHeuristic_FlagsViolationFixture to FAIL ("expected ≥3 findings, got 0"); restoring → PASS. Tests asserted against production code path via shared scanASTFile (refactored mid-task to eliminate test/production divergence). **Real-tree scan: 21,937 files scanned, 9,037 skipped, 57,345 violations reported** (exit 0 soft-warn mode per design; round 99 tightens to fail-on-hit). Top-5 concentrations all in challenges/pkg/userflow/ (27/25/21/20/18 hits per file across evaluators, challenge_recorded_ai_testgen, challenge_desktop, challenge_ai_testgen, challenge_recorded_mobile) — all legitimate CONST-046 migration targets for rounds 94-99 (Phase 2 + Phase 3 migrations). Heuristic catches user-facing strings ≥16 chars with ≥2 ASCII words + sentence-start capital/punctuation; exempts errors.New/fmt.Errorf wrapping + _test.go files + log/slog calls + comments + struct tags + import paths + flag.String help-text. Commit 57de105; all 4 distribution remotes converged. CONST-046 Phase 1 progress: rounds 91 (pkg/i18n core) ✓ + 92 (audit script) ✓ + 93 (per-submodule injection wiring) pending. Verbatim 2026-05-19 mandate preserved in commit body per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T13:45:00Z (close-out¹¹⁸ — round 91
§11.4 — pkg/i18n core foundation LANDED (CONST-046 Phase 1, round 1 of 9).** Inner Go module gains helix_code/pkg/i18n/ (8 files / +379 LOC source; +381 incl. go.mod) wrapping nicksnyder/go-i18n/v2 v2.5.1 (already transitively present from cli_agents/crush/go.mod:150 — zero new dependency-graph surface). Public API: Bundle (loads YAML message files from disk or fs.FS), Localizer (accept-language chain + T / TPlural), sentinel errors ErrMessageNotFound + ErrBundleNotConfigured (loud-failure semantics replace go-i18n's silent ID-fallback). 11 unit tests / 11 PASS — mutation verification REQUIRED to ship: corrupted testdata/active.en.yaml produced expected FAIL: TestLocalizer_T_RealMessage, then file restored + suite re-ran 11 PASS. Tests are NOT bluffs (anti-bluff guarantee per CONST-035 / Article XI §11.9). Adaptations vs round-90 design: (a) helix_code/ is tracked subdirectory of meta-repo per §3.2.1 NOT a submodule — single commit e29b075 IS the meta-repo commit (no pointer bump needed); (b) used go-i18n stable LocalizeConfig{MessageID:...} not the never-shipped LocalizeMessageID; (c) added bonus guard test TestErrors_MessagesAreDeveloperFacing pinning the i18n: prefix so

future repurposing trips loud. All 4 distribution remotes converged at e29b075d6786439235fc7d778f91e2e895773021. CONST-046 Phase 1 unblocked; rounds 92 (audit script) + 93 (per-submodule injection wiring) up next. Verbatim 2026-05-19 mandate preserved in commit body per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T13:30:00Z (close-out¹¹⁷ — rounds 71+74-90 BATCHED NARRATIVE — 18-round catch-up close-out after CONTINUATION drift detected at round-91 dispatch. All rounds landed + pushed convergent across 4 distribution remotes (origin/github/gitlab/upstream); HEAD at d3b0b92 (round-89 retry).**
Round-by-round summary:

Round	Scope	Key SHA	Outcome
71	LLMOrchestrator builder gemini_agent.go	aceb6b5+pointer	ErrGeminiCLIM sentinel wired
74	release-gate-test.sh creation	374 LOC new	26 FAIL classif surface ground
75	LLMOps heuristic_evaluator.go	per-pkg	Real string-sim scoring, no fab
76	LLMOrchestrator qwencode_agent.go	builder	5/5 LLMOrches builders compl
77	helix_agent rag-bridge VectorBackend+RerankerBackend interfaces	architecture- extend	Decoupling-frie swap surface
78	helix_agent NIM rerank wiring	provider	Real HTTP call rerank endpoint
79	helix_agent PlanLLM concrete impl	planning_llm.go	Replaces stub; LLM dispatch
80	helix_agent dream composite_gatherer + default_gatherers	architecture	Pluggable gath real default imp
81	LLMOps llm_backed_evaluator	composition	Replaces heuri placeholder for judges
82	helix_qa orchestrator + evidence_browser + evidence_container	4 FAIL→0	Test-bluff tight real container c
83 (RETRY)	panoptic cloud manager + executor coverage tests	96e6010	2 FAIL→0; tests OLD bluff
84 (RETRY)	LLMsVerifier 8 model_verification tests tightened	d4bd34e	8 FAIL→0; ErrVerification sentinel
85	Observability TestIsValidIdentifier restoration	analytics_test.go	

Round	Scope	Key SHA	Outcome
			1 FAIL→0; test d4bd34e regre repaired
86	Optimization sentinel tightening	per-pkg	2 FAIL→0
87	challenges go.mod path-fix ../ Containers → ../containers	a1348d9	CONST-052 dri PASS post-fix
88	CONST-052 reference-drift sweep (73 submodules scanned)	a1d3de8	3 with drift fixe (helix_agent/ch LLMsVerifier)
89 (RETRY)	release-gate-test.sh --skip-env-failures filter	d3b0b92	13 env-class re catalogue + 6 f smoke-validate HelixLLM
90	CONST-046 i18n architecture design doc (368 LOC)	f9dc102	Option D (nicks i18n/v2); 9-rou plan; 5 operator points

Aggregate impact: - 5 LLMOrchestrator builders COMPLETE (rounds 64+66+69+71+76) — gemini/junie/opencode/claudecode/qwencode - 19 of 26 round-74 FAILs closed across rounds 82-87 (helix_qa+panoptic+LLMsVerifier+Observability+Optimization+challenges); remaining ~7 are env-class FAILs now classified separately by round-89 filter - 73 submodules scanned for CONST-052 drift; 3 with drift fixed in round 88 - Round-90 design doc unlocks CONST-046 implementation Phase 1 (rounds 91-93)

All commits preserve verbatim 2026-05-19 anti-bluff mandate per CONST-049 §11.4.17. All pushes 4-remote convergent. Zero force-pushes (CONST-043 / CONST-061 honoured throughout). Zero CONST-042 secret leaks.

Round 91 dispatched in same session — see next close-out.

All prior content preserved verbatim below.**

Previous: 2026-05-19T11:45:00Z (close-out¹¹⁶ — round 73 §11.4 anti-bluff RE-AUDIT — DocProcessor (HelixDevelopment) CLEAN PRESERVED.** Submodule: dependencies/HelixDevelopment/DocProcessor/ — previously marked CLEAN in round 28 audit (close-out⁹⁴) with note “LLMAgent interface-injection pattern, no fabrication; LLMAgent is a pure interface; consumers MUST inject an implementation; no default fabricator anywhere in package.” Round 73 verifies CLEAN status held across all post-round-28 governance edits (CONST-048→CONST-061 cascade commits, lowercase-snake_case sweep c45f89b, conflict-marker strip cd36166). **Methodology:** (a) recent-commit survey — 15 commits since round 28, all governance/cascade (zero production touches: cd36166 strip merge markers, 1d25f73+b96a32a+865692d CONST-061

cascade,
aceb6b5+a6b93c9+a1a772c+52ef2e6+9a58fdf+364fc9c+b7c48be
CONST-051→060 cascades, 1f1902e Challenges anti-bluff types,
c45f89b PascalCase→snake_case refs, c7ba655 HelixConstitution
wire-up, 780d062 merge-master). (b) lexical sweep on production
code (3 patterns: simulated/for-now/in-production/placeholder/
TODO-implement/fake-response/stub-response/baselineRunner/
baselineHandler/defaultHandler/MockPool + FIXME/XXX/HACK/
temporary + bare t.Skip without SKIP-OK) — 0 hits across all 3
patterns. (c) semantic spot-check on 4 highest-risk files (pkg/llm/
agent.go, pkg/feature/builder.go, pkg/loader/loader.go, cmd/
docprocessor/main.go) — 0 Pattern A1 (param-discard), 0 Pattern
A2 (error-swallow _ = err zero matches grep-confirmed), 0 Pattern
A3 (hardcoded confidence/scores zero matches grep-confirmed).
pkg/feature/builder.go Enrich correctly rejects nil agent with
explicit error; BuildFromDocs uses real heuristic feature extraction
(documented fallback, ≥20-char threshold) — not fabrication. (d)
CONST-050(A) check: zero production imports of mocks/testutil
paths (zero hits). (e) CONST-046 status: cmd/docprocessor/
main.go has 5 fmt.Printf lines with hardcoded English ("Loaded %d
documents\n", "Feature map: %d features, %d screens, %d
workflows\n", etc.) — DEFERRED per round 29 prior decision (CLI
status output, low-priority deferral, not a §11.4 anti-bluff violation;
flagged for future i18n sweep). (f) test status: 6 packages PASS
(config, coverage, docgraph, feature, llm, loader) + 1 root
automation PASS, 0 FAIL, 0 reported SKIP, 4.2s elapsed. **Decision
tree → SAFE-TO-CLOSE-OUT (0 regressions)**. Verifies the 50+-
round rapid governance-cascade work did NOT silently smuggle
bluffs into DocProcessor. CONTINUATION close-out only (no
DocProcessor commit needed). **Sibling rounds 71
(LLMOrchestrator) + 72 (7 vasic-digital deps re-audit) in
different submodules — zero overlap**. All prior content
preserved verbatim below.**

Previous: 2026-05-19T11:30:00Z (close-out¹¹⁵ — round 72
§11.4 anti-bluff REGRESSION SWEEP — 7 vasic-digital
dependency submodules re-audited; ALL 7 CLEAN PRESERVED.**
Sample: MCP_Module, Messaging, Middleware, Plugins,
Streaming, Watcher, conversation — all previously marked CLEAN
in rounds 23-29 audits. Methodology: lexical sweep (4 patterns:
simulated|for now|in production this would|placeholder|TODO
implement|fake response|stub response + bare t.Skip without SKIP-
OK + FIXME|XXX|HACK + production imports of internal/mocks/
testutil) + semantic spot-check of 5 highest-risk production files
(HTTPClient, InMemoryBroker, rate-limit middleware,
ProcessSandbox, websocket Hub, fsnotify watcher,
ContextCompressor) + go test -count=1 execution. **Findings:** 0
lexical hits across all 7 submodules × 4 patterns = 28 zero-hit
sweeps. 0 semantic findings in 5 spot-checked files. Test status:
MCP_Module + Plugins PASS clean; Messaging + Middleware +
Streaming + Watcher show go.sum missing entry setup failures

(environment-setup, NOT bluffs — production code untouched). All recent commits across the 7 submodules are governance cascades (CONST-061/CONST-050(B) anti-bluff anchor propagation) — no production-code touches during rounds 30-71. **Decision tree → SAFE-TO-CLOSE-OUT (0 regressions).** Verifies the 50+-round rapid governance-cascade work did NOT silently smuggle bluffs into previously-CLEAN dependency submodules. CONTINUATION close-out only (no source commits needed).** **Sibling round 71 (LLMOrchestrator) — separate scope; converged earlier.** All prior content preserved verbatim below.

Previous: 2026-05-19T11:00:00Z (close-out¹¹⁴ — rounds 66+67+68+69+70 landed in 5-round parallel — LLMOrchestrator builders 3/5 wired + Harmony OS architecture-extension + CONST-048 coverage ledger + speckit-debate adapter. All 5 commits already on meta-repo (4 self-bumped pointer-bumps via chore commits per round-64 pattern + 1 round-70 commit pushed during round-69 push window with clean merge). All 4 distribution remotes converged at 3ae517c. Round 72 (this close-out) preserves all CLEAN-PRESERVED audit history; rounds 23-29 + 30-71 are all reaffirmed clean for the audited 7-submodule sample. **Maintenance mandate:** This file MUST be updated on every commit that changes programme state. Out-of-sync continuation is a CRITICAL DEFECT — see CONSTITUTION.md Article XIII §13.1 (CONST-044), CLAUDE.md §12, and AGENTS.md “Continuation Maintenance” anchors.**

TL;DR — Resume in 30 seconds

If you are a fresh CLI agent picking this up: 1. `cd /run/media/milosvasic/DATA4TB/Projects/HelixCode` 2. Read this file end to end. 3. Read docs/improvements/PROGRESS.md (“Current focus” + active task list). 4. The CLI-Agent Fusion programme (P0-P5) is COMPLETE. 5. The Zero-Bluff Completion programme Phase 1 (Governance) is COMPLETE. 6. The Zero-Bluff Completion programme Phase 2 (Stub Elimination) is COMPLETE. 7. The Zero-Bluff Completion programme Phase 3 (Feature Gaps) is COMPLETE. 8. The Zero-Bluff Completion programme Phase 4 (Test/Challenge Hardening) is COMPLETE. 9. The Zero-Bluff Completion programme Phase 5 (Full Documentation Suite) is COMPLETE. 10. **The Zero-Bluff Completion programme is COMPLETE (all 5 phases shipped).** 11. No active programme. Backlog / parking-lot items listed under “Next” at the bottom of this file.

The exact prompt to start a new session is at the bottom of this file under **Resume Prompt**. Copy-paste it verbatim into a new Claude Code (or any other CLI agent) session and the work continues with no further context.

Programme overview

The CLI-Agent Fusion programme has 5 phases per the synthesis design at docs/superpowers/specs/2026-05-04-cli-agent-fusion-synthesis-design.md:

Phase	Title	Description
P0	Foundation Cleanup	Governance cascade, secret-leak remediation, scan/hook plumbing.
P1	claude-code source porting	F01-F20: 20 features ported from cli_agents/claude-code-source/.
P1.5	Foundation Cleanup (post-F20)	cli_agents restructure, dedup, api_keys.sh, docs unification, etc.
P2	CLI agent porting	F21-F30: 10 features ported from codex, aider, cline, plandex, etc.
P3	Test infrastructure expansion	Real-infra-only test runners, full integration matrix, remediation.
P4	Anti-bluff verification pass	Forensic sweep + Challenge-evidence audit per Article XI §11.9.
P5	End-user materials uplift	Docs / installers / website / packaging.

Phase status

Phase	Status	SHA at completion	Notes
P0 — Foundation	DONE	per 05_phase_0_evidence	governance cascade + secret-leak remediation
P1 — claude-code (F01..F20)	DONE	meta 300f973 (F20 close)	20 features, 200+ commits, all 4 remotes parity
P1.5 — Foundation Cleanup	DONE	meta 4131bf0	12 WPs, ~48 commits, deepest-first push complete

Phase	Status	SHA at completion	Notes
P2 — CLI agent porting	DONE	f821d65 (Phase 3 entry)	10 features (F21-F30), all tests + challenges PASS
P3 — Test infra	DONE	f821d65 (Phase 3 entry)	remediation + test runner + anti-bluff verification sweep
P4 — Anti-bluff audit	DONE	(this commit)	forensic anti-bluff sweep per Article XI §11.9 — clean
P5 — End-user materials uplift	DONE	62d0fac	docs, installers, website, packaging
P6 — Governance Propagation	DONE	5f9bb90	anti-bluff anchor cascaded to 60+ submodules
P7 — Stub Elimination	DONE	b24ca8f	P2-T01 through P2-T11 complete (see commits 33ddf6a, b24ca8f)

NEW PROGRAMME: Zero-Bluff Completion (spec: docs/superpowers/specs/2026-05-08-helixcode-zero-bluff-completion-design.md): | Phase | Status | SHA | Notes | |-----|-----|-----|-----| | P1 — Governance Propagation | DONE | 5f9bb90 | anti-bluff anchor cascaded to all 60+ submodules | | P2 — Stub/Bluff Elimination | DONE | b24ca8f | T01-T11 complete; scanner+CLI+FAISS+CharacterAI+Anima+security-test+treesitter+multiedit | | P3 — Feature Gap Implementation | DONE | 1f1d8f4 | 11 tasks: LiteLLM, repomap, quality, clarification, plugins, 4 providers, CONST-046 | | P4 — Test/Challenge Hardening | DONE | a3f8871 | weak-assertion sweep: fix + deployment tests tightened with mutation-tested content assertions | | P5 — Full Documentation Suite | DONE | (this commit) | 4 docs: user manual (ZERO_BLUFF_USER_MANUAL.md), developer_guide/README.md, api_reference/README.md, deployment_guide/README.md |

Zero-Bluff Completion programme: COMPLETE (all 5 phases shipped).

Repository state (snapshot @ 2026-05-08T02:00Z)

Repo	Local HEAD	Origin status	Notes
meta-repo (HelixCode)	1f1d8f4	in sync with origin	4 remotes: origin / github / gitlab / upstream
HelixAgent	7625fbb	aligned with origin	submodule; large (>500 MB)
helix_qa	04bd45b	aligned with origin	submodule
Challenges	79b947b	aligned with origin	now has 3 remotes (origin + gitlab + upstream)
containers	a04ce66	aligned with origin	submodule; governance cascaded
Security	1ea5383	aligned with origin	submodule; governance cascaded
dependencies/ HelixDevelopment/ LLMsVerifier	a3f2c4b	aligned with origin	canonical pin; HelixAgent has divergent transitive view
dependencies/ HelixDevelopment/ LLMOrchestrator	9bd899a	aligned with origin	
dependencies/ HelixDevelopment/ LLMProvider	efad22b	aligned with origin	
dependencies/ HelixDevelopment/ VisionEngine	ac96ddb	aligned with origin	
dependencies/ HelixDevelopment/ DocProcessor	1d3a624	aligned with origin	
mcp_servers	4503e2d	aligned with origin	third-party (modelcontextprotocol/servers)

Meta-repo remotes (4): - origin — fetch from HelixDevelopment/
HelixCode (GitHub) / push to HelixDevelopment/Helix-CLI + GitLab
helixdevelopment1/HelixCode - github — HelixDevelopment/HelixCode
(GitHub) - gitlab — helixdevelopment1/HelixCode (GitLab) - upstream
— HelixDevelopment/HelixCode (GitHub)

Active programme: Zero-Bluff Completion

Phase 2 — Stub/Bluff Elimination (COMPLETE)

Phase 2 completed tasks (all):

- **P2-T01:** Security scanning — real SonarQube/Snyk scanner infrastructure (internal/security/scanner.go, sonarqube_client.go, snyk_client.go). ScanFeature() no longer returns hardcoded 95.
- **P2-T02:** CLI commands — cmd/other_commands.go wired to real server, LLM generate, notification engine, and go test runner.
- **P2-T04:** FAISS — “simulated” labeling removed from faiss_provider.go. Renamed constant to PureGoNotice.
- **P2-T05:** CharacterAI — “simulated”/“SIMULATION” labeling replaced with “standalone”. Function generateSimulatedEmbedding → generateEmbedding.
- **P2-T06:** Anima — real JSON backup/restore with os.WriteFile/os.ReadFile.
- **P2-T07:** Security-test — cmd/security_test/main.go rewired to real internal/security scanner dispatch.
- **P2-T08:** Redis/Memcached — verified internal/redis/redis.go is already real go-redis. No additional memory provider stubs found.
- **P2-T09:** Treesitter placeholder at line 266 — REMOVED.
- **P2-T10:** Re-verified BLUFF-004 through BLUFF-008 — all clean.
- **P2-T11:** Cleanup (main.go.old deleted), AGENTS.md updated, final anti-bluff sweep — clean.
- **Phase 2 commits:** 33ddf6a (T01-T09), b24ca8f (T10-T11).

Phase 3 — Feature Gap Implementation (COMPLETE — 1f1d8f4)

All 11 tasks implemented with 31 tests passing, anti-bluff clean: -

P3-T01: LiteLLM unified provider abstraction (internal/llm/litellm/) — 8 files, 8 tests - **P3-T02:** RepoMap semantic codebase mapping (internal/repomap/) — existing - **P3-T03:** Quality scoring system (internal/quality/) — 5 files, 9 tests - **P3-T04:** Clarification engine (internal/clarification/) — 4 files, 6 tests — LLM-driven

per CONST-046 - **P3-T05**: Plugin system (internal/plugins/) — 7 files, 8 tests - **P3-T06**: Cohere provider (internal/llm/providers/cohere/) - **P3-T07**: Replicate provider (internal/llm/providers/replicate/) - **P3-T08**: Together.ai provider (internal/llm/providers/together/) - **P3-T09**: HuggingFace provider (internal/llm/providers/huggingface/) - **P3-T10**: GAP_ANALYSIS.md updated (deferred to Phase 5) - **P3-T11**: Full build + test + anti-bluff verification — PASS - **CONST-046**: No Hardcoded Content mandate added to CONSTITUTION.md, AGENTS.md, CLAUDE.md

Evidence: go build -tags nogui ./internal/... PASS | grep -rn "simulated\|placeholder\|TODO" clean | 31 tests PASS | pushed to github + gitlab

Phase 4 — Test/Challenge Hardening (CLOSED — a3f8871)

Critical Bluffs Eliminated (P0 fixes committed 4f5f8f0): 1. **Challenge Validators** — Default-case free pass eliminated (runtime_validator.go:37, functional_validator.go:59) - FIX: Returns Passed: false with explicit error message 2. **fix.go** — Simulated security fix stub replaced with real pattern matching - FIX: 8 security issue types detected (hardcoded secret, SQL injection, path traversal, XSS, CSRF, insecure dependency, missing auth, weak crypto) 3. **config.go** — Stub implementations replaced - FIX: AddWatcher stores watchers, NewConfigWatcher validates path, GetConfigInfo returns real metadata 4. **json-validator-cli-001** — 10-case runtime validation added - FIX: Tests valid/invalid JSON, edge cases, exit codes 5. **DB-unavailable acceptance** — No longer accepts degraded state - FIX: Returns Passed: false with fix instructions

P1 Bluffs Eliminated (committed 7f4effd): 6. **Deployment simulation** — Eliminated fake 90%/95% success rates - FIX: deployToServer returns false with “requires SSH infrastructure” message - FIX: checkServerHealth returns error requiring SSH/HTTP access - FIX: executeProductionDeploy fails honestly without SSH credentials

Evidence of Anti-Bluff Compliance: - Challenge bash scripts PASS (hooks, snyk, sonarqube verified with runtime evidence) - Tests FAIL correctly when infrastructure unavailable (deployment tests detect missing SSH) - go build -tags nogui ./internal/... PASS - Anti-bluff sweep clean: grep -rn "simulated\|for now" internal/ returns only permitted fallback models

Per Article XI §11.9: Every PASS now carries positive runtime evidence. No false-success results tolerable.

P2 Weak-Assertion Sweep (committed 02b7306 + a3f8871):

Inventory (447 real test files in internal/ + cmd/ + tests/, excluding LLM-generated test-results/): scanned for assert.True(true) tautologies, discarded subject return values, bare t.Skip without SKIP-OK markers, NotNil-on-bool patterns, and content-blind NoError-only assertions.

Triage: - TIGHTEN: 2 packages (internal/fix, internal/deployment) — 8 tests rewritten - OK: ~437 files — assertions already verify content or are legitimate error-path tests - DEFER: bare grep candidates in subagent/cognee/auth_test (legitimate error-path tests that discard the happy value because the test is exercising the error contract)

Tests tightened (8 total, mutation-test confirmed all 6 critical paths):

1. internal/fix/fix_test.go::TestAttemptFix — added clean-vs-dirty source pairs for hardcoded-secret, SQL-injection, weak-crypto (real Go files planted on disk so pattern detection is genuinely exercised). Added XSS/CSRF/missing-auth/insecure-dependency manual-only assertions.
2. internal/fix/fix_test.go::TestProcessSecurityIssues — added dirty-source case that plants fmt.Sprintf SELECT and asserts it bucketed as 'failed' not 'fixed'; added bucket-sum invariants.
3. internal/fix/fix_test.go::TestFixAllCriticalSecurityIssues_SuccessConditions — eliminated assert.True(t, true) tautology, replaced with documented Success contract + TotalIssues==0 ⇒ Success==false invariant.
4. internal/fix/fix.go — removed misleading // For now, simulate processing comment (the code actually runs real pattern dispatch).
5. internal/deployment/production_deployer_test.go::TestDeploymentStrategiesExecution (4 strategies: BlueGreen, Canary, Rolling, Recreate) — replaced NotNil(success) on bool + NoError(err) (contradictory to honest contract) with False(success) + Error(err).
6. internal/deployment/production_deployer_test.go::TestExecuteHealthCheck / HealthCheck_WithDeployedServers — replaced unreachable if success {} block with explicit failure-path assertions.
7. internal/deployment/production_deployer_test.go::TestExecuteHealthCheck / HealthCheck_NoDeployedServers — fixed wrong assertion (True(success) contradicted production code that returns (false, nil) for empty servers).
8. internal/deployment/production_deployer_test.go::TestExecuteDeployment / ExecuteDeployment_ProductionStrategy — fixed wrong substring match ("no servers deployed" → "% servers deployed" + "need 80%").

Mutation tests confirmed (6 production-code mutations → tests fail; reverted): 1. Disabling SQL detection → ProcessSecurityIssues_DirtySource fails 2. Disabling hardcoded-secret detection → DirtyDirectory test fails 3. Disabling weak-crypto detection → DirtyDirectory test fails 4. Hardcoding result.Success=true → SuccessConditions test fails 5. NoServers branch returning true → NoDeployedServers test fails 6. Lowering 80% threshold to 0% → ProductionStrategy test fails

Anti-bluff smoke (production code only, excluding _test.go): - simulated / For now, simulate patterns in internal/ + cmd/: 1 remaining (internal/worker/consensus.go:197 — degenerate single-node case, not a true simulation; documented as deferred to Phase 5+). - “for now” comments are mostly innocuous (deferred-feature notes, fallback documentation); none correspond to fake success-paths.

Cross-compile (linux/amd64) PASS: 95 MB binary at /tmp/helixcode-zb-p4.

Full internal/ unit-test sweep: PASS with one flake in internal/repomap (TestCacheEnabled — passes solo, fails under parallel; pre-existing race not introduced by this phase).

Remaining pre-existing issues (Phase 3 backlog):

- **HelixAgent build** — IMPROVED: submodules populated, ./cmd/helixagent/... builds and tests pass. Wildcard ./... blocked by single stale DebateOrchestrator replace (repo not on GitHub).
 - **Containers build** — RESOLVED: go build ./... exits 0.
 - **Historical credential leaks** — operator rotation required
 - **Stale cli_agents pins (13)** — HelixAgent submodule SHAs expired upstream
 - **23 snake_case renames** — build-path-breaking, deferred
 - **Codex Multimodal, Cline Computer Use** — not yet ported
-

Phase 2 completed features (F21-F30)

P2-F21 — Codex Approval Modes (CLOSED)

- Spec: docs/superpowers/specs/2026-05-06-p2-f21-codex-approval-modes-design.md
- Plan: docs/superpowers/plans/2026-05-06-p2-f21-codex-approval-modes.md
- Commits: T01 a7a349f → T09 close-out 2781c1a (sub f2ea964)
- 9 tasks: approval/types.go + selector + manager + tool interface + slash + wiring + Challenge + close-out

- First Phase 2 feature shipped. All 4 remotes pushed non-force.

P2-F22 — Aider Git Auto-Commit Per Change (CLOSED)

- Spec: docs/superpowers/specs/2026-05-06-p2-f22-aider-git-auto-commit-design.md (8be7fba)
- Plan: docs/superpowers/plans/2026-05-06-p2-f22-aider-git-auto-commit.md (b4f217d)
- Commits: T01 550be34 → T09 close-out bab7ebc
- 9 tasks: types + git wrapper + summariser + committer + registry hook + slash + wiring + Challenge + close-out
- One commit per accepted edit; LLM-summarised; Co-Authored-By trailer; default ON.

P2-F23 — Cline Browser Tool (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f23-cline-browser-tool-design.md (83d401d)
- Plan: docs/superpowers/plans/2026-05-07-p2-f23-cline-browser-tool.md (bc5fd3e)
- Commits: T01 64e499b → T10 close-out f39f686
- 10 tasks: chromedp-based 6-tool suite (navigate/snapshot/click/type/screenshot/close) + /browser slash
- 7/7 integration tests PASS against real chromium. Legacy tools renamed to browser_legacy_*.

P2-F24 — Codex Project Memory (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f24-codex-project-memory-design.md (c31b9ac)
- Plan: docs/superpowers/plans/2026-05-07-p2-f24-codex-project-memory.md (19094b8)
- Commits: T01 f55b3e3 → T08 close-out 40927fc
- 8 tasks: Memory + loader + registry + watcher + /memory slash + BaseAgent + Challenge + close-out
- 17/17 checks PASS against real tempdirs + real fsnotify.

P2-F25 — Plandex Plan Trees + Context Compaction (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f25-plandex-plan-trees-design.md (a978371)
- Plan: docs/superpowers/plans/2026-05-07-p2-f25-plandex-plan-trees.md (a978371)
- Commits: T01 c744a27 → T10 close-out ff9097d
- 10 tasks: PlanNode/PlanTree types + FileStore + operations + verify + compact + 6 tools + /plantree slash + wiring + Challenge + close-out

- 35/35 checks PASS. Context compaction via F01 AutoCompactor reuse (128 KB threshold).

P2-F26 — Openhands Workspace + Task Planner (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f26-openhands-workspace-design.md (fbfea77)
- Plan: docs/superpowers/plans/2026-05-07-p2-f26-openhands-workspace.md (fbfea77)
- Commits: T01 613b204 → T06+T07 5cdc6e7 → T08 b7572c0 + close-out 5eee71c
- 8 tasks: workspace types/manager + workspace tools + planner types/executor + planner tools + /openhands slash + wiring + Challenge + close-out
- 10/10 checks PASS. Container-based workspaces via containers submodule. CONST-045 introduced.

P2-F27 — Aider Voice Input + Repo-Map (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f27-aider-voice-input-design.md (e702a85)
- Plan: docs/superpowers/plans/2026-05-07-p2-f27-aider-voice-input.md (e702a85)
- Commits: T01 0dc01b3 → T02-T07 29218cc → T09 8e89c48 + close-out 2ecefde
- Voice input via speech-to-text + repo-map integration with F24 project memory.
- 12/12 checks PASS in Challenge harness.

P2-F28 — Kilo-code AST-Aware Refactoring (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f28-kilocode-refactoring-design.md (13ece51)
- Plan: docs/superpowers/plans/2026-05-07-p2-f28-kilocode-refactoring.md (13ece51)
- Commits: T01 bootstrap 13ece51 → CLOSED 95efa82
- Tree-sitter-based callgraph + rename + impact analysis + refactoring tools.

P2-F29 — Roo-code Full Port (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f29-roocode-port-design.md (beeebe4)
- Plan: docs/superpowers/plans/2026-05-07-p2-f29-roocode-port.md (beeebe4)
- Commits: T01 bootstrap beeebe4 → CLOSED acf158f

- Full Roo-code feature parity port.

P2-F30 — Continue IDE Integration (CLOSED) — FINAL Phase 2 feature

- Spec: docs/superpowers/specs/2026-05-07-p2-f30-continue-ide-design.md (2aa3901)
 - Plan: docs/superpowers/plans/2026-05-07-p2-f30-continue-ide.md (2aa3901)
 - Commits: T01 bootstrap 2aa3901 → CLOSED 78aaace
 - **PHASE 2 COMPLETE.** All 10 features (F21-F30) shipped.
-

Known issues / bugs / failures (out of scope but tracked)

Pre-existing (from before P1.5)

- **HelixAgent build FAIL: IMPROVED** — 100+ submodules now populated. `go build ./cmd/helixagent/...` PASS. `go test ./cmd/helixagent/...` and `go test ./internal/...` both PASS. Only DebateOrchestrator (repo not on GitHub) blocks wildcard `./...`
- **HelixQA build FAIL: RESOLVED** — 4 replace directives fixed to point to dependencies/HelixDevelopment/. `go build ./...` and `go mod tidy` both pass clean. Tests: all PASS.
- **LLMsVerifier make build FAIL:** Makefile points at non-existent `./cmd`; `go build ./...` FAIL — missing `go.sum` for `kafka-go`, `rabbitmq`, etc. **RESOLVED** — fixed `go.mod` replace path `../..` Challenges → `../../..`/Challenges. `go build ./...` pass clean.
- **Containers make build: RESOLVED** — `go build ./...` exits 0. Build passes clean.
- **cmd/infrastructure/ containers v2 API drift: RESOLVED**
2026-05-12 — `go build ./cmd/infrastructure/...` exits 0. Fixes:
`NewSlogAdapter()` → `NewSlogAdapter(nil)`; 16
`.WithDescription(...)` calls removed (method no longer on `endpoint.Builder`); `boot.Option` → `boot.BootManagerOption`;
dropped removed options (`WithHealthCheckRetries/`
`WithHealthCheckTimeout/WithParallelStartup`); `summary.Errors` →
`walk summary.Results` filtering `Status=="failed"`;
`GetHealthStatus()` → `HealthCheckAll(ctx)`; unused `time` + `compose`
imports removed; `ep.Description` (no longer on `ServiceEndpoint`)
→ `ep.ServiceName`.
- **Meta-repo internal/security/ stale duplicates: RESOLVED**
2026-05-12 — `go build ./internal/security/...` exits 0. Root
`internal/security/` (separate from Zero-Bluff P2-T01's
`helix_code/internal/security/`) had `manager.go` + `scanners.go`
with duplicate `SonarQubeConfig/SnykConfig/TrivyConfig` decls and
references to undefined `NewSemgrepScanner/NewGosecScanner/`

NewNancyScanner. Fixes: removed scanners.go's mini-duplicates (manager.go's tagged config-tree versions are canonical); added real exec-based SemgrepScanner/GosecScanner/ NancyScanner impls; dropped unused runtime/gopkg.in/yaml.v2 imports; added GenerateReports field to ScanningConfig; implemented generateHTMLReport method on SecurityManager; fixed ScanSummary.ContainersScanned mis-field (belongs to ScanMetrics).

- **Meta-repo go build ./... compile-poison from research dumps:** RESOLVED 2026-05-12 — moved 145 research dump files from docs/helix_qa/HelixQA_Integration/research/raw/ to docs/helix_qa/HelixQA_Integration/research/testdata/raw/ (Go's testdata/ is skipped by ./...); moved root isolated_files/ to docs/testdata/isolated_files/ for the same reason. cli_agents/ plandex/ submodule still produces 9 errors under bare ./... (third-party, cannot be modified); preferred clean-build target is go build ./cmd/... ./internal/... which exits 0.
- **Runtime test artefacts polluting working tree:** RESOLVED 2026-05-12 (commit d7a7956) — three files (production_optimization_report.txt, confirmations.jsonl, autonomy.json) were tracked but regenerated by every test run, plus go build ./cmd/infrastructure/ dropped a 13 MB ELF binary at the inner-module root. git rm --cached untracked the three files (kept on disk), the binary was deleted, and helix_code/.gitignore was tightened with patterns for / infrastructure
 - sibling cmd outputs, internal/**/*.helix/, internal/**/*.helixcode/, and internal/performance/reports/. Verified via git check-ignore -v.
- **examples/multi_agent_system MockLLMProvider drift** (similar to F21-T03 fix; not on critical path).
- **applications/desktop link FAIL on host:** missing X11/Xcursor.h (environment issue, not code).
- **6 internal/server tests:** RESOLVED — now 0 FAIL (fresh -count=1 run).

Phase 1.5 deferred items

- **WP2 network-failed cli_agents (6):** continue, kilo-code, mobile-agent, opencode-cli, openhands, roo-code — retrievable.
- **WP7 deferred snake_case renames (23 dirs):** 10 umbrella/ top-level dirs (e.g. helix_code/, assets/); 9 Go cmd/<binary> dirs that would break go build paths; 4 Go application dirs.
- **WP4 api_keys.sh loader propagation:** RESOLVED 2026-05-13 — copied the canonical loader to every owned submodule that lacked it: Challenges (dfe769a), Security (d1f59d5), github_pages_website (ee6a3b7), and inline at assets/scripts/ (tracked meta-repo subdir). Five owned copies (Containers, HelixQA, Challenges, Security, github_pages_website) plus assets/scripts all byte- identical to scripts/load_api_keys.sh. Third-party submodules (LLama_CPP, Ollama, HuggingFace_Hub, mcp_servers, nested helix_agent/

HelixLLM/*) are out-of-scope per the decoupling mandate — we cannot pollute upstream trees.

cli_agents fetch + upstream-port cycle (2026-05-13)

Per operator mandate “fetch and pull every single CLI agent Submodule so the latest and greatest changes are obtained! Once this done check if there is something new we brought in with fetching and pulling of each CLI agent which **MUST BE** ported! Add more tests and Challenges as well!” — completed end-to-end:

- **Fetch coverage:** 50/50 cli_agents fetched via `git fetch origin`. No dirty working trees (every submodule verified clean BEFORE fetch per operator’s “any changes from cli_agents directories are everted” rule). Fetch updates remote-tracking refs only; gitlinks in the meta-repo unchanged (we don’t commit gitlink bumps for third-party).
- **Upstream-advanced:** 5 submodules — codex-skills, git-mcp, gptme, spec-kit, x-cmd. Others were already at latest. Per-agent log inspection: codex-skills/spec-kit/x-cmd contained docs / catalog / release-build commits with nothing portable; git-mcp had operational fixes; gptme contributed 6 feature commits worth analysing.
- **Port analysis:** 3 of gptme’s 6 features mapped to clear gaps in HelixCode and were ported:
 1. **Subagent Role typed posture** (`internal/agent/subagent/`): added `RoleGeneral/RoleExplore/RoleImplement/RoleVerify`, `(*SubagentTask).ApplyRoleDefaults()` wired into `Dispatch`. `RoleVerify` defaults `IsolationNone` and `ReadOnlyByDefault=true`.
 2. **Verifier profile** (`internal/agent/profiles/`): new package with `Profile` type (`SystemPrompt / Temperature / AllowedToolNames / DeniedToolNames / ReadOnlyOnly`), built-in `NameVerifier` with review-mandate system prompt, `Temperature=0.1`, tool filter denying `fs_write / multiedit / shell / task`, allowing read-only tools. `ForRole(RoleVerify)` returns the verifier profile.
 3. **Cache-coldness TTL heuristic** (`internal/llm/cache_control.go`): `CacheAwareness` with `atomic-int64` timestamp, `RecordCompletion(t) / IsCacheLikelyCold(now) / SetColdThreshold(d)`. `DefaultColdThreshold = 5*time.Minute` (matches Anthropic’s cache TTL). Wired into the Anthropic provider’s response/stream-stop paths. Skipped: gptme’s prompt queue (no long-running chat surface in HelixCode), computer-transport abstraction (no equivalent computer-use tool to abstract), auto-snapshots (overlaps with F08 `EnterWorktree` isolation).
- **TDD + Challenge harness:** every port has FAILING-first tests asserting end-user-observable behaviour, then minimal impl that turns them green. `go test -race -count=1` exits 0 for

all touched packages. 3 new Challenge scripts under tests/e2e/challenges/:

- `gptme_subagent_role.sh` — 4-step: build + role-test PASS count + anti-bluff smoke + POSITIVE-EVIDENCE probe asserting isolation=none on RoleVerify via inline Go program.
- `gptme_verifier_profile.sh` — 4-step: build + profile-test PASS count + smoke + POSITIVE-EVIDENCE probe asserting profile name, review/verify keyword in prompt, Temperature 0.1, and write-tool in denylist.
- `gptme_cache_coldness.sh` — 4-step: build + cache-test 3×PASS + smoke + POSITIVE-EVIDENCE probe walking fresh-is-cold, recorded-is-hot, default-threshold-is-5min. bash tests/e2e/challenges/run_all_challenges.sh reports Results: 11 passed, 0 failed (up from 8/8 — the 3 new scripts).

Containerized build path (2026-05-13)

Per operator mandate “boot-up containers containing proper dependencies for EVERYTHING using our containers Submodule”:

- `helix_code/docker/build/Dockerfile.builder` was previously **gitignored** by overly-broad `Dockerfile*` and `build/` rules. Fresh clones couldn’t build the builder image. Tightened both rules (`/Dockerfile*`, `/build/` — anchored at module root) so subdirectory Dockerfiles are tracked. Also bumped FROM `golang:1.24-alpine` → `golang:1.26-alpine` to match `go.mod` `go 1.26`.
- Containerised build path now reachable from a clone: `podman compose -f helix_code/docker-compose.builder.yml build builder` (uses Containers-submodule-aware compose semantics; substitute `docker compose` if available). The builder image carries Go 1.26 + `golangci-lint` + alpine apk deps + `postgres-client`; mounts the workspace and reuses Go module/build caches between runs.
- Five more Dockerfiles also became trackable and were checked in: `tests/e2e/mocks/Dockerfile.{llm,slack}-mock`, `tests/infrastructure/Dockerfile.ssh-{server,worker}`, `docker/security/snyk/Dockerfile`.

Dual API-key loader (2026-05-13)

Per operator mandate “We shall have API keys in `.env` file in the root of the project or all of them as the part of `api_keys.sh` in our host’s home dir! Both of it MUST BE fully supported!” — VERIFIED end-to-end:

- `scripts/load_api_keys.sh` prefers `$HOME/api_keys.sh` when present, falls back to `.env` at meta-repo root.

- Test 1 — `$HOME/api_keys.sh` path: loader sourced from project root, HOME pointing at real home dir → 42 secret-like vars in env.
- Test 2 — `.env` fallback path: loader sourced with HOME pointed at empty fakehome → 46 secret-like vars in env from project `.env`.
- Both paths individually load full secret set; neither silently drops. The loader's "prefer `$HOME/api_keys.sh`, fallback to `.env`" precedence matches the operator's stated semantics.
- **HelixLLM/.gitmodules has stale submodules/HelixQA declaration** (directory absent on disk; only declaration remains).

Constitutional debt (open since P0)

- **LLMsVerifier dual-pin divergence** (P0-04): RESOLVED in P1.5-WP2; doc had been stale. The transitive duplicate `helix_agent/LLMsVerifier` was eliminated in WP2 (only a backup remains at `helix_agent/.container-caps-backup-20260425-151947/LLMsVerifier`). `helix_agent/go.mod:232` now replaces `digital.vasic.llmsverifier => ../dependencies/HelixDevelopment/LLMsVerifier/llm-verifier`, so HelixAgent transitively consumes the canonical pin. The pin-parity gate `scripts/verify-llmsverifier-pin-parity.sh` is degenerate (regression-tripwire only, header confirms WP2 elimination). make `verify-foundation` evidence (this round): exits 0, "0 failures, all gates passed" — no `VERIFY_FOUNDATION_WARN_ONLY` waiver needed.
- **Historical SSH key + helix.security.json leaks** (P0-T08.5): material is immortal in git history. Mitigated; rotation required by operator.
- **SonarQube + Snyk live-scan deferral** (P0-T08.7): infrastructure wired, awaiting credential rotation by operator.

Phase 3 remaining items (carried forward, non-blocking for Phase 4)

- **HelixAgent build** — IMPROVED: submodules populated, `./cmd/helixagent/...` builds and tests pass. Wildcard `./...` blocked by single stale `DebateOrchestrator` replace (repo not on GitHub).
- **Containers build** — RESOLVED: `go build ./...` exits 0.
- **Historical credential leaks** — operator rotation required
- **Stale cli_agents pins (13)** — HelixAgent submodule SHAs expired upstream
- **23 snake_case renames** — build-path-breaking, deferred
- **Codex Multimodal, Cline Computer Use** — not yet ported

CONST-045 — No Hardcoded Distribution Hosts

ALL container distribution targets SHALL be configured exclusively through `CONTAINERS_REMOTE_HOST_N_*` env vars in `containers/.env` ($N=1..100$; iteration stops at first absent `_NAME`; the `containers` module `pkg/envconfig/parser.go` is the authoritative loader). The `.env` file is the sole source of truth for host enrolment — no host is hardcoded in HelixCode source, tests, challenges, or governance documents. Every non-unit test run and every production deployment MUST use whichever hosts are currently configured when `CONTAINERS_REMOTE_ENABLED=true`. Adding, removing, or modifying a host means editing `containers/.env`; no code change is required. The CURRENT configured set can be audited with `grep '^CONTAINERS_REMOTE_HOST_' containers/.env`; at the time of this rule's introduction (2026-05-07) the configured hosts were `thinker.local`, but the rule applies to whatever set is in `.env` at any future point ($N \geq 1$). Direct `docker/podman` commands, manual container start/stop, and ad-hoc remote hosts outside the `.env` mechanism are strictly prohibited.

How to resume

From a new CLI agent / LLM session

Type the **Resume Prompt** at the end of this file verbatim. It triggers continuation without further user context.

Programme conventions to apply (verbatim list)

1. **Subagent-driven-development always.** Never inline-implement multi-task features. Skip approval gates per the user's auto-approve memory (`memory/auto_approve_designs.md`).
2. **Commit on main.** All work flows through main. No feature branches.
3. **Push to 4 remotes (non-force only):** origin, github, gitlab, upstream for the meta-repo. Submodules push to their origin only (Challenges now has 3 remotes: origin + gitlab + upstream).
4. **Deepest-first push order.** Submodules → meta-repo. If meta-repo's gitlinks reference unpushed submodule SHAs, the meta-repo push will succeed but cloners will fail to resolve submodule pointers.
5. **Each feature has:** spec → plan → per-task TDD commits → Challenge harness commit → close-out commit. No exceptions.
6. **Anti-bluff smoke must always be clean.** Run before each commit:

```
grep -rn "simulated\|for now\|TODO implement\|placeholder" helix_code/internal helix_code/cmd && echo BLUFF  
|| echo clean.
```

7. **Runtime evidence required for every PASS** per CONST-035 / Article XI §11.9. No metadata-only / configuration-only / absence-of-error PASS.
8. **api_keys.sh > .env precedence.** Any tool that needs API keys sources them via scripts/lib/api_keys.sh first; falls back to .env.
9. **Non-FF push = STOP.** Never force, never --force-with-lease. If a push is rejected, investigate before retrying.
10. **No CI/CD pipelines.** All gates run via Makefile / scripts. Per CLAUDE.md Rule 1.
11. **No HTTPS for git.** SSH only.
12. **Every claim of “done” carries pasted terminal output** from a real run against real artefacts. Per CLAUDE.md Rule 8.

Picking up Phase 3 work

If Phase 3 is the active phase when you resume: 1. Verify state: `git log --oneline -3` should show the latest Phase 3 commits. 2. Read docs/improvements/PROGRESS.md §Phase 3 — Issue remediation section. 3. Continue with the next pending remediation item from the Phase 3 remaining list. 4. Run `make test` to verify current state before making changes.

Picking up Phase 4 work

If Phase 4 is the active phase when you resume: 1. Verify state: `git log --oneline -3` should show commits 4f5f8f0 (P0 fixes) and 7f4effd (P1 fixes). 2. Read docs/improvements/PROGRESS.md §Phase 4 section. 3. Anti-bluff sweep is COMPLETE for critical packages (validators, fix, config, deployment). 4. Verify remaining test files for weak assertions (only nil/err checks without content verification). 5. Run `make test` to verify current state before making changes. 6. Once all test suites validated, advance to Phase 5 (end-user materials).

Picking up Phase 5 work

Maintenance mandate

This document MUST be updated when:

- Any task is completed (update T-status table + add commit SHA).
- Any feature is closed out (update Phase status table + repository SHAs).
- Any known issue is discovered (add to “Known issues” section).
- Any phase boundary is crossed.

- Any deferred item is fixed or further deferred.
- Any new remote/submodule is added or removed.
- Any constitutional clause is added or amended.

If this document is out-of-sync with the actual state of the work, the inconsistency is a **CRITICAL DEFECT** — same severity as a false-success test result (CONST-035). See:

- CONSTITUTION.md Article XIII §13.1 (CONST-044) — Continuation Document Maintenance Mandate
- CLAUDE.md §12 — Continuation Maintenance
- AGENTS.md — “Continuation Maintenance” anchor

Verification (TBD): scripts/verify_continuation_sync.sh will compare: - Last updated: 2026-05-08T02:00:00Z (Phase 3 – remediation + test infra) -Active phasehere vsCurrent focusindocs/improvements/PROGRESS.md. - Tasks-done count here vs ticked-tasks count inPROGRESS.md`. - Known-issue list here covers all documented failures in evidence files.

Non-zero exit = sync violation → blocking pre-push.

Resume Prompt

Copy-paste this verbatim into a new CLI-agent session to continue:

Read /run/media/milosvasic/DATA4TB/Projects/helix_code/docs/CONTINUATION.md and continue all work. Use subagent-driven-development. Skip approval gates per the project's auto-approve memory. Push all submodules + meta-repo to all configured remotes (non-force only) when each work package or feature is closed out.

Document version log

Date	Updater	What changed
2026-05-06	Initial create	Captures state through P2-F21-T04 (5ef13b8
2026-05-06	T06 update	T06 (/approval slash command) closed; 6 of
2026-05-06	T07 update	T07 (main.go wiring + registry hook + integ
2026-05-06	T08 update	T08 (Challenge harness 5 phases, meta 2781 out + push 4 remotes) is next.
2026-05-06	T09 close-out	F21 (Codex Approval Modes) CLOSED — fir
2026-05-06	F22 docs	F22 (Aider Git Auto-Commit Per Change) sp
2026-05-07	F23 docs	F23 (Cline Browser Tool) spec + plan lande
2026-05-07	F24 docs	F24 (Codex Project Memory) spec + plan la

Date	Updater	What changed
2026-05-07	F25 docs	F25 (Plandex Plan Trees + Context Compact)
2026-05-07	F26 docs	F26 (Openhands Workspace + Task Planner)
2026-05-07	F27 docs	F27 (Aider Voice Input + Repo-Map) spec +
2026-05-07	F28 docs	F28 (Kilo-code AST-Aware Refactoring) spec
2026-05-07	F29 docs	F29 (Roo-code Full Port) spec + plan landed
2026-05-07	F30 docs	F30 (Continue IDE Integration) spec + plan
2026-05-07	Phase 3 entry	Phase 2 CLOSED (F21-F30 complete). Phase
2026-05-08	Full sync	F26-F30 close-out sections added; Phase 3 a synced with PROGRESS.md.
2026-05-08	Phase 3 sync	LLMsVerifier build, helix_qa build, 6 interna pruned. CONTINUATION.md synced with PI
2026-05-08	Phase 3 close	Phase 3 CLOSED. All code-actionable remed containers build fixed. Phase 4 started (anti
2026-05-08	ZB Phase 2-3	Zero-Bluff Phase 2 (Stub Elimination) and P CONST-046 added. Anti-bluff clean. Pushed
2026-05-08	ZB Phase 4	Zero-Bluff Phase 4 (Test/Challenge Hardenin config, deployment). Article XI §11.9 compli
2026-05-12	ZB Phase 4 close	Zero-Bluff Phase 4 CLOSED. Weak-assertion + internal/deployment with mutation-test co Commits 02b7306 + a3f8871. Cross-compile li
2026-05-12	Hygiene	Untracked 3 runtime test artefacts + tighter binary. Commit d7a7956; all 4 remotes synced
2026-05-12	Anti-bluff sweep	Re-running go test -short ./internal/... su internal/discovery/client.go discoverByDNS h net.DefaultResolver.LookupHost(ctx, ...) bou Engine.StartSession goroutine raced with t.T + t.Cleanup hook in setupQATestServer; (3) in tests starved under suite parallelism — gate consecutive full short-suite runs PASS clean
2026-05-12	Challenge sweep	bash tests/e2e/challenges/run_all_challenges cascade.sh exit 1 with 61 third-party submod unreachable in practice (can't commit into t paths). Relaxed verify script to accept prese deliberate ACK (in-submodule marker still h 8/8 PASS. Also untracked 2 more build binar extended .gitignore. Commit 099f06a.
2026-05-12	Challenges submodule sweep	cd Challenges && make test-short failed at Tes test hardcoded a consumer-Yole APK path th violating its CLAUDE.md “100% decoupled” YOLE_ANDROID_ID_APK_PATH) env var with SKIP-OK Challenges submodule commit 8024463 push repo pin bump 12904e4. make test-short 17/1
2026-05-12	Zero-bluff sweep	Per operator mandate “do not skip any tests testing.Short() skips in internal/tools/brows serializeChromiumLaunches(t) — exclusive flo

Date	Updater	What changed
2026-05-12	Skip-annotation sweep	launches across the entire go test process t. switched from runtime t.Skip to //go:build default test binary; Challenges submodule r 278b617); (3) fixed 12 real data races at prod internal/helixqa, internal/llm, internal/memo telemetry, internal/tools/mapping, internal/t Verifications: go test -race ./internal/... (f warnings, 3 consecutive runs; go vet clean. now) comment removed). Commit 49936c0 + Per the no-silent-skips governance gate: 101 (Containers 25, helix_qa 75, LLMsVerifier 1) (no adb/scrcpy/GStreamer/Tesseract/Paddle testcontainers, missing posix utils, etc.). NO code vet warning in helix_code/internal/llm, into the EOF-decode branch where the for-l four layered fixes: auto-exclude listed third- markers (#short-mode, P1-F14-T10), tighten JS (), path-based post-filter for nested-vendore opensource/). go vet ./cmd/... ./internal/.. e16bfeb, LLMsVerifier 98758126. Meta-repo co Per “fix and polish everything”: ran go test failures surfaced and resolved at root cause digital.vasic.models => ../Models pointed at dependencies/HelixDevelopment/Models as a sib decoupled by not relying on HelixAgent. (2) checksums for transitive testify/go-spew). F TestGetCapabilities asserted hardcoded mod Venice API; Venice renamed its catalogue, s substring assertions (strings.Contains(m, "l remain CONST-036 compliant. Submodule p Security eae0cec; new Models pin added. Ev commit 7c81a40.
2026-05-12	Submodule test-parity sweep	Continuing the per-submodule sweep: Vision [setup failed] due to stale go.sum drift (mis additionally had pkg/remote/deployer_test.go orphaned by the Ollama→LlamaCpp deploye “undefined” errors and was phantom-covera path tests in remote_test.go already cover th VisionEngine a092195, DocProcessor 3d11e41.
2026-05-12	Submodule test-parity round 2	Operator paused autonomous loop for sever (a) Discovered mistborn’s podman MACHIN back — podman ps reported “Cannot connect start”. Started VM via podman machine start power state is untouched). Re-ran scripts/m reconnected → database.connected:true, redi
2026-05-14	Mistborn restore + integration matrix probe (round 37)	integration matrix. Started local postgres sourced .env.full-test, ran go test -tags=in Surfaced ONE legitimate anti-bluff-correct t with "0" is not greater than "0" — the test c

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2026-05-13	Race-clean verification round 33	len(RetrieveMemory(...).Results) > 0, and ge catching the BUG #14 territory (cognee pro implement the real cognee API client — maj CORRECT by failing rather than fake-passin mistborn-tunneled stack; run_all_challenges evidence files / 107 counted PASS). All 4 ren (infrastructure work + finding documentatio Ran go test -short -race -count=1 across all task (1.0s), internal/worker (31.6s), internal All packages: race-clean. Zero race conc Manager fix shipped across rounds 1-32 host went idle mid-round (SCP closed during SKIP-OK: #env-mistborn-offline rather than fa gates. run_all_challenges: 12/12 PASS (one c round). No code changes this round — verifi
2026-05-13	Production-bug sweep round 32 (operator mandate re-asserted 4th time — broader surfaces)	Operator re-asserted the anti-bluff mandate beyond the saturated API matrix. Re-verified Broader-surface probes: (i) Challenges subn userflow 182s). (ii) helix_qa submodule shor (iv) containers — 8 packages PASS. (v) Helix — fails on missing X11/Xcursor (system-lib c build ./... which greedy-built the Fyne-GU headless environment WITH X11/Xcursor.h: main_nogui.go build-tagged variant AND a se compile-gate was lying about portability — i environments with optional system libs insta adding -tags=nogui to the verify-compile Mal in this environment without X11 libs. helix-q round — build-system fix). run_all_challenge
2026-05-13	Positive-coverage round 31 (final-corner + anti-bluff source scan)	Round 31 closed the final-corner gaps: (a) G (gorilla's reject-without-Upgrade — route IS both 503 with "QA engine is disabled" — ho 503 saying "disabled" is preferable to a 200 grep hits in helix_code/internal + helix_code substitution variable names (placeholder as t model_download_manager.go), (iii) low-prior refactor opportunities, no runtime bluff). Cl NO fake responses. helix-qa expanded 135 - screenshot/engines 503 with anti-leak asser same). helix-qa: 107/107 PASS (138 evidenc secrets.sh: clean. Session summary: 32 re rounds; helix-qa harness 6 → 138 eviden
2026-05-13	Positive-coverage round 30 (helix-bridge + account-deactivation)	Round 30: (a) helix-bridge live LLM smoke - deepseek/openrouter); 3/5 respond OK (gro (env key revoked or restricted); openrouter codebase bugs. (b) DELETE /users/me lifecy (200), then attempted to log in as that user VerifyJWTWithDB fix correctly enforces the "marked deleted" stub. (c) /system/status re

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2026-05-13	Positive-coverage round 29 (schema fidelity, 100-counted milestone)	("QA engine is disabled" — intentional config change). HCQA-098-PRE-REGISTER + HCQA-098-PRE-REGISTER + HCQA-099 (deleted-user login → 401 with deleted user after delete would be a critical CONST-035 vulnerability). helix-qa: 104/104 PASS (135 evidence files). secrets.sh: clean. Round 29: schema-fidelity static probes for type mismatches: every entry must have id/name/status/type per provider; llm/models — every entry must have id/provider; context length, which would mislead callers; count must be 0 (matches /memory/stats.systems); cross-endpoint guard). All clean. Probed /llm/models/registered; the single /providers/:id endpoint was expanded 127 → 130 evidence files: HCQA-098-PRE-REGISTER static-checks block (run early, in deterministic order) crossed the 100-counted-checks milestone). All clean.
2026-05-13	Positive-coverage round 28 (robustness)	Round 28: probed concurrent requests (20 per provider, duplicate-id collisions), large payloads (1 MB), empty body shapes (empty {} for required-field endpoints). Pagination is a feature gap, not a bluff: the limit (return ALL 315 tasks / 300 workers) was never claimed pagination support; the limit was for future product work. helix-qa expanded 124 evidence files (empty body), HCQA-093 (1MB-payload-201) expanded 124 evidence files. PASS (127 evidence files). run_all_challenges: 12/127 PASS. secrets.sh: clean.
2026-05-13	Production-bug sweep round 27	Round 27 probed auth/register input validation (correctly 400 — auth path is clean), then swapped to a more bug: (32) GET /workers/<malformed>/metadata respondInvalidID branch. Fixed. Also: SQL-injection in workers;--") — pgx prepared statements parser table survived intact. Positive-coverage probes for concatenated SQL. helix-qa expanded 120 → 124 evidence files (malformed-400), HCQA-090 (register-overload-201) expanded 124 evidence files, HCQA-091 (SQL-injection defense — worker uses per-run unique hostname to avoid duplicate hostnames) (124 evidence files). run_all_challenges: 12/127 PASS. secrets.sh: clean.
2026-05-13	Production-bug sweep round 26	Round 26 found 3 more bugs in sessions + workers: project_id (well-formed UUID but no such project exists) + existent projects. Root cause: in-memory session manager declares. Fix: createSession handler now validates projectManager.GetProject(req.ProjectID) BEFORE returning ErrProjectNotFound for missing project. (30) Session created with non-UUID project_id (value too long) — session created with non-UUID project_id respondInvalidID for malformed UUIDs → 400. 500 leaking pg SQLSTATE 22001 "value too long for character" leakage (column type visible) + CONST-035 (worker.ErrWorkerHostnameTooLong sentinel + hostname too long).

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2026-05-13	Production-bug sweep round 25 (mandate re-asserted 3rd time)	RegisterWorker pre-validates BEFORE the I expanded 117 → 120 evidence files: HCQA-0 malformed-project_id→400), HCQA-088 (wo “SQLSTATE”/“22001”/“character varying”). run_all_challenges: 12/12 PASS. go test -sho clean. Operator re-asserted the anti-bluff mandate cascade (0 failures) + anchor in all 6 root do more real bug — (28) PUT /projects/<malform HTTP 500 leaking raw postgres "invalid inp Same CONST-042 + CONST-035 double-viol DeleteProject in manager_db.go passed the ra validation; pg rejected with the SQLSTATE l err != nil { return ... ErrInvalidProjectID additional handlers that were missing respon createTaskCheckpoint, getTaskCheckpoints, TestDatabaseManager_DeleteProjectInvalidID u — the old test mocked a postgres SQLSTATE wrap (which leaked the pg detail); new test malformed input. helix-qa expanded 112 → 1 DELETE 400 with anti-leak check for “SQLS POST/GET 400), HCQA-085 (session PUT-on run_all_challenges: 12/12 PASS. go test -sho clean.
2026-05-13	BUG #27 fix generalized — round 24	Round 24 extended the round-23 fix (task.Err wired) across the full handler matrix. Added project.ErrInvalidOwnerID sentinels (sed-repl fixed the cross-package AssignTask call site with the task sentinel since the failure surfa over all 4 sentinels. Wired 7 additional hand retryTask, deleteWorker, updateWorker, wor through. Verified with curl matrix: 7 endpoi 400. helix-qa expanded 105 → 112 evidence complete/retry/fail + worker-delete/update/ evidence files). run_all_challenges: 12/12 PA scan-secrets.sh: clean.
2026-05-13	Production-bug sweep round 23 (operator re-asserted mandate)	Operator re-asserted the anti-bluff mandate cascade requirement. Re-confirmed: verify- + listed third-party submodule; CONST-035 docs (CONSTITUTION.md / CLAUDE.md / A password login (correctly 401), duplicate-en GET /tasks/ (correctly 404 via getTask’s gen returned HTTP 500 with “invalid task ID: in malformed UUID is CLIENT input error, sho task-manager call sites that do uuid.Parse(ic unstructured returns replaced via sed across err, what) writes 400 when the sentinel mat for the remaining 7 task-id handlers as a fut

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2026-05-13	Production-bug sweep round 22	evidence files: HCQA-073 (DELETE-malform run_all_challenges: 12/12 PASS. go test -sho Round 22 probed duplicate-create + workflo Found 2 more bugs: (25) duplicate-hostnam "workers_hostname_unique" (SQLSTATE 23505). I isUniqueViolation(err) helper that parses pg unique-violations BEFORE leaking; handler workflows/* returned 500 with "project not f project.ErrProjectNotFound sentinel (7 unstru manager_db.go); 4 workflow handlers conso Bonus fix: the 3 non-planning workflow hanc memory manager) instead of s.projectManag s.projectManager. helix-qa expanded 101 → 1 hostname-409-no-leak with explicit anti-leak "workers_hostname_unique"/"SQLSTATE"/" helix-qa: 76/76 PASS (104 evidence files). ru SSH tunnels that dropped mid-session — sin mistborn-up.sh re-run brings them back). scri Round 21 probed remaining endpoints for si <bogus>/heartbeat returned HTTP 500 with r \"worker_metrics\" violates foreign key cons — TWO violations in one response: (a) CON SQL state visible to API users); (b) CONST-0 resource error). Root cause: UpdateWorkerHea even after the workers-table UPDATE match UPDATE FIRST; if zero, surface ErrWorkerNot the sentinel to 404 with clean "Worker not f bogus task IDs — already correctly return 4 since the WHERE-id-and-status filter can't d query; 422 + clear message is acceptable). I bogus-404 with explicit anti-leak check (resp against CONST-042 regression). helix-qa: 74 PASS. go test -short server+worker: ok. scri
2026-05-13	Production-bug sweep round 21	Round 20 source-scanned for the broader "E antipattern. Found 5 occurrences across tas resource id caused HTTP 500 instead of 404 being a server error when the cause is "clie exported sentinels task.ErrTaskNotFound (cov (covers all 6 session manager.go not-found r worker.ErrWorkerNotFound already in memory_ UpdateWorker. The 5 handlers (deleteTask, errors.Is-check and return 404. helix-qa exp DELETE-task-404, HCQA-066 fail-task-404, worker-404, HCQA-069 DELETE-session-40 run_all_challenges: 12/12 PASS. go test -sho clean.
2026-05-13	Production-bug sweep round 20 (100-evidence milestone)	Round 19: probed task dependencies (forwa reassign behavior, project-delete cascade. F already-assigned task returned HTTP 500 (s

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2026-05-13	Production-bug sweep round 18	AssignTask wrap task.ErrTaskInvalidStateTra RowsAffected==0; assignTask handler now message. Decided NOT to fix: dependency f validates deps lazily) and soft-delete with or status='deleted', sessions can legitimately r HCQA-064-PRE-TASK + HCQA-064-PRE-W1 HCQA-064 (reassign-already-assigned → 422 run_all_challenges: 12/12 PASS. go test -sho Round 18 found 1 more bug while probing t on a non-running task, POST /tasks/:id/start already-completed task all returned HTTP 5 retry-on-non-failed. Each was a client-side s broken” when really the request was incomple introduced exported sentinel task.ErrTaskInv ErrTaskNotRetryable). StartTask and Complet handlers errors.Is-check and return 422 Un
2026-05-13	Positive-coverage round 17	Round 17: scanned for more silently-discard Probed three deeper state-accuracy question complete with {result: ...} body — YES (a request key result, response key result_data BY created_at DESC — YES, first entry is the exactly 1 when a new task+worker is create hardcoded-zero or stale-cache bluff). No nev that mechanically lock these invariants in pl PRE-CREATE + HCQA-058-PRE-START (set HCQA-059-PRE-CREATE + HCQA-059 (list I CREATE-T + HCQA-060-PRE-CREATE-W + I (84 evidence files). run_all_challenges: 12/1
2026-05-13	Production-bug sweep round 16	Round 16: scanned for more silently-discard beyond the comment documenting the fix. T #20: (20) POST /api/v1/workers/:id/heartbeat workers.cpu_usage_percent / memory_usage_perc forever. The handler updated only workers.la the worker’s snapshot columns (which the C were never touched. Time-series data WAS p resource’s own snapshot fields were frozen worker.DatabaseManager.UpdateWorkerHeartbeat (CASE WHEN > 0 preserves existing on om evidence files: HCQA-HB-PRE-W + HCQA-H workers/:id snapshot reflects heartbeat — c run_all_challenges: 12/12 PASS. go test -sho
2026-05-13	Production-bug sweep round 15 (governance reinforced)	Operator re-asserted the anti-bluff mandate <i>bluff manner — they MUST confirm that all</i> CONST-035 / Article XI §11.9 anchor IS pres CLAUDE.md / AGENTS.md + helix_code/ tri across every owned + listed third-party sub a TRIPLE bluff: (19) POST /api/v1/tasks/:id/ while task_checkpoints history table stayed p for the history INSERT — silently discarding

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		the schema requires worker_id NOT NULL REFERENCE Triple bluff: (a) discarded error, (b) fabricated error, (c) fabricated success Subsequent GET /tasks/:id/checkpoints always returns a success result HONEST but the underlying bluff (his sentinel task.ErrCheckpointRequiresAssignment) is not + supply worker_id in INSERT + surface request HCQA-CKPT-PRE-W / HCQA-CKPT-PRE-T (see HCQA-CKPT-PRE-ASSIGN, HCQA-055 (assigned worker_id just-created checkpoint — proves real persistence) PASS (72 evidence files). run_all_challenges secrets.sh: clean.
2026-05-13	Positive-coverage round 14	Round 14 added positive-coverage probes for new bug surfaced — but locking the behavior llamacpp + /llm/providers/openrouter: all 40 constitutional cap at 7 entries (FallbackModel) provider coverage is achieved via the live verification retry full round-trip and /projects/:projectId/61 → 66 evidence files: HCQA-051-CREATE + error_message), HCQA-052 (retry transition) round-8 sentinel distinguishes legitimate retry round-trips). helix-qa: 57/57 PASS (66 evidence files) secrets.sh: clean.
2026-05-13	Production-bug sweep round 13	Round 13 uncovered 1 more real bug while assigning, the response showed "assigned_worker_id" column was correctly populated. Same regression test column into local pointers assignedWorkerID/columns assigned them to the returned Task struct's assignedWorkerID silently lied about assignment state for the error (which calls GetTask post-update). Fixed in HCQA-ASSIGN-PRE-W + HCQA-ASSIGN-PRE-W assigned_worker matching requested worker_id trips assigned_worker — proves the fix works HCQA-050 (heartbeat endpoint). helix-qa: 54/54 PASS. go test -short server+task+worker: ok
2026-05-13	Production-bug sweep round 12	Round 12 uncovered 1 more real bug (5th in session-action lifecycle probes: (17) GET /api/v1/workers/:id/metrics empty-history case — same nil-slice→null blank metrics. Fixed in getWorkerMetrics with []*WorkerMetric) → 56 evidence files: HCQA-044 (worker-metrics) + HCQA-044-PRE-SESS (project+session for worker for the probe — decouples HCQA-044) + HCQA-044-PRE-SESS (project+session for worker for the probe — decouples HCQA-044) HCQA-046 (action=complete → status:complete) error). helix-qa: 51/51 PASS (56 evidence files) scripts/scan-secrets.sh: clean.
2026-05-13	Production-bug sweep round 11	Round 11 uncovered 1 more real bug plus a new bug surfaced — but locking the behavior PUT /api/v1/workers/:id returned HTTP 500 "constraint" — same TEXT[] NOT NULL pattern related defect: the UPDATE unconditionally updates hostname (a partial update) would overwrite the

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2026-05-13	Production-bug sweep round 10	fixed in worker.DatabaseManager.UpdateWorker: bare SET col = \$1 with COALESCE(NULLIF(\$1, ' THEN \$4 ELSE max_concurrent_tasks END for ma the input is zero-value. helix-qa expanded 43 HCQA-042 (PUT renames display_name + m HCQA-043-DEL (DELETE 200), HCQA-043 (run_all_challenges: 12/12 PASS. go test -sho Round 10 uncovered 1 more real bug while returned HTTP 500 “ERROR: column path d update attempt. Root cause: internal/project referenced path and type columns that DON database.go:333). Real schema columns: work (which holds the project type under config[“rename a project” endpoint was therefore u GetProject’s column-mapping pattern: RETU config[“type”] and metadata from config[“met HCQA-039 (create-for-lifecycle), HCQA-040 round-trip), HCQA-041-DEL (DELETE 200), evidence files). run_all_challenges: 12/12 PA secrets.sh: clean.
2026-05-13	Production-bug sweep round 9	Round 9 uncovered 1 deep bluff in the mem systems all hardcoded as status: “available 0 — direct contradiction. Reading the handle metadata with NO connection probe of any l any of the 6 listed providers (Cognitive/Weavi handler claims “available” for 6 systems wh connections. Fix: introduced a memoryStatus(manager is bound + a reachability probe ru (matches stats). The catalogue itself (id/nam of supported backends — that’s real metadata “HCQA-MEM-BLUFF” probe asserts NO ent future regression to hardcoded “available” v PASS (41 evidence files). run_all_challenges secrets.sh: clean.
2026-05-13	Production-bug sweep round 8	Round 8 uncovered 2 more real bugs by ext tasks/:id/checkpoints returned “checkpoints” JSON contract bluff (tasks, projects, worker []map[string]interface{}{}) in getTaskCheckpoi “task not found, not in failed state, or max r error code that lies about the nature of the p side state mismatch, not an internal bug). In internal/task/manager_db.go; handler now er state error, 500 only for genuine DB faults. I create-for-lifecycle, HCQA-037 checkpoints- HCQA-038-PRE-START / HCQA-038-PRE-CC internal-step probes write evidence but don test -short server+task: ok. scripts/scan-sec
2026-05-13		Coverage expansion: probed POST /api/v1/se the round-4 HCQA-031 create-project probe

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	Sessions-create probe addition (round 7)	surfaced — Mode validation correctly reject building/testing/refactoring/debugging/depl HCQA-035 probe stitches the just-created p asserts 201 + session.project_id round-trip run_all_challenges: 12/12 PASS.
2026-05-13	Production-bug sweep round 6	Round 6 uncovered 1 more real bug, same p returned HTTP 401 “User not authenticated” Root cause: refreshToken handler called c.Ge (must stay public for register/login). Fixed b VerifyJWTWithDB, mirroring the logout handle JWT (200) for valid Bearer, 401 with “Author “Invalid or expired token” for malformed/exp test updated to TestRefreshToken_WithBearer (unverifiable Bearer returns 401 which is the HCQA-033 (valid Bearer issues new JWT — : iat is per-second and within-1-sec refresh ca Bearer returns 401). helix-qa: 34/34 PASS. r ok. scripts/scan-secrets.sh: clean.
2026-05-13	Production-bug sweep round 5	Round 5 uncovered 1 more real bug, same p returned HTTP 500 with ssh_config NOT NU optional ssh_config map. Same fix as task_da sshConfig to map[string]interface{}{} and nil — pgx serializes nil slice as NULL; the colum explicitly provides the value). helix-qa expan body and asserts 201 + UUID + hostname-r PASS. go test -short server+worker: ok. scri
2026-05-13	Production-bug sweep round 4	Round 4 uncovered 2 more real bugs: (8) PO invalid UUID length: 12” — internal/project "default-user") literally hardcoded as the ov matching the error. Handler now extracts th CreateProjectWithUser directly with user.ID.S publicProjects Gin group with comment // n create projects or trigger workflow executio projectId. This was a real production securit now requires c.Get("user") which is only set projects group; publicProjects removed enti. POST /projects must 401), HCQA-031 (auth Existing unit test TestCreateProject_ValidReq (mirrors TestListProjects pattern). helix-qa: internal/server/...: ok. scripts/scan-secrets.s
2026-05-13	Production-bug sweep round 3	Continuing the anti-bluff push uncovered 2 r api/v1/users/me returned a User stub with is authMiddleware was calling the cheap VerifyJ JWT claims; every other field defaulted to ze (fetches the full user + enforces is_active). bonus: deactivated accounts can no longer c api/v1/workers returned "workers": null — th tasks and projects); fixed by defaulting to [] (workers as array), HCQA-028 (system stats

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2026-05-13	Production-bug sweep round 2	AND created_at non-zero — would have caused PASS. run_all_challenges: 12/12 PASS. go test Continuing anti-bluff push uncovered 3 more checks with tasks-CRUD lifecycle and project 500 “null value in column task_data violates internal/task/manager_db.go:CreateTask forward fixed by defaulting parameters to empty map task_data JSONB NOT NULL now upheld). (4) GET empty list — JSON contract bluff (Go’s nil-slice projects returned 401 “user_id not found in c.GetString("user_id") while authMiddleware’s entire projects-list endpoint was unreachable established c.Get("user") pattern. (6) Same HCQA-019..027: tasks-CRUD full lifecycle (1 deleted-404), plus list-projects-as-array and 27/27 PASS. run_all_challenges: 12 passed. secrets.sh: clean. Stood up the mistborn.local distribution stack operator mandate “all tests and Challenges tested codebase really works as expected”. (gitignored, key-based auth only — password compose.mistborn.yml 7-service podman stack grafana); (3) scripts/mistborn-{up,down}.sh b scripts/bridge/main.go (helix-bridge) multi-p loader contract — 3/5 providers respond live anti-bluff QA harness with captured wire evi duration_ms in per-check JSON); (6) tests/e run_all_challenges.sh (12/12 PASS); (7) BLU returned a fabricated status: available stub returns 404 for unknown providers, sourced TestGetLLMProvider_UnknownIDReturns404 repro deep content (count>0, database.connected login → /users/me → garbage-bearer-401 → l from registration, user.username === register in HCQA-004/005/006 (assertions of expecta discovered + fixed: helix_qa orchestrator 0- tests at integration_test.go/orchestrator_edg assert.True(t, result.Success) to anti-bluff a 180s. Local verification: HelixCode locally s tunnel; helix-qa: 19/19 PASS with no FAIL a HelixCode internal -race short: 91 packages bluff smoke: clean (5 pre-existing benign ma commits 6e58ab6 → 30a1e2b → 5c44ba3, all 4 re
2026-05-13	Mistborn distribution + anti-bluff QA harness	(1) Cascade Challenge runtime evidence (DDoS, stress, chaos) × 5 submodules (Helix PASS against stub on port 8081 with captur (helix_qa/containers/EventBus/HelixLLM DL 19. The structural-only concern is now subs submodule cascade with consistent latency
2026-05-15	Cascade Challenge runtime-validation widened + Task #274 isolation infeasibility documented (round 41 close-out ⁴⁴)	

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2026-05-15	CONST-051(B)↔(C) tension documented + go.work attempt + stability re-evidence (round 41 close-out ⁴³)	enumerated 19 files importing digital.vasic lines with debate constants (ModelHelixDebat fields deeply interwoven with non-debate ha into debate vs non-debate halves — multi-da Documented as truly infeasible in single-ses upstream repo OR refactor HelixAgent to re Operator-directed continuation. (1) CONST docs/architecture/CONST-051-tension.md: docu HelixLLM post-Tasks #254/#273 (relative-p Attempted go.work workspace fix at HelixCo digital.vasic.docprocessor module identifier AND dependencies/vasic-digital/DocProcesso LLMOrchestrator, LLMProvider, VisionEngin Documented 4 real-fix options for operator: [recommended], (C) accept current state, (D tension persists by design. (2) Stability re- PASSED (one transient 19/1 caught early — Challenge, recovered next run, same transie cascade coverage. HelixCode inner make ver authoritative — 7 alternate org-name URLs Operator-directed deep close. (1) CONST-0 + constitution + 64/68 owned at parity; depe cascade commit) pulled via FF; 3 third-party flagged informational. (2) Cascade Challen port 8081 — proves structural cascade is NO p95=6.1ms, containers DDoS 50/50 100% 50/50 100% p50=2.2ms, dependencies/He Captured wire evidence — partial address o cascade Challenges now have proven happy template structure). (3) Task #274 forensi DebateOrchestrator, vasic-digital/debate, Hel Debate_Orchestrator, vasic-digital/debate-orch github. 12 sub-packages required (agents, a testing, tools, topology, validation, root). Bu feature fake”). Operator action genuinely debate-dependent code (~6 files in internal, module source. Task #274 stays pending. Operator-directed deep close of all actionab per-submodule capture: ran git submodule party (LLama_CPP 187 ahead, Ollama 33, H vasic-digital/Models 1 ahead — pulled via FF originated upstream). (2) CONST-050(B) c coverage.sh regression gate — exposed that received governance anchors but NOT the 6 cascade (10 new in close-out⁴⁰ + 44 from cl HelixCode inner module verify-compile: (the CONST-051(C) Task #254/#273 refactor regenerated: 9 platforms × 30 features × 6 VERIFIED cells preserved; governance verifi
2026-05-15	Address remaining unfinished items — cascade runtime- validated + Task #274 forensics + Models pulled (round 41 close-out ⁴²)	Operator-directed deep close. (1) CONST-0 + constitution + 64/68 owned at parity; depe cascade commit) pulled via FF; 3 third-party flagged informational. (2) Cascade Challen port 8081 — proves structural cascade is NO p95=6.1ms, containers DDoS 50/50 100% 50/50 100% p50=2.2ms, dependencies/He Captured wire evidence — partial address o cascade Challenges now have proven happy template structure). (3) Task #274 forensi DebateOrchestrator, vasic-digital/debate, Hel Debate_Orchestrator, vasic-digital/debate-orch github. 12 sub-packages required (agents, a testing, tools, topology, validation, root). Bu feature fake”). Operator action genuinely debate-dependent code (~6 files in internal, module source. Task #274 stays pending. Operator-directed deep close of all actionab per-submodule capture: ran git submodule party (LLama_CPP 187 ahead, Ollama 33, H vasic-digital/Models 1 ahead — pulled via FF originated upstream). (2) CONST-050(B) c coverage.sh regression gate — exposed that received governance anchors but NOT the 6 cascade (10 new in close-out⁴⁰ + 44 from cl HelixCode inner module verify-compile: (the CONST-051(C) Task #254/#273 refactor regenerated: 9 platforms × 30 features × 6 VERIFIED cells preserved; governance verifi
2026-05-15	Complete cascade: 54 submodules + regression gate + verify-compile + ledger refresh (round 41 close-out ⁴¹)	Operator-directed deep close of all actionab per-submodule capture: ran git submodule party (LLama_CPP 187 ahead, Ollama 33, H vasic-digital/Models 1 ahead — pulled via FF originated upstream). (2) CONST-050(B) c coverage.sh regression gate — exposed that received governance anchors but NOT the 6 cascade (10 new in close-out⁴⁰ + 44 from cl HelixCode inner module verify-compile: (the CONST-051(C) Task #254/#273 refactor regenerated: 9 platforms × 30 features × 6 VERIFIED cells preserved; governance verifi

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2026-05-15	Task #273 COMPLETE — HelixLLM 34 nested submodules migrated; full cascade closed (round 41 close-out ⁴⁰)	coverage.sh asserts every owned submodule (executable, ≥1KB, contains PASS + SKIP-OK per github_pages_website). Final anti-bluff capture GREEN + verify-cascade-coverage.sh 66/66 HelixCode inner compile clean. Remaining helix #274 (DebateOrchestrator missing repo, open services running), helix_qa autonomous session design). All 6 verifier gates GREEN against the full using the identical pattern from completed Task own-org submodules + updated go.mod repository → ../../vasic-digital/<Name> for vasic-digital for parent-root residents). 10 new vasic-digital Watcher, Middleware, RateLimiter, I18n, Re full CONST-048..060 anchor block + .gitignore Lazy 5869b9a, Watcher 7850f10, Middleware e Document ad85bb4, Filesystem d13a5f2, TOOM → +10 new; -1 constitution removed per CO cascade target). Anti-bluff captured evidence implementable gate green – anti-bluff covenant submodules), G2 anti-bluff smoke, G3 mock-audit, G6 case-conformance all GREEN. Two coordinated cross-org rename), Task #274 (upstream). Operator-directed comprehensive audit of u submodule_owned.txt had only 12 entries while 5/6+0/6+20/20 sweeps in close-outs^{35,38} were CONST-035 false-success in the verifier's submodules (3 HelixDevelopment + 42 vasic missing same anchors. (3) 4 submodules (Ba missing .gitignore per CONST-053. (4) Helix style problem (tracked as Task #273, multi-on digital.vasic.debate → ./DebateOrchestrator doesn't exist on GitHub; production code at imports it). Fixes shipped: 45+1 submodule insertions per file × 3 files × 46 submodules pushed to their origin remotes (panoptic f5d830aecc, and 41 vasic-digital submodules with 45 included in this commit. Final verifier s tracked as Task #273. Pre-existing breaka (Task #274 candidate — operator needs to r code).
2026-05-15	Address unfinished features + issues — uncover real cascade gaps, fix 45+1 ungoverned submodules (round 41 close-out ³⁹)	Operator re-invocation of canonical anti-bluff CONST-060 (Fetch-before-edit Mandate), the all --prune (all 4 HelixCode remotes fetched upstream — no drift since close-out ³⁷), git -- → empty (constitution still at 1b0b03a, the \$1 evidence per §11.4.2 of its own anti-bluff inv evidence — the actual HEAD.@{u} output, even
2026-05-15	Anti-bluff re-invocation — CONST-060 protocol self-exercised + 6/6+20/20 re-evidence (round 41 close-out ³⁸)	

Date	Updater	What changed
2026-05-15	CONST-060 cascade — Fetch-before-edit Mandate codified + anti-bluff re-evidence capture (round 41 close-out ³⁷)	evidence requirement): re-ran the verification of <code>verify-all-constitution-rules.sh</code> 6/6 GATES “bluff covenant honoured”; (2) <code>run_all_challenges.sh</code> CHALLENGES PASSED! Anti-Bluff Verification: COMPLETE (CONST-035 / §11.9 + CONST-047..060) ×3. The bluff covenant is honoured every time the operator is captured during execution. No false-successes. Operator re-invocation of canonical anti-bluff verification fetch+pull per CONST-049/060 (which is the upstream added §11.4.37 (Fetch-before-edit) to prevent session redundant-work incident on Project-CONSTITUTION.md + CLAUDE.md + AGENT.md (36 governance files updated this round). Each round: (403603d), Challenges (af4fba1, 4 remotes), canonical github_pages_website (b0ccb17), DocProcessor (5af6b6b), VisionEngine (f229b64), LLMsVerification (f229b64). <code>.gitmodules</code> constitution pointer bumped to 1. evidence per §11.4.2: (1) <code>verify-all-constitution-rules.sh</code> Task #254 close); (2) <code>run_all_challenges.sh</code> Anti-Bluff Verification: COMPLETE; (3) verification present across every owned submodule. The codebase carries positive runtime evidence G4 verifier-gate FAIL eliminated; verifier-gate batches (close-out³⁵ + this commit’s 8 follow-up from HelixAgent to canonical parent-root local vasic-digital/<name>, 3 HelixDevelopment resolves to ../Challenges (already at HelixCo 55c9a9d1) on its origin (HelixDevelopment/Helix go.mod replace directives. Replace directive HelixDevelopment for the 3 HelixDevelopment documented import/SDK/runtime resolver” per surprise: HelixAgent’s ConversationContext s digital/conversation.git (lowercase, different Task #254. Remaining single-session-block orgs — operator-coordination required). First measurable progress on Task #254 (the submodules migrated from <code>helix_agent/<name></code> canonical parent root at <code>dependencies/vasic-digital/</code> Observability, Auth, Storage. Workflow valid <name>.git dependencies/vasic-digital/<name> rm <name> in HelixAgent; (3) update HelixAgent vasic-digital/<Name> — this is the “document names. HelixAgent commit 3b97d1b1 pushed pointer-bumps for 5 new submodules + Helix 41 nested own-org chains (5 fewer). Remaining proven pattern. Per-submodule pattern repeated.
2026-05-15	Task #254 COMPLETE — all 46 HelixAgent nested own-org submodules migrated to parent root (round 41 close-out ³⁶)	Final capture after the 12-submodule CONSGates GREEN, 1 FAIL = G4 known Task #254 G1 governance cascade PASS (14 anchors ×
2026-05-15	Task #254 pilot: CONST-051(C) HelixAgent migration — 5 of 46 nested submodules (round 41 close-out ³⁵)	Final capture after the 12-submodule CONSGates GREEN, 1 FAIL = G4 known Task #254 G1 governance cascade PASS (14 anchors ×
2026-05-15	Round-41 cascade capstone — post-cascade verifier	Final capture after the 12-submodule CONSGates GREEN, 1 FAIL = G4 known Task #254 G1 governance cascade PASS (14 anchors ×

Date	Updater	What changed
	sweep + ledger refresh (round 41 close-out ³⁴)	PASS (zero production bluff markers), G3 m audit PASS, G6 case conformance PASS. (2) failed — released evidence still holds. (3) r platforms × 30 features × 12 owned submo preserved). Round-41 totals: 13 close-outs HelixCode + 11 owned submodules; 12 of 1 matrix (HelixQA, Challenges, Containers, S VisionEngine, LLMsVerifier, Models, panopt reasons (HelixAgent multi-session Task #25 remotes (github + gitlab + origin fanout + u anti-bluff covenant is honoured: 20/20 C owned-submodule cascades + 72 new te CONST-035 / Article XI §11.9.
2026-05-15	CONST-051(A) panoptic + github_pages_website cascade — 12 of 14 owned submodules now covered (round 41 close-out ³³)	Final two non-blocked owned submodules ca panoptic) adds the 6 CONST-050(B) Challen github_pages_website (commit pushed to challenges/scripts/ directory with the 6 Cha gracefully — structural contract still capture bumped per CONST-049 step 7. 12 of 14 ov coverage (HelixQA + Challenges + contain LLMProvider + VisionEngine + LLMsVerifie Challenge files this session: 72 (6 HelixCode submodules excluded for valid reasons: Hel chains need refactoring first per CONST-051 surface — would be redundant). Cumulative + ³¹ Ch/Co/Sec + ³² 6 Deps + ³³ Pano+GhPw
2026-05-15	CONST-051(A) dependencies/ HelixDevelopment cascade — 6 submodules × 6 test types (round 41 close-out ³²)	Mass cascade continuing operator's "do not submodules cascaded with the 6 missing CC cascade_const050b.sh) — same anatomy, per-s commit pushed), LLMOrchestrator (LLMORC (VISIONENGINE_*), LLMsVerifier (LLMSVERIFIER new Challenge files added across 6 submo respective origins. HelixCode .gitmodules po 7. Cumulative round-41 cascade: HelixQA + dependencies/HelixDevelopment = 10 o matrix (60 new Challenge files this session) panoptic, HelixAgent (multi-session Task #2
2026-05-15	CONST-051(A) Challenges + containers + Security cascade — 6 new test types each (round 41 close-out ³¹)	Continuing the operator's "do not stop until Three more owned submodules cascaded wi Challenges (commit 873c0b1, pushed to all 4 fanout + upstream); Containers (commit 7f Security (commit 56fdc16, pushed to origin bluff Challenges (ddos / stress / chaos / scali (CHALLENGES_*, CONTAINERS_*, SECURITY_*). All 6 they SKIP-OK cleanly when their target env evidence when wired. Challenge counts per 5→11. HelixCode meta-repo .gitmodules poin Cumulative round-41 submodule cascade: H owned submodules now have full CONST

Date	Updater	What changed
2026-05-15	CONST-051(A) helix_qa submodule cascade — 6 new test types in helix_qa (round 41 close-out ³⁰)	round). Remaining 8 owned submodules: As mcp_servers, panoptic, etc. — to follow in su Per operator’s “do not stop until ABSOLUTE submodule-side CONST-050(B) cascade. heli challenges/scripts/ matching the round-41 E stress_sustained_load_challenge.sh + chaos_f scaling_horizontal_challenge.sh + ui_termina Each ports the HelixCode template (5-6 step honest SKIP-OK per §11.4.3 topology-dispat localhost:8081/health, HELIXQA_BIN auto-disco against Python stub: 200/200 (100%) pass Other 5 SKIP-OK cleanly when helix_qa infr to its origin remote (github.com:HelixDevel .gitmodules pointer bumped to 1f03882 in thi submodules to cascade: Challenges, Contain
2026-05-15	CONST-050(B) matrix-completion sweep + ledger regeneration (round 41 close-out ²⁹)	Post-#266-closure verification capture. (1) : governance cascade PASS (14 anchors × 36 markers in helix_code/{internal,cmd}), G3 r internal/mocks), G4 nested own-org submod session), G5 .gitignore + sensitive-file PASS, candidates / 12 protected). 5/6 green; 1 FA run_all_challenges.sh end-to-end: 20/20 PAS Verification: COMPLETE. This is the release (3) scripts/regenerate-coverage-ledger.sh: re features × 12 owned submodules; 46 nested cells preserved). The Anti-Bluff Verification : scaling ✓ UI ✓ UX ✓ (all 6 of Task #266’s m runner).
2026-05-15	UX End-to-End Flow Challenge — FINAL 6th of CONST-050(B) missing test types — closes Tasks #266 + #256 (round 41 close-out ²⁸)	Task #270 closed; Tasks #266 + #256 (6 challenges/ux_end_to_end_flow.sh — drives th across the chain: discover → health → list-m post-liveness. 6-step structure: (1) binary-j hints (wizard/provider/key/list-models/health exposes ≥1 selectable model (pick first); (5) xyz -prompt ping → assert error is user-com name / “model” / “provider” / “not found” / “ hostile (panic / goroutine / runtime error / s EVERY step’s output); (6) post-error -health journey, no leaked state). Anti-bluff value: from journey coherence — what UI proves a hostile blacklist (panic:, goroutine [0-9]+ \[catches the “stack-trace leaks to user surfac Configurable knobs: HELIX_CLI_BIN, UX_CLI_ captured evidence: binary OK, 4/5 recovery llama-3.2-3b), bogus model → exit 1 + 4 use now 20/20 PASS (was 19/19 after UI). Task commit. Anti-bluff matrix per CONST-050(B) ✓ scaling ✓ UI ✓ UX ✓.
2026-05-15		

Date	Updater	What changed
	UI Terminal-Interaction Challenge — 5th of CONST-050(B) missing test types (round 41 close-out ²⁷)	Task #269 closed (Task #266 5-of-6 installm drives the built helix_code/bin/cli binary no (1) binary-presence dispatch (SKIP-OK if hel (all 8 documented flags present: -approval - interactive -notify); (3) -health output as containing operational /  / ); (4) -list-m of Provider: + Context Size: + Status: lines - (5) post-invocation re-runability (second temp” classes); (6) invalid-flag graceful-ex timeout). Configurable knobs: HELIX_CLI_B UI_MIN_MODELS (1). Anti-bluff value: catches return-code gating misses. Validated end-to- model entries with all 3 labels matched, re now 19/19 PASS (was 18/18 after scaling). installment.
2026-05-15	Horizontal-Scaling Challenge — 4th of CONST-050(B) missing test types (round 41 close-out ²⁶)	Task #268 closed (Task #266 4-of-6 installm operator-safe horizontal-scaling Challenge v SCALING_REPLICA_URLS env var resolving to ≥2 dispatch (reachable-replica detection + SKI load test (50 reqs × N replicas at concurren (±50% tolerance, informational since direct replicas (modulo uptime/timestamp via sed DB/cache” misconfiguration class which is a probe. Configurable knobs: SCALING_REPLIC SCALING_CONCURRENCY (10), SCALING_MIN_PASS_PO OK path on bare run (default empty REPLIC Validated happy path against 3-replica Pytho reachable, schema valid each, 150/150 (10 sha256 matched across all 3 replicas (eh 18/18 PASS (was 17/17 after chaos). Remain
2026-05-15	Chaos Failure-Injection Challenge — 3rd of CONST-050(B) missing test types (round 41 close-out ²⁵)	Task #267 closed (Task #266 3-of-6 installm operator-safe client-side fault injection (NO power-management ban). Differs from DDoS is the chaos. 5 injection vectors: (1) 5 ma bad HTTP version, double-slash path, null by Huge: <8 KiB>); (3) 5 concurrent slow-loris co abrupt connection close mid-request; (5) mi reqs at concurrency 20 to /api/v1/health wh HELIX_CHAOS_HOST/PORT, CHAOS_LEGIT_REQUESTS (6-step structure: pre-chaos probe → malfo header + post-huge liveness → slow-loris sp chaos liveness + schema sanity. Anti-bluff crash vs clean 4xx reject); 000 on oversized users (pass-rate < threshold); zombie post-o SKIP-OK when server unreachable. Validat 200, 5 malformed shapes survived, oversized Huge per design), 5 slow-loris spawned, 100 chaos 200+valid status. curl-code formatting

Date	Updater	What changed
2026-05-15	CONST-052 rename-safety guardrails + G6 strengthening (round 41 close-out ²⁴)	“000000” when curl’s -w %{http_code} already runner now 17/17 PASS (was 16/16). Remaining 1 test type (scalability). Operator follow-up safety mandate 2026-05-15 to codebase which is by convention non-snapshots. System and working building process!” Strengthened rules.sh with explicit NEVER_RENAME_PATTERNS and manager files (Makefile, Cargo.toml, go.mod, Kotlin/Android/Apple/C#/Swift; Android AOSP vendor/, kernel-5.10/, packages/, prebuilts/, breaks the AOSP build); AOSP-mandated bootstrap.bash, BUILD, OWNERS); build/cargo (.git/, .github/); coordinated owned-project rename atomic rename across GitHub+GitLab+all code documents the hard rule + per-rename-batch run_all_challenges.sh + verify-all-constitution resolution regression test). Task #252 described protected + 6 unprotected rename candidate stability check: 3 sequential runs × 16/16 on live-LLM Challenge). Anti-bluff sweep verified known Task #254 (HelixAgent 46 nested sub (internal/llm, cmd/cli).
2026-05-15	Stress Sustained-Load Challenge — 2nd of CONST-050(B) missing test types (round 41 close-out ²³)	Task #266 2-of-6 installment delivered. New server’s /api/v1/health endpoint with sustained rps, per-second pass-rate snapshot, latency-knobs: STRESS_DURATION_SEC (default 30), STRESS_TIMEOUT_SEC (5), STRESS_MIN_PASS_PCT (95). structure: (1) pre-stress probe + 10-sample with-empty-body bluff; (3) sustained loop with aggregate analysis + worst-second pass-rate latency vs baseline comparison — catches alone would miss; (6) post-stress liveness + overall + worst-second pass-rates, p50/p95/SKIP-OK pattern. Validated end-to-end against worst-second, p50=0.5ms, degradation=-20 fractional-seconds bug caught + fixed (was FAIL PASS (was 15). Remaining 4 test types (scalability).
2026-05-15	DDoS Health-Flood Challenge — first installment of CONST-050(B) missing test types (round 41 close-out ²²)	Task #256 first installment delivered (1 of 6). New tests/e2e/challenges/ddos_health_flood. concurrent requests + asserts pass-rate through vars: HELIX_HEALTH_URL, DDOS_REQUESTS (default DDOS_MIN_PASS_PCT (95). Captured wire evidence seconds, latency p50/p95/p99. 5-step structure sanity (real JSON with status field, not empty liveness (catches “fell-over-during-flood” class topology-dispatch. Validated end-to-end against concurrency → 100% pass rate, p50=1.0ms Full runner now 15/15 PASS (was 14/15). as Task #266 — each follows the same skeleton

Date	Updater	What changed
2026-05-15	scripts/verify-all-constitution-rules.sh — CONST-055 enforcement engine (round 41 close-out ²¹)	Task #265 closed. Implemented the canonical names as the enforcement engine for even. New scripts/verify-all-constitution-rules.sh. G1 governance-cascade (§11.9 + CONST-04 smoke (production code grep for “simulated G3 CONST-050(A) mock-from-production (for CONST-051(C) nested-own-org submodule of tracked-sensitive-file scan; G6 CONST-052 candidates without failing since rename is p canonical-fix on violation. Paired meta-test s anti-bluff requirement, §1.1 mutation pair): FAIL, reverts cleanly. 9 sub-tests PASS (G2 surfaced + fixed two real findings: (a) dependo CONST-053 — fixed inline via Models comm was being flagged as a tracked sensitive file (matches .env.full-test / .env.test / .env.ci 5/6 gates GREEN, 1 FAIL = G4 HelixAgent’s (multi-session). Meta-test caught a real bluff which.gitignore correctly rejected — defen past .gitignore, which is the actual failure m Operator re-invoked the canonical CONST-0 itself is already in place (14 anchors × 36 ov Operative answer per §11.4.2 (captured-evid Challenges and captured runtime evidence. a214f0c → 464ada1 (upstream merge “drop post-merge. CONST-035 anti-bluff smoke production this would" helix_code/internal h Challenge (6/6 PASS): live LLM round-trip probe + token stats 📊 tokens: in=... out=. Challenge (5/5 PASS): live sentinel-echo pr in a temp file — proves the @-mention conte (tests/e2e/challenges/run_all_challenges.sh) PASSED! Anti-Bluff Verification: COMPLETE + cmd/cli/...): 7 packages green (ok dev.hel Constitution pointer bumped 464ada1 in sar verifier: 0 failures. The anti-bluff covenant is produced positive captured evidence of user
2026-05-15	Anti-bluff verification sweep — CONST-035 captured runtime evidence (round 41 close-out ²⁰)	Task #263 closed — last genuine-violation in resource-policy/apply_caps.py no longer hard helix_code/helix_code/, /helix_code/mcp-serve others). Refactor: SKIP_PATH_FRAGMENTS_DEFAULT vendor/, external/, .container-caps-backup-, paths-file <path> flag (+ RESOURCE_POLICY_SKIP from a one-per-line file at runtime via new l project-file ∪ -exclude. Example file contains paths.example.list shipped alongside the scr resource-policy-skip-paths.list (16 fragmen --help shows the new flag; grep for helix_co returns 0 matches. containers submodule co
2026-05-15	containers/ apply_caps.py CONST-051(B) decoupling refactor (round 41 close-out ¹⁹)	

Date	Updater	What changed
2026-05-15	scripts/regenerate-coverage-ledger.sh + first mechanical regeneration (round 41 close-out ¹⁸)	Task #261 closed. New generator script scr the coverage ledger per CONST-048 + \$11.4 supported platforms from helix_code/Makefil harmony-os/containers/headless); (2) extrac 30); (3) inventories test-type presence + Ma owned submodule's .gitmodules for nested o verify-governance-cascade.sh and parses anc every prior VERIFIED cell (anti-bluff: never au remain operator-controlled per CONST-035 30 features, 12 owned submodules, 30 VERI own-org submodules (CONST-051(C) viola bash bugs in early-version (grep -c exit-1 wi semantics on multiline reads) — fixed defen tested with ROUND_TAG="close-out-18-reg preserve VERIFIED markers and refresh me of script) + \$11.4.22 (lightweight doc-sync v
2026-05-15	load_api_keys.sh genericisation + github_pages_website sub-audit (round 41 close-out ¹⁷)	Two closures bundled. Task #262 (cosmet scripts/load_api_keys.sh header + function-c (project-agnostic per CONST-051(B))" + "do identically to 4 submodule copies (HelixQA committed + pushed to its origin). Function alias). Behavior unchanged; smoke-tested. T enumerated 12 flagged files; 1 was the cosm legitimate-cross-project content/tooling — tl file table added to decoupling_review.md. No github_pages_website beyond the load_a submodule pointer bumps + root edit) and t cosmetic-only decoupling branch is now full
2026-05-15	CONST-051(B) decoupling review first pass (round 41 close-out ¹⁶)	Task #255 first pass delivered. Published do remediation decision for every owned-subm #250 audit. Decision taxonomy (3 classes): l Challenge-bank entries / website content) → Task #262 to genericise; genuine-violation (paths) → Task #263 medium-scope refactor larger Task #254 remediation. github_pages Tally: 38 files legitimate-cross-project (no a violation (1 follow-up task), 12 files pending Task #254. The honest decision document is up tasks per the operator's "per-file review"
2026-05-15	@-file mentions in REPL (round 41 close-out ¹⁵)	Task #244 closed. Modern-CLI-agent parity in the REPL and the CLI auto-attaches that helix_code/cmd/cli/main.go: new helpers atMe text or after non-identifier char so user@host match) and expandAtMentions(prompt) append attached_files> block to the prompt for each not embedded — anti-bluff: never silently tr silently skipped. Missing-file mentions stay v CLI surfaces a  attached: <path> line per a at_mentions_test.go (PASS). New anti-bluff C

Date	Updater	What changed
2026-05-15	CONST-048 coverage ledger first publication (round 41 close-out ¹⁴)	paired structural + runtime gates including a temp file and verifies the LLM quotes it back (not a wire-level bluff). Verified live with Grok LLM correctly described the file contents in 100 tps=93.3. Wired into run_all_challenges.sh new Task #257 surface delivered. Published docs for HelixCode. Lists every F01-F30 feature + documented promise, no open bugs, docs in honest UNCONFIRMED: markers per §11.4.6 no-path). Includes test-type matrix (CONST-050 benchmarking/challenges/helixqa + partial chaos/stress/ux + partial ui/performance). In governance anchor cascade snapshot (all 14 submodules = 36 files). Generator script scr (first edition authored manually) — tracked features carry UNCONFIRMED: anti-bluff p file decoupling review; HelixAgent's 46 nest remediation. Audit trail row added inside th
2026-05-15	First CONST-051(C) remediation: panoptic moved out of Challenges to HelixCode root (round 41 close-out ¹³)	First concrete remediation of a CONST-051(C) Task #250's audit. challenges/.gitmodules was submodule — forbidden by CONST-051(C). I panoptic inside Challenges (committed as Ch github + gitlab + origin (fans to github+gitl repo to add panoptic at <root>/panoptic/ per (the same SHA Challenges had been using — needed in Challenges: its pkg/panoptic/cli_a as a constructor parameter + the workdir is config-injection was already in place. Verific removal. Governance cascade verifier 14-an closes the first CONST-051(C) violation. Tas much larger remaining CONST-051(C) work
2026-05-15	CONST-057/058/059 HelixCode-side cascade (round 41 close-out ¹²)	Closes the cascade-side gap from close-out¹¹ (§11.4.33/34/35 added by another agent dur Type-aware Closure-Status Vocabulary (Bug (→ Fixed.md) suffix; same migration-disciplin Mandate (every Reopened item carries **Re vocabulary Reason values; reopens without Canonical-Root Inheritance Clarity (constitu ARE extensions opening with inheritance po Cascaded through HelixCode root governan CRUSH.md/QWEN.md), inner helix_code/ m AGENTS.md), and all 12 owned submodules containers 6453dfd8; DocProcessor 364fc9c LLMsVerifier 060306b1; Models d8793574; 3 remotes; HelixAgent aca15473; helix_qa 2 check (§11.9 + CONST-047..059) — 0 failure
2026-05-15	CONST-054/055/056 codified + cascaded (round 41 close-out ¹¹)	Three new universal anchors codified per op Constitution submodule HEAD advanced: 6f added §11.4.33 type-aware closure status +

Date	Updater	What changed
2026-05-15	CONST-053 codified + cascaded into 12 owned submodules + CONST-051 audit complete (round 41 close-out ¹⁰)	root inheritance clarity by another agent) → Dependency-Manifest: every owned-by-us dependencies; tooling incorporate-submodule manifest, recurses for each dep, aborts on c §11.4.32 / CONST-055 Post-Constitution fetched + pulled, the project MUST run scri HEAD is canonical — the enforcement engine install_upstreams on Clone/Add: every cl invocation from the repo root IF upstreams/ (silently breaks §2.1 multi-upstream-push no (CONSTITUTION.md/CLAUDE.md/AGENTS (helix_code/CONSTITUTION.md/CLAUDE.m CONST-047 in two passes: CONST-054+055 dd12c5db — pushed to 4 remotes; container 200d1d73; LLMProvider 5b6cfb5c; LLMsVer github_pages_website 71d3dabf — pushed t Security fbda6e2c). Verifier extended to 11-36 governance files. Constitution submodule tasks (multi-session): #252 phased renames #254 HelixAgent’s 46 nested own-org subm 6 missing test types; #257 coverage ledger; incorporate-submodule tool; CONST-055 ver (§11.4.33/34/35) still need HelixCode-side c §11.4.30 / CONST-053 .gitignore + No-V submodule (commit 6f30ce7 pushed to origin session). Six forbidden-from-version-control generator products), cache files (__pycache_ *.key, id_rsa*, secrets/, api_keys.sh), genera an ignore-line alone is not sufficient — no tr attention required: inspect git diff --staged commit. CONST-053 cascaded through Helix submodules (Challenges 24e69a15 to 4 rem LLMOrchestrator 5531dc8a; LLMProvider 8 VisionEngine 11ff7071; github_pages_websi 53a534f3; Security a5125e2f). Verifier exten CONST-047/048/049/050/051/052/053) — 0 compliance audit (Task #250) complete submodule chains — Challenges has 1 (panc CONST-051(B) hardcoded HelixCode refs — (3) CONST-050(B) missing test types — DDo CONST-048 coverage ledger ABSENT (Task per operator’s phased-execution directive. Per operator mandate 2026-05-15 (5th majo Lowercase-Snake_Case-Naming Mandat rule itself, 45d3678 for install_upstreams.sh B directory/submodule/file MUST use lowerca (helix_code/, challenges/, containers/, helix_ upstreams/, dependencies/) MUST be renam references. Common-sense exceptions: lang
2026-05-15	CONST-052 codified + cascaded + install_upstreams BOTH-dir support (round 41 close-out ⁹)	

Date	Updater	What changed
2026-05-15	CONST-051 codified + cascaded + 2 production bluffs fixed (round 41 close-out ⁸)	inside language-roots, vendor third-party su CONST-050(B) (regression test + full test-ty transition supported via install_upstreams.s <dir>/Upstreams (smoke test passed: syntax C to legacy path with WARN). Cascaded CONS (CONSTITUTION.md/CLAUDE.md/AGENTS (helix_code/CONSTITUTION.md/CLAUDE.m CONST-047 — Challenges 5eab467d (pushe 52ef2e6a; LLMOrchestrator baa35eb7; LLM c309cf84; VisionEngine debecfe4; github_pa 5ee00da4; helix_qa 70c7a55a; Security 323 to 7-anchor check (§11.9 + CONST-047/048/ Actual rename execution deferred to a s explicit instruction: comprehensive brainsto full test coverage per change. Spurious addi during earlier pwd confusion) were reverted CONST-052 cascade anchor; the constitution Three workstreams. (A) Two CONST-050 internal/memory/providers/chromadb_provider. bluff “for now ... we’d need to know which c collection via new listCollectionNamesLocked fix — the existing ListCollections parser only [{id,name,metadata}] objects, new helper is t unknown/empty bodies. (2) helix_code/inter longer claims “Simple equality check for nov complex filtering”; now supports documente \$contains via filter-value-as-operator-map sy operators fail closed. Both unit tests PASS. C manager.go removed (commit 8df3376). (B) §1 2026-05-15: every owned-by-us submodule (C ATMOSphere1234321, Bear-Suite, BoatOS1 part of the consuming project’s codebase. T full decoupling/reusability — NEVER inject injection; (C) dependency-layout — all subm <root>/submodules/<name>/; nested own-org su exempt. Codified in constitution submodule cascade into 12 owned submodules per CO submodules at new heads (Challenges dded. pushed to 3 remotes; others to origin). Verif walk CONST-051 — now requires all 6 ancho CONST-050 + CONST-051) per file in every files. Per CONST-047 (Recursive Submodule Appl universal anchors codified in round 41 close that for each of the 12 owned submodules: (C (2) appends CONST-048, CONST-049, CONS AGENTS.md if missing (grep-checks ^## CON markers introduced; (4) commits + pushes t submodule commits landed (Challenges l
2026-05-15	CONST-048/049/050 cascaded into 12 owned submodules + verifier extended (round 41 close-out ⁷)	

Date	Updater	What changed
2026-05-15	Groq stream double-emit fix + CONST-048/049/050 governance cascade (round 41 close-out ⁶)	LLMOrchestrator 531d5924; LLMProvider e VisionEngine 3e2ebaa; github_pages_website Security 951180e), all pushed to their respec github+gitlab+origin+upstream per its mul github+origin+upstream; others to origin). committed merge-conflict markers (lines 82 92dcec1ef9fa07529e955946c60c205efb2b90f2 pat CLAUDE.md). Resolved by removing the 3 n (\$11.4.26 step 5). Also extended scripts/ver addition to existing \$11.9 + CONST-047 — n submodule. Verifier result: 0 failures across 12 submodule pointers bumped + verifier ex atomic close-out. Two distinct workstreams in one round. (A) the UX layer): internal/llm/groq_provider.go (Content=delta) AND a final-completion chu accumulated content). The renderers (stream final full-content chunk was rendered on top for a “1 2 3” model response. Fix: at FinishF (metadata fields Usage/FinishReason/Provid final chunk; rendered content not duplicated "Reply with only: 1 2 3" --stream → clean “1 anchors per operator mandates 2026-05-15 (every feature/functionality/flow/use-case/ec MUST ship with full automation tests provin matching documented promise + no open b CONST-049 Constitution-Submodule Up with \$11.4.17 classification + verbatim man conflict resolution preserving union → casca SAME commit); \$11.4.27 / CONST-050 No (mocks/stubs/placeholders/TODOs/FIXMEs p fully-implemented system; production code + integration + E2E + full-automation + se benchmarking + UI + UX + Challenges + h incorporated recursively). All three codified AGENTS.md — commits cdc7b17 for \$11.4.25 constitution origin); cascaded into HelixCod CRUSH.md, QWEN.md AND inner helix_cod CONST-048/049/050. constitution submodul repo commit (CONST-049 step 7 verified cor authoring (step 1 fetch+pull pulled 7 upstre cascade verifier: 0 failures. Cascade to owne round.
2026-05-15	OpenRouter live / models + Mistral/ DeepSeek F12 fast- path (round 41 close- out ⁵)	Continued the F12 readiness sweep from ro stale-default-model bluff closed (CONST internal/llm/openrouter_provider.go led with not a valid model ID”). End-to-end probe cor other 4 returned 404/429). Replaced with a preferredOpenRouterDefaults list (round-41-pr

Date	Updater	What changed
2026-05-14	Modern-CLI-agent UX complete: conversational REPL + canonical-loader normalisation (round 41 close-out ⁴)	unreachable, falls back to the verified seed 1 → 364 models loaded from live catalog, default “four” with token stats (in=74 out=21 time=) providers — added MistralProvider and Deep XAIProvider; ProviderTypeMistral existed in wired into NewCloudProvider + parseCloudProv Removed them from the Path-B disallow list verification: HELIX_LLM_PROVIDER=mistral → mi HELIX_LLM_PROVIDER=deepseek → deepseek-chat TestSelector_RejectsNonCloudType pivoted from conversational_repl Challenge: 6/6 PASS un providers explicitly. Gemini probe re-confirm Ollama/Llama.cpp not running locally so un Operator’s “is HelixCode ready for practical end-to-end probe that surfaced and fixed FC extension: (1) env-var fallback for OpenAI manual claimed all do); (2) default-model llama-3-8b which doesn’t exist on any cloud word-only + 6 hardcoded commands — does context + slash commands /help//exit//clea extended to all 13 providers (was claimin → <PROVIDER>_API_KEY for 28 names + survive silently). New Challenge tests/e2e/challenge gates (catches both source-regression AND HELIX_LLM_PROVIDER=groq ./bin/cli → real con Groq + token stats 📊 in=40 out=2 total=42 modern-CLI-agent UX parity (Claude Co provider + distributed-execution + gove Per operator’s “is HelixCode ready like mod recommendations” directive: extended the F providers. Cloud (11): anthropic, bedrock, v qwen, copilot. Local (2): ollama, llamacpp adapters bridging the generic ProviderConf verified each via rebuilt CLI: HELIX_LLM_P missing fallback. Path B narrowed to 5 (deep compatible passthrough via server). Pre-exis (TestSelector_RejectsNonCloudType→deeps TestNewCloudProvider_RejectsLocalProvide User manual §2.4 updated. TWO of my own referenced nonexistent docs/COMPLETE_H ZERO_BLUFF_USER_MANUAL.md §2.4; (2) -model llama3.2” — that UX now works afte just-plug-in-API-key-and-go parity with Cline) for the 11 most popular cloud pro superset of any single competitor’s prov remotes.
2026-05-14	F12 fast-path 4→13 providers + readiness UX cleanup (round 41 close-out ³ — modern CLI agent parity)	Operator standing mandate (8th re-assertion including recursive cascade. Surveyed all 12 CLAUDE.md/AGENTS.md = 36 files) and pr
2026-05-14	Constitution- inheritance cascade across 12 owned	

Date	Updater	What changed
	submodules (round 41 close-out ²)	file missing it. 23 files received the prepend earlier in HelixQA, Challenges, Containers, VisionEngine). Each submodule's commit pushed owned-submodule pointers + pushed to all 4 family project (e.g. HelixCode, HelixAgent, consumed by multiple parent projects. Project (GitHub+GitLab-only remotes) is preserved project-narrower rule wins per inheritance FULLY satisfied for the constitution-inheritance cascade.
2026-05-14	HelixLLM/ submodules/HelixQA wire-up — round-39 deferred close-out (round 41)	Closed the last deferred item from the round HelixQA was declared in .gitmodules but never 'submodules/HelixQA' did not match any file(git@github.com:HelixDevelopment/HelixQA.git Cascade close-out: HelixLLM 8a504f9 → Helix All gates GREEN (verify-governance-cascade:
2026-05-14	challenges/ CLAUDE.md merge- conflict resolution + panoptic screenshot bluff (round 41)	While auditing for further bluffs, surfaced a conflict markers (<<<<<<< HEAD at line 651, == 92dcec1ef9fa07529e955946c60c205efb2b90f2 at 1 manual. Resolved by removing only the mar §11.4.15 forensic anchors; other side: §6.O- tightened Testpanoptic_WebAppIntegration { t.Logf } else { t.Logf } (passed regardless cleanly. panoptic 73b13b0 → Challenges 186
2026-05-14	HelixConstitution submodule incorporated at constitution/ (round 41 continued)	Operator direct in-conversation mandate (20 fully the HelixConstitution (git@github.com. responsible for root definitions of the Constr further!" — distinct from the earlier suspect GitFlic + GitVerse upstreams contradicting a Git submodule at ./constitution/, pinned a root governance trio (CONSTITUTION.md, CLAUDE unconditionally, (b) project rules EXTEND n universal scope, project tightenings (e.g. §6. narrow. Deliberately NOT done for safety: violate §6.W), did NOT add GitFlic/GitVerse submodules, did NOT modify the constitution repo remotes.
2026-05-14	Round-41 begin: 5 production-code ctx- honour fixes (CONST-035)	Continued anti-bluff sweep into the notificat Telegram, PagerDuty) all had Send(ctx conte deadline support but the implementations c context.WithTimeout or context.WithCancel ha at the API-contract level. Fixed all 5 with ht http.DefaultClient.Do(req). Discord fix muta tightened TestDiscordChannel_Send_ContextCar context.Canceled)) FAILs; with the fix in plac (Slack+Teams+Telegram+PagerDuty), both round-40+41 ledger: 68+ anti-bluff impro skips gate GREEN; verify-governance-casca internal/notification/... PASSes with all 5

Date	Updater	What changed
2026-05-14	Anti-bluff round-40 final ledger: 23 fixes + 1 production-code defect surfaced & fixed	Final round-40 totals after the operator re-a bluff improvements + 1 real production providers/character_ai_provider.go:702+733 — unconditionally and panicked on nil config. 7 TestCharacterAIProvider_CreateCollectionWith { recover() }() + t.Logf'd the panic + disca gracefully". TRIPLE bluff. Tightening the tes empty error message) FAILED loudly with th production: treat nil config as “use defaults’ prod fix, passes with it. Other tightenings (multi_provider GetModels nil-slice contract error rather than silently running tests with commits pushed to all 4 meta-repo remotes, tightenings + 4× integration-test tightening hardening + 1× production-code panic fix = helix-qa SKIP-OK’d on #env-db-unreachable as Prompt-injection flag from earlier in round h Final round-40 batch surfacing the biggest h form go test ./... \ grep -q "FAIL" && (ec silently PASS on “no test files” output, comp doesn’t print the expected grep token. Repla which uses Go’s actual exit code. Files tigh (workflow+session), phase5 (mcp+memory- TRIPLE BLUFF surfaced and fixed in ph && exit 1 — script already cd’d to HelixCode such file or directory”), the && shortcut skip empty input and returned non-zero, and the years without actually running the unit tests honestly FAIL once (proving the gate is now runs. Commit 1aaa09d. Cumulative round-40 (ee4ba5a) + 9 unit test tightenings across 4 c tightenings (5c771fe) + 6 phase challenge tig verified on representative test (sed-replace : Operator re-asserted CONST-035 verbatim 6 on our plate”. Audited helix_code/internal/ f NOTHING (zero assert.* / require.* / t.Fata Surfaced 9 zero-assertion-with-discards can round): TestAIIntegration_Get{Conversation assert.Nil pinning the documented “Initializ TestNewClient_WithDatabase (replaced _, _ nil-client + “Redis” in error, success-path re TestHealthMonitor_PerformHealthChecks (s Health/Metrics are NOT mutated — pins the more in this commit: TestCharacterAIProv cancelled ctx — now pins in-memory Health TestChromaDBProvider_Store_DefaultCollec = usedDefaultCollection — now require.NoEr TestDiscordChannel_Send_ContextCancellat http.Post ignores ctx; a future promotion to
2026-05-14	Anti-bluff phase-challenge gates: grep-based → exit-code-based (round 40 close)	Final round-40 batch surfacing the biggest h form go test ./... \ grep -q "FAIL" && (ec silently PASS on “no test files” output, comp doesn’t print the expected grep token. Repla which uses Go’s actual exit code. Files tigh (workflow+session), phase5 (mcp+memory- TRIPLE BLUFF surfaced and fixed in ph && exit 1 — script already cd’d to HelixCode such file or directory”), the && shortcut skip empty input and returned non-zero, and the years without actually running the unit tests honestly FAIL once (proving the gate is now runs. Commit 1aaa09d. Cumulative round-40 (ee4ba5a) + 9 unit test tightenings across 4 c tightenings (5c771fe) + 6 phase challenge tig verified on representative test (sed-replace : Operator re-asserted CONST-035 verbatim 6 on our plate”. Audited helix_code/internal/ f NOTHING (zero assert.* / require.* / t.Fata Surfaced 9 zero-assertion-with-discards can round): TestAIIntegration_Get{Conversation assert.Nil pinning the documented “Initializ TestNewClient_WithDatabase (replaced _, _ nil-client + “Redis” in error, success-path re TestHealthMonitor_PerformHealthChecks (s Health/Metrics are NOT mutated — pins the more in this commit: TestCharacterAIProv cancelled ctx — now pins in-memory Health TestChromaDBProvider_Store_DefaultCollec = usedDefaultCollection — now require.NoEr TestDiscordChannel_Send_ContextCancellat http.Post ignores ctx; a future promotion to
2026-05-14	Anti-bluff zero-assertion test sweep (round 40 continued)	Operator re-asserted CONST-035 verbatim 6 on our plate”. Audited helix_code/internal/ f NOTHING (zero assert.* / require.* / t.Fata Surfaced 9 zero-assertion-with-discards can round): TestAIIntegration_Get{Conversation assert.Nil pinning the documented “Initializ TestNewClient_WithDatabase (replaced _, _ nil-client + “Redis” in error, success-path re TestHealthMonitor_PerformHealthChecks (s Health/Metrics are NOT mutated — pins the more in this commit: TestCharacterAIProv cancelled ctx — now pins in-memory Health TestChromaDBProvider_Store_DefaultCollec = usedDefaultCollection — now require.NoEr TestDiscordChannel_Send_ContextCancellat http.Post ignores ctx; a future promotion to

Date	Updater	What changed
2026-05-14	Anti-bluff Step-3b: SKIP-OK when DB unreachable (round 40)	explicit contract update); TestAllIntegration, the comment's claim: empty without Initializ remotes at 08746b8. Prompt-injection flag: a constitution/ submodule with 4 upstreams i ("GitFlic, GitVerse, and all other providers a added, no installer was executed. Operator re-asserted CONST-035 verbatim 5 in anti-bluff manner. Verified the anchor cas governance files PASS via verify-governance the live-server Challenge to distinguish "fea tests/e2e/challenges/helix_qa_live_anti_bluff inspects database.connected. If false (or the b failed signatures), emit SKIP-OK: #env-db-unr and exit 0, mirroring the existing Step-1 #en previous FAIL cascade (HCQA-015/016/017 postgres credential drift was created by my only database.connected=false in /server/info validation errors, real auth defects, or featu degraded HC: challenge correctly SKIP-OK's After completing the round-39 CONST-047 3 expand helix_qa_live_anti_bluff evidence be deployment-configuration drift, not a co via scripts/mistborn-up.sh, accepts user heli helixpass with SASL auth FATAL: password aut HELIX_POSTGRES_PASSWORD:-helixpass and the p helixcode-r38) used some other secret that i this troubleshooting and the correct passwo correctly override viper config (verified: HC the bug is purely about the unknown-to-this cascade work (12 + 44 + 11 = 67 newly-cas across 3 recursion layers) is complete and server probe will return to SKIP-OK: #env-mis harness expects DB on register/login probes when DB is unreachable — these regression mistborn-postgres password is provided to a Extended the recursive cascade one more la grandchildren) + helix_agent/challenges/par cascaded, 23 idempotent skip (shared s (HelixLLM/submodules/HelixQA not init owned-grandchild pointers + pushed to orig pushed. HelixAgent commit c5ecc279 bumps commit 64d2632 bumps HelixAgent pointer + upstream). Round 39 cumulative cascade = 67 newly-cascaded submodules + 25 i covered across 3 layers of recursion unc Per CONST-047 the verbatim mandate "fully with the only deferred item being the uninit when its submodule is initialized in a future
2026-05-14	Known-issue: mistborn-postgres SASL credential drift (round 39 follow-up)	
2026-05-14	CONST-047 recursive cascade — 3-deep close-out (round 39 continued ²)	
2026-05-14		

Date	Updater	What changed
	CONST-047 recursive cascade — HelixAgent layer (round 39 continued)	Continued the recursive cascade per CONST-044 (digital + HelixDevelopment orgs). Per submodules anchor (if missing) + CONST-047 anchor (if missing) 44/46 cascaded this round, 2 already had one other), 0 failures. HelixAgent commit 4971fe7 bumps HelixAgent gitlab + origin + upstream). Cascade-verification \$11.9 + CONST-047 in every owned-submodule layer not yet covered by the script — it walks (recursion) : HelixAgent → HelixLLM (34 grandchild — 36 additional submodules at the 3rd layer helixagent-nested-cascade.sh) reusable for the regression (post-cascade run_all_challenges; build clean).
2026-05-14	CONST-047 recursive submodule cascade (round 39)	Operator re-asserted verbatim (2026-05-14) <i>Submodules we control under our organization everywhere with full bluff-proofing and complete tests and Challenges coverage! If it is not already AGENTS.MD and CLAUDE.MD of the main project follow this rule (mandatory constraint) ALWAYS.</i> Application Mandate. Authored full anchor operative rules + owned-submodule baseline cascade-referenced in root CLAUDE.md and AGENTS.MD {CONSTITUTION, CLAUDE, AGENTS}.md. Then cascaded owned submodule's CONSTITUTION.md / CLAUDE.md + 36 governance-file additions + 6 at root/inner submodule cascade commits pushed to evergreen github+gitlab+origin+upstream; the others resolved. Conflict resolution: 9 submodules had non-Fork prior session — reset to upstream tip, re-applied bumps 12 submodule pointers + 6 root/inner and is now at parity across all 4 meta-repo repositories. production code touched this round — pure surfaces unaffected (no probe changes; the verify-governance-cascade.sh augmentation a

close-out¹³¹ — rounds 191-305 catch-up (constitution cascade + helix_code/ internal i18n + per-repo deep-doc + PAN-001 fix)

Date: 2026-05-19 **Last commit at close-out time:** 2d95892 (gov: round-304 — bump challenges pointer) **Theme:** Single batched CONST-044 catch-up narrative for ~115 rounds executed since close-out¹³⁰ (round 192). Per CONST-044 (Continuation Document Maintenance Mandate) the documentation drift across rounds

191-305 is itself a CRITICAL DEFECT of equivalent severity to a CONST-035 false-success result; this close-out remediates that drift in a single condensed entry. Per-round narration cadence resumes next round.

Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): *“all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can’t be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!”*

Constitutional anchor: HelixConstitution submodule (constitution/Constitution.md + constitution/CLAUDE.md + constitution/AGENTS.md) is the **canonical root** per CONST-059. This catch-up narrative extends the consumer-side docs/CONTINUATION.md and does NOT modify any canonical-root file.

What happened (thematic, not per-round)

1. **Constitution cascade rounds** — rounds 191 / 207 / 227 landed constitution-submodule §11.4.41-66 anchors as CONST-054 through CONST-067 (per CONST-049 fetch-before-edit pipeline + CONST-047 recursive submodule application). All anchors cascaded verbatim or by ID-reference into every owned-submodule CONSTITUTION.md / CLAUDE.md / AGENTS.md across vasic-digital + HelixDevelopment orgs. Verifier scripts/verify-governance-cascade.sh GREEN at each round’s close.
2. **helix_code/internal i18n migration completed** — rounds 220-241 covered all 60+ packages of helix_code/internal/ per CONST-046 (no hardcoded user-facing content). Each migrated package shipped: (a) i18n/ resource bundles in 5 locales (en, sr, ja, es, de — fixture set), (b) i18n.Get(locale, key) call-sites replacing static literals, (c) audit-gate scripts/lint-hardcoded-strings.sh exit 0 across whole tree at round-241 close. Packages with no user-facing strings (pure internal infra) received NO-OP marker comments. helix_code/cmd subtree + LLMsVerifier deferred to HXC-003 backlog.
3. **Per-repo deep-doc enrichment** — rounds 242-305 applied the enrichment template across 52 vasic-digital + 9 HelixDevelopment + 5 top-level submodules (~65 enrichment rounds). Per-repo template (Template C): README ~150-210 LOC (purpose + architecture + quickstart + capability matrix + Challenges section + governance pointer), docs/test-

coverage.md (per-package coverage matrix + Challenge wire-evidence pointers), challenges/runner/main.go (binary that executes the bank), challenges/*_describe_challenge.sh paired-mutation scripts (assertion + intentional-mutation revert proving the assertion would fail without the fix), 5-locale fixture set under challenges/fixtures/i18n/. Pushed per-submodule across all configured upstream remotes per CONST-056.

4. **Real production bugs discovered + fixed during enrichment:**

- **PAN-001** (round 298 discovery, round 302 fix): vasic-digital/Panoptic UTF-8 truncation bug — truncateAt(s, n) sliced at byte offset n splitting multi-byte runes mid-codepoint (Cyrillic/CJK fixtures rendered as mojibake). Fix: rune-aware truncation via []rune(s)[:n] + size budget. Paired-mutation Challenge truncation_describe_challenge.sh proves it.
- **Veritas stale CLAUDE.md** (round 288 discovery): documents removed RemoveSource API.Deferred — not blocking.
- **LLMOps structural-bluff Challenge** (round 290): existing Challenge asserted on file-presence (grep), replaced with exit-code-driven go test invocation per the round-40 phase-challenge tightening pattern.

5. **Submodule deduplication finding** (round 277):

HelixDevelopment/Models and vasic-digital/Models point at the same upstream repo. Documented; no consolidation work this round.

6. **Sibling-session commit-message race pattern** (rounds 228-285): repeated non-FF rejections on submodule pushes because sibling subagent sessions had landed concurrent commits. Mitigation per CONST-043 (no force-push, no --force-with-lease without explicit per-operation approval): fetch + rebase or re-apply onto upstream tip + re-push. Pattern documented in close-out narrative.

What's still open (rolled forward into the live backlog)

- **VEN-002, HXA-002, HXQ-001** — still BLOCKED on operator decisions
- **HXC-001** — In progress (round 343). Operator approved “agent defaults” for all 12 decisions; round 343 executed 3 build-verified batches renaming 13 owned-org leaf submodule dirs to snake_case: HelixDevelopment/{Models→models, DebateOrchestrator→debate_orchestrator} + 11 vasic-digital/* zero-go.mod-consumer leaves (auto_temp, claritas,

doc_processor, gandalf_solutions, hyper_tune, i_llm, leak_hub, ouroboros, plinius_common, veritas, vision_engine). Deferred (submodule-go.mod-entanglement / parent-dir / cluster-C): ~37 remaining owned-org leaves consumed by helix_agent/helix_qa/HelixLLM go.mod, parent dirs HelixDevelopment/→helix_development/ + vasic-digital/ kept (GitHub-org handle), 59 Upstreams/→upstreams/ dirs inside submodule trees. Each deferred batch needs dedicated per-submodule rounds to avoid entangling pre-existing uncommitted work.

- **HXC-003** CONST-046 migration backlog — helix_code/internal COMPLETE; remaining pending: LLMsVerifier ~1804 strings + cmd/helix_config ~241 strings + cmd/cli ~7 strings
- **Veritas CLAUDE.md** stale RemoveSource reference — deferred from round 288
- **Nested third-party submodule pointer drift** in helix_qa/tools/opensource/* — deferred per CONST-051(C) (third-party submodules exempt from own-org dependency-layout rules)

Next

- Resume per-round narration cadence (CONST-044 compliance)
- HXC-003 remaining CONST-046 migrations (LLMsVerifier is the biggest single chunk)
- Cascade detection follow-up for any newly-pulled constitution anchors (CONST-055 sweep on next constitution-submodule pull)
- Operator-blocked item triage (VEN-002 / HXA-002 / HXQ-001 / HXC-001)

Posture

- **Constitutional posture:** CONST-035 anti-bluff + CONST-044 continuation-maintenance + CONST-046 i18n + CONST-047 recursive-submodule-application + CONST-049 verbatim-mandate-preservation + CONST-050(A) no-fakes-beyond-unit-tests + CONST-051(B/C) decoupling + dependency-layout + CONST-052 snake_case + CONST-053 .gitignore-hygiene + CONST-055 post-pull-validation + CONST-056 install_upstreams-on-clone + CONST-057 Type-aware closure vocab + CONST-058 reopened-source-attribution + CONST-059 canonical-root inheritance + CONST-060 fetch-before-edit all honoured per-round. This close-out¹³¹ remediates the CONST-044 drift across rounds 191-305 in a single batched narrative; per-round cadence resumes next round.
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close-out¹³² — round 398: Speed Programme launch + constitution §11.4.68/70-74 cascade + mirror reconciliation

Date: 2026-05-20 **Theme:** Three concurrent working points active this session, recorded per CONST-044 in the same commit that advances state (operator docs-sync mandate: *“Keep Issues and relevant documentation up to date and in sync ... All working points we are doing MUST BE there as well!”*).

Working points (filed into the live backlog as Issues)

1. **HXC-006 — HelixCode Speed Programme** (Feature, In progress). Operator mandate (2026-05-20) to make all HelixCode + owned-submodule code 3-5x faster than competitor AI CLI agents without breaking features or weakening anti-bluff posture. Research corpus complete under docs/research/speed/ (5 docs — bottleneck audit, competitive analysis, optimization techniques, overview, 6-phase / 31-task phased plan). **Phase 0 (measurement baseline) execution started** — establishing reproducible per-operation latency/throughput baselines so every later speed-up carries before/after captured evidence per CONST-035.
2. **HXC-007 — Constitution §11.4.68/70-74 cascade** (Task, In progress). Constitution submodule fetched + pulled first per CONST-049 (584b3ee → 34a82b3); 6 new rules cascaded verbatim/by-ID into meta-repo governance files + 67 owned submodules per CONST-047; meta .gitmodules constitution pointer bumped. Stays open until the CONST-049 step-4 multi-upstream reconciliation converges all mirrors.
3. **HXC-008 — CONST-055 G1 governance gaps** (Bug, In progress). Post-constitution-pull validation sweep surfaced two **pre-existing** (not regression) gaps: (a) scripts/verify-all-constitution-rules.sh references a non-existent dependencies/HelixDevelopment/Models path — actual dirs are lowercase models + vasic-digital/Models; (b) helix_qa/CONSTITUTION.md missing CONST-047..057, VisionEngine/CONSTITUTION.md missing §11.4.69.
4. **HXC-009 — Owned-submodule mirror-divergence reconciliation** (Task, In progress). Systemic divergence between vasic-digital ↔ HelixDevelopment GitHub/GitLab mirrors of several owned submodules. helix_qa reconciled via

merge-first commit a0397a4; VisionEngine, LLMProvider, others under reconciliation per CONST-061 / §11.4.71 merge-first — NO force-push, NO history rewrite.

State

- 4 new Issues filed (HXC-006/007/008/009); none closed this session; Issues_Summary.md regenerated — 6 open (1 Bug, 3 Task, 2 Feature, 0 BLOCKED).
- Tracker HTML+PDF exports regenerated for all touched docs per §11.4.12/53/60/65.
- No production code touched in this docs-sync commit; submodule pointer changes from parallel sessions deliberately NOT staged (parallel sessions share the meta worktree).

Next

- HXC-006 Phase 0 baseline capture → Phase 1 of the speed phased plan.
 - HXC-009 reconcile remaining divergent owned-submodule mirrors (VisionEngine, LLMProvider).
 - HXC-008 fix the verifier path reference + cascade the missing CONST-047..057 / §11.4.69 anchors; re-run verify-all-constitution-rules.sh G1.
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close-out¹³³ — CodeGraph incorporation Phase D (CG14-CG16)

Date: 2026-05-20 **Theme:** CodeGraph (MCP code-graph server) incorporation Phase D — governance covenant sync + documentation + CONST-047 recursive sweep. Phases A+B (install, scan, five-agent registration) already landed; this close-out completes the governance/docs phase per docs/research/codegraph/incorporation-plan.md §6.

Work performed

1. **CG14 — anti-bluff covenant gap fix.** Dified root QWEN.md and CRUSH.md against the current CLAUDE.md/AGENTS.md CONST-0xx anchor set. Both were stale (last modified 2026-05-16, before the 2026-05-20 CLAUDE.md/AGENTS.md revisions): each carried CONST-035..059 but was **missing CONST-060 (Fetch-before-edit, §11.4.37) and the §11.4.68/70-74 anchor block** (Positive Sink-Side Evidence, Subagent-Driven-Default, Pre-Push Fetch+Investigate, Audio Top-Priority, Main-Spec Versioning, Submodule-Catalogue-First). Appended all six

anchors verbatim from the canonical CLAUDE.md text to both files. Classification: universal (per §11.4.17) — consumer-side extension files mirroring the project-level anchors.

2. **CG15 — documentation.** Updated tools/codegraph/README.md (Revision 2) — added per-agent registration section with the cross-agent config table and the Catalogue-Check: no-match 2026-05-20 metadata line (§11.4.74 — CodeGraph is third-party, not in the vasic-digital/HelixDevelopment catalogue). Added a CodeGraph integration section to docs/COMPLETE_CLI_REFERENCE.md (install/init/scan/per-agent registration) and removed a stray closing-tag artefact at the end of that file. This close-out note added per CONST-044.
3. **CG16 — CONST-047 recursive sweep.** Ran scripts/verify-governance-cascade.sh — **2 failures, both pre-existing and already tracked as HXC-008:** (a) dependencies/HelixDevelopment/Models verifier path-list entry out of sync with on-disk lowercase models dir; (b) helix_qa/CONSTITUTION.md missing CONST-047..057. Neither failure is caused by the QWEN.md/CRUSH.md/codegraph Phase-D changes — the covenant-sync edits introduce no new cascade gap. Owned-submodule CodeGraph-config assessment: no owned submodule (helix_qa, challenges, containers, security) currently warrants its own CodeGraph MCP wiring — the meta-repo + inner helix_code/ graphs already cover the first-party Go code the agents query; a per-submodule graph would add maintenance cost for marginal benefit. Recorded as assessed-no-follow-up; revisit if a submodule grows a substantial standalone codebase queried independently.

State

- QWEN.md + CRUSH.md now carry the complete CONST-035..060 + §11.4.68/70-74 anti-bluff covenant — fully in sync with CLAUDE.md/AGENTS.md.
- tools/codegraph/README.md Revision 2; docs/COMPLETE_CLI_REFERENCE.md gains a CodeGraph section.
- Cascade verifier: 2 pre-existing failures (HXC-008) unchanged — Phase D introduced no regression.
- HTML+PDF exports regenerated for touched docs per §11.4.65.

Next

- HXC-008 remains the path to clearing the 2 cascade-verifier failures.
 - CodeGraph Phase C anti-bluff Challenges (CG10-CG13) remain the outstanding incorporation work if not yet landed by a sibling session.
-

close-out¹³⁴ — CodeGraph incorporation Phase C (CG10-CG13)

Date: 2026-05-20 **Theme:** CodeGraph (MCP code-graph server) incorporation Phase C — anti-bluff verification per docs/research/codegraph/incorporation-plan.md §5 + §6. Proves CodeGraph genuinely works end-to-end on the real HelixCode repo and reaches all five CLI agents (CONST-035 / Article XI §11.9).

Work performed

1. **CG10 — Layer A.** Replaced the tools/codegraph/verify.sh stub with the real end-to-end proof: codegraph status (asserts non-zero files/nodes/edges — 39 022 / 624 092 / 1 644 454 captured), codegraph query Provider (asserts real Go symbols), codegraph context (asserts real repo files), plus an anti-bluff token guard. Wrapped as Challenge CG-CHALLENGE-01. **PASS** — evidence docs/research/codegraph/evidence/phase-c/cg-challenge-01-*.
2. **CG11 — Layer B.** Challenge CG-CHALLENGE-02 drives codegraph serve --mcp over stdio with real JSON-RPC: initialize → serverInfo, tools/list → 9 codegraph_* tools, tools/call codegraph_search → real graph data. JSON-RPC wire transcript captured (cg-challenge-02-jsonrpc-wire.jsonl). **PASS**.
3. **CG12 — Layer C.** Challenges CG-CHALLENGE-03..07, one per agent, with a shared lib-agent-challenge.sh driver: tier-1 drives the agent CLI non-interactively and asserts a real .go symbol path; tier-2 (§11.4.3 topology dispatch) falls back HONESTLY to connect-only + tool-discovery proof when an agent cannot be driven to a real answer. **Results:** Claude Code (primary) — **true end-to-end PASS**; OpenCode — true end-to-end PASS; Crush — true end-to-end PASS; Kimi CLI — connect-only PASS (LLM monthly-quota 429 blocked the answer; its MCP loader did enumerate all 9 codegraph tools); Qwen Code — connect-only PASS (bundled OAuth free tier discontinued upstream 2026-04-15). 3/5 agents true end-to-end, 2/5 connect-only — reported honestly, no false end-to-end claims.
4. **CG13 — helix_qa bank.** Registered the 7 Challenges as a helix_qa test bank tools/codegraph/challenges/codegraph-integration.bank.yaml. Per CONST-051(B) the bank lives in the HelixCode tree — NOT inside the project-not-aware helix_qa submodule (the bank hardcodes HelixCode Challenge paths, which would couple the reusable submodule to HelixCode). The helix_qa runner accepts arbitrary bank paths via -banks, so it is fully consumable: helixqa run -banks tools/codegraph/

challenges/codegraph-integration.bank.yaml. Runner tools/codegraph/challenges/run-all.sh executes all 7 with per-Challenge captured evidence.

State

- 7 Challenges under tools/codegraph/challenges/ — all bash -n-clean with §11.4.18 doc blocks; bank run **7/7 PASS** after the CG-CHALLENGE-03 arg-shift fix.
- tools/codegraph/verify.sh is now a real anti-bluff proof (no longer a stub).
- All runtime evidence under docs/research/codegraph/evidence/phase-c/ — status JSON, query JSON, JSON-RPC wire transcript, 5 agent transcripts, per-Challenge run logs.
- helix_qa autonomous-session integration: bank registered + discoverable; direct execution via run-all.sh captures per-check evidence. Full autonomous-session orchestration left to a helix_qa QA run.

Next

- CodeGraph Phase D (CG14-CG16) already landed (close-out¹³³).
- A helix_qa autonomous QA session may execute the codegraph-integration bank for orchestrated per-check evidence capture.

close-out¹³⁵ — HelixCode Speed Programme COMPLETE (P5-T04 — all 6 phases / 31 tasks landed)

Date: 2026-05-20 **Theme:** Speed-programme final task P5-T04 — CONST-048 coverage ledger, release-gate sweep, and programme close-out. Closes **HXC-006** (HelixCode Speed Programme, Feature). Operator mandate 2026-05-20: make HelixCode + owned-submodule code 3-5× faster than competitor AI CLI agents without breaking any feature or weakening anti-bluff posture.

Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): *“all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can’t be used! This MUST NOT be the*

case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

Work performed (P5-T04)

1. **CONST-048 coverage ledger.** Created docs/research/speed/05-coverage-ledger.md (constitution §11.4.44 metadata header) — one row per task P0-T01..P5-T04 (mapped to opportunities O1-O21), each carrying the commit SHA, the captured before→after measurement, and the six CONST-048 invariants ([AB][WC][MD][NB][DS][TF]). Roll-up: **29 PASS + 2 PARTIAL + 0 DEFERRED** across all 31 tasks.
2. **Release-gate sweep.** scripts/audit-const046-hardcoded-content.sh ran exit 0 (speed work added no user-facing strings — no new hardcoded content). scripts/verify-governance-cascade.sh reported 2 failures = the already-tracked **HXC-008** pre-existing drift (stale Models path + helix_qa/CONSTITUTION.md missing CONST-047..057), NOT speed-programme regressions. Anti-bluff smoke grep -rn "simulated\\|for now\\|TODO implement\\|placeholder" helix_code/internal helix_code/cmd (prod, non_test.go) = clean. make verify-compile not re-run (P5-T04 is docs-only; compile state unchanged from 98315a14).
3. **Two deferred findings filed** per §11.4.15/16: **HXC-011** (Bug — helix_qa runner emits hollow sub-µs PASSED metadata rows for desktop-platform bank cases without executing them — a §11.4 PASS-bluff in the QA runner) and **HXC-012** (Bug — data race in helix_code/internal/llm/load_balancer.go stat-collector goroutine under full-package -race). Both pre-existing; surfaced during the speed programme; reported honestly.
4. **HXC-006 closed.** All 31 tasks genuinely landed → HXC-006 updated to Implemented (→ Fixed.md) per CONST-057/§11.4.33 and migrated to docs/Fixed.md per §11.4.19. Issues_Summary.md + Fixed_Summary.md regenerated; HTML/PDF exports refreshed.

Speed-programme outcome

All 6 phases / 31 tasks landed across rounds — P0 (baseline harness), P1 (LLM & startup wins), P2 (context-build speed), P3 (interactive & agent-loop levers), P4 (profile-gated tuning), P5 (dev-experience + submodule cascade + close-out). Headline measured wins (each with pasted in-session evidence per CONST-035): lazy Ollama discovery 67µs→2.7ns, lazy CLI startup ~8.85×, cache pre-warm ~7.6×, regexp hoist ~7.4×, repo-map cache ~10.6× warm, parallel search ~4.39×, incremental tree-

sitter $\sim 21\times$, small-model routing $\sim 5.87\times$, diff edits 94-99% token cut, fast-apply $\sim 516\times$, tool parallelism $\sim 5.99\times$, PGO -46% CPU-bound.

Honest PARTIALs (no green PASS claimed): P5-T01 build/test parallelism tuning is landed and correct but the suite-wall-time before/after delta was not captured as a pasted benchmark; P5-T02 is a partial single-provider internal/llm split (Cerebras extracted to internal/llm/providers/cerebras/) — a full 18-provider extraction is infeasible without an import cycle.

State

- docs/research/speed/05-coverage-ledger.md committed (+ HTML/PDF exports).
- HXC-006 closed in docs/Issues.md (migration tombstone) → full record in docs/Fixed.md.
- HXC-011 + HXC-012 filed Queued in docs/Issues.md.
- Issues_Summary.md / Fixed_Summary.md regenerated; tracker exports refreshed.

Next

- HXC-011 (helix_qa desktop-platform hollow-PASS) closed round 402 — Fixed (→ Fixed.md): the runner's run path on the desktop platform now genuinely executes a bank case's shell: action via os/exec and scores PASS/FAIL on the real exit code (reproduce-before-fix RED→GREEN; helix_qa commit 6b46df0, meta pointer bumped). HXC-012 (load_balancer.go race) closed round 401. Both pre-existing speed-programme findings now resolved.
- The speed programme is complete; no further speed tasks queued.

close-out¹³⁶ — round 403: HXC-008 governance-gap fix + HXC-007/HXC-009 verified-complete closure

What landed

1. **HXC-008 closed Fixed (→ Fixed.md)** (Bug). The two pre-existing CONST-055 G1 cascade-drift gaps surfaced in close-out¹³² / CG16 are fixed:
 - 1. docs/improvements/submodule_owned.txt line 10 referenced dependencies/HelixDevelopment/Models — a path that does not exist on disk; the CONST-052 batch-1 rename (a1ea3c8) had lowercased it to models (.gitmodules

- registers it lowercase). Corrected to dependencies/
HelixDevelopment/models. The cascade verifier (scripts/
verify-governance-cascade.sh) reads this file, so the false
FAIL: dependencies/HelixDevelopment/Models – path does
not exist on disk line is now gone.
- 1. helix_qa/CONSTITUTION.md was missing anchors
CONST-047..057 (CLAUDE.md/AGENTS.md already
carried them); VisionEngine/CONSTITUTION.md was missing
§11.4.69 (likewise). The contiguous CONST-047..057
block (217 lines) was cascaded into helix_qa/
CONSTITUTION.md from the meta CONSTITUTION.md; the
§11.4.69 anchor (80 lines) into VisionEngine/
CONSTITUTION.md. Both submodules committed + pushed
to all upstreams (helix_qa 1364d23; VisionEngine b3a13d8,
integrating upstream docs commit bfe3b06 first per
§11.4.71).
 - Verification: scripts/verify-governance-cascade.sh → ===
Result: 0 failures === PASS; scripts/verify-all-
constitution-rules.sh → Gates run: 6 / Failures: 0 (G1-G6
all PASS).
2. **HXC-007 closed Completed (→ Fixed.md)** (Task). Constitution
§11.4.68/70-74 cascade verified genuinely complete: meta
.gitmodules constitution pointer confirmed at 34a82b3; spot-
check `grep -c "11.4.70\|11.4.74"=6` across helix_qa,
dependencies/HelixDevelopment/LLMProvider, challenges
CLAUDE.md.
 3. **HXC-009 closed Completed (→ Fixed.md)** (Task). Owned-
submodule mirror-divergence reconciliation verified complete:
VisionEngine carries convergence merge 3485f5f and pushes
FF-clean to all 4 upstreams; helix_qa, LLMProvider,
challenges, containers, DocProcessor all converged per
CONST-061/§11.4.71 merge-first.

State

- All three items migrated to docs/Fixed.md per §11.4.19; docs/
Issues.md sections retained as migration tombstones.
- docs/Issues_Summary.md + docs/Fixed_Summary.md updated (open
count 6→3); all 8 tracker exports (4 HTML + 4 PDF)
regenerated via scripts/regenerate-tracker-exports.sh.
- Meta-repo: verifier-input fix + tracker migrations + exports
committed; helix_qa + VisionEngine meta pointers bumped.

Next

- No open Bug items remain. Open: HXC-001 (Task), HXC-003
(Feature), HXC-010 (Operator-blocked Task — CodeGraph end-
to-end verification awaiting LLM-backend credentials).
-

close-out¹³⁷ — round 463: HXC-003 closed Implemented (→ Fixed.md) — CONST-046 i18n migration campaign concluded

What landed

1. **HXC-003 closed Implemented (→ Fixed.md)** (Type: Feature, per CONST-057/§11.4.33). The CONST-046 i18n migration campaign — the longest-running Feature in docs/Issues.md — is **concluded**. The genuine user-facing (C) string-literal surface (UI text, prompts shown to the operator, error messages surfaced to end users, labels, helper text) is **exhausted across all 7 CONST-046 scope areas**:
 - 1. helix_code internal/, (2) cmd/, (3) applications/ — confirmed exhausted rounds 461/462 (applications final-sweep × 110 literals at meta HEAD 72389451).
 - 1. LLMsVerifier — round 452.
 - 1. helix_qa.
 - 1. all owned vasic-digital/* submodules + (7) all owned HelixDevelopment/* submodules — rounds 413/441.
2. **Scale + anti-bluff posture**. Across ~91-462 rounds, **tens of thousands of literals** were migrated through i18n seams — nicksnyder/go-i18n/v2 Bundle/Localizer (Option D, design f9dc102, pkg/i18n foundation e29b075) plus locale-aware .toml/.yaml resource files. **Every migration round shipped paired-mutation anti-bluff tests** (per §1.1): a known unmigrated literal is planted and the test asserts the audit-gate reports FAIL — so a PASS certifies real i18n coverage rather than absence-of-error. The audit-gate scripts/audit-const046-hardcoded-content.sh --fail-on-new is enforced; each round reran --update-baseline so the snapshot shrank monotonically.
3. **Remaining audit hits are out of scope**. The ~55k audit-baseline hits that remain are **all OUT of CONST-046 scope** — (A) LLM prompt templates (strings addressed to a model, not a human), (B) wrapped-error developer-facing tech strings, identifier tokens, struct-tag keys, format-spec tokens, and test fixtures — classified file-by-file in docs/audits/2026-05-20-internal-const046-classification.md (Revision 2). CONST-046's invariant is satisfied: no user-facing text is hardcoded as a static literal; every such string is LLM-generated, i18n-loaded, or metadata-composed.

State

- HXC-003 migrated to docs/Fixed.md per §11.4.19; docs/Issues.md section retained as a migration tombstone.

- docs/Issues_Summary.md updated (open count 3→2; Feature open 1→0); docs/Fixed_Summary.md updated (Feature count 67→68; total closed 95→96; open snapshot 3→2).
- All 8 tracker exports (4 HTML + 4 PDF) regenerated via scripts/regenerate-tracker-exports.sh.

Next

- Open set is now exactly 2 items: **HXC-001** (CONST-052 rename programme — Task, In progress; ~37 submodule-entangled leaf renames + parent dirs + 59 Upstreams dirs deferred) and **HXC-010** (Kimi CLI + Qwen Code CodeGraph end-to-end verification — Operator-blocked Task; awaiting LLM-backend credentials). No open Bug or Feature items remain.
-

close-out¹³⁸ — round 464: HXC-010 closed Completed (→ Fixed.md) — Kimi/Qwen CodeGraph end-to-end verified with operator-provided credentials

Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): *“all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can’t be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!”*

What landed

1. **HXC-010 closed Completed (→ Fixed.md)** (Type: Task, per CONST-057/§11.4.33). The operator supplied OpenAI-compatible router credentials (/home/milosvasic/api_keys.sh), unblocking the two CodeGraph per-agent Challenges that had been Operator-blocked since round 463.
2. **§11.4.10.A pre-use leak-audit: CLEAN.** git grep + git log -S of the three relevant key values (KIMI_API_KEY, OPENROUTER_API_KEY, SILICONFLOW_API_KEY) confirmed none has ever been committed to a tracked file or git history.
3. **The originally-blocking backends remain blocked.** KIMI_API_KEY shares the **same exhausted account-level monthly billing-cycle quota** as Kimi’s bundled OAuth (exceeded_current_quota_error on api.kimi.com/coding/v1);

Qwen's bundled OAuth free tier is still discontinued; OPENROUTER_API_KEY had insufficient credit (~\$0.0007). Resolution: both agents driven against the **SiliconFlow** OpenAI-compatible router, which has credit and serves both target models with working tool-calling.

4. **Kimi CLI** — driven via an openai_legacy-type provider (config-file ~/.kimi/config-codegraph-or.toml carrying only a placeholder api_key); the real key injected at runtime via the OPENAI_API_KEY env var (kimi_cli/llm.py augment_provider_with_env_vars), model moonshotai/Kimi-K2.6.
Qwen Code — driven via --auth-type openai with OPENAI_API_KEY/OPENAI_BASE_URL/OPENAI_MODEL env vars (key NEVER written into the tracked .qwen/settings.json), model Qwen/Qwen3-Coder-30B-A3B-Instruct.
5. **Both Challenge scripts updated** (tools/codegraph/challenges/cg-challenge-05-kimi.sh + cg-challenge-07-qwen.sh) to honour HELIX_CG_OPENAI_API_KEY + HELIX_CG_OPENAI_BASE_URL (+ optional HELIX_CG_QWEN_MODEL) for the credentialed path, falling back to the bundled quota-gated provider when those env vars are absent — anti-bluff preserved (the connect-only fallback is still reported honestly when no creds are present).
6. **Result — both true tier-1 PASS.** cg-challenge-05-kimi.sh → CG-CHALLENGE-05: PASS (true end-to-end — agent invoked codegraph_* and returned real graph data); cg-challenge-07-qwen.sh → CG-CHALLENGE-07: PASS (true end-to-end ...). Each transcript shows the MCP loader connecting to the codegraph server (9 codegraph_* tools), the agent invoking codegraph_search for symbol Provider, the ToolResult/tool_result returning 10 real .go symbol paths from the scanned HelixCode graph (first: docs/helix_qa/HelixQA_Integration/research/testdata/raw/pkg_llm_provider.go:40), and the agent answering with a real file path.

State

- HXC-010 migrated to docs/Fixed.md per §11.4.19; docs/Issues.md section retained as a migration tombstone.
- Evidence captured under docs/research/codegraph/evidence/hxc010/ (both transcripts + README); secret-scan of every transcript confirms **no API-key value present**.
- docs/Issues_Summary.md updated (open count 2→1; Operator-blocked Task 1→0); docs/Fixed_Summary.md updated (Task count 7→8; total closed 96→97; open snapshot 2→1).
- All 8 tracker exports (4 HTML + 4 PDF) regenerated via scripts/regenerate-tracker-exports.sh.
- All 7 CodeGraph anti-bluff Challenges (CG-CHALLENGE-01..07) now true-end-to-end verified across Claude Code, OpenCode, Crush, Kimi CLI, and Qwen Code.

Next

- Open set is now exactly 1 item: **HXC-001** (CONST-052 rename programme — Task, In progress). No open Bug, Feature, or Operator-blocked items remain.